T3 group factors: general UV

One-loop renormalisable RGBeta structure prefactors across T3 models.

$$j = \frac{d-1}{2}, \qquad C_2(d) = \frac{d^2-1}{4}, \qquad T(d) = \frac{d(d^2-1)}{12}.$$

In every comparison table, $\mathcal{S}$ denotes the RGE structure multiplying the displayed group factor in $16\pi^2\beta$.

Coupling and interaction notation

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2-1)/4$
$T(d)$	$d(d^2-1)/12$
$C_{LLS_1S_2}^{ij;AB}$	$L_i L_j S_1^A S_2^B$
C_{ijab}	$L_i L_j \phi_a \phi_b$
C_5^{ij}	$(L_i^T C L_j) H H$
R_W	$16\pi^2 \beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}$

Additional general-interaction couplings

Symbol	Field content	Stored coupling / contraction
λ_7	$(H^\dagger)^2 (S_1^\dagger)^2$	`lambdaHHdagS1barS1bar`; invariant contraction model-dependent

λ_{12}	$H H^\dagger S_1^\dagger S_2^\dagger$	`lambdaHHdagS1barS2barCross`; invariant contraction model-dependent
λ_8	$(H^\dagger)^2 S_2^2$	`lambdaHHdagS2S2`; invariant contraction model-dependent
λ_{11}	$S_1 (S_1^\dagger)^2 S_2^\dagger$	`lambdaS1S1bar2S2bar`; invariant contraction model-dependent
λ_9	$(S_1^\dagger)^2 (S_2^\dagger)^2$	`lambdaS1bar2S2bar2`; invariant contraction model-dependent
λ_{10}	$S_1^\dagger S_2 (S_2^\dagger)^2$	`lambdaS1barS2S2bar2`; invariant contraction model-dependent
λ_{X_1}	$(S_2^\dagger)^4$	`lambdaT3ExtraQ000004I1`; contraction: `((1/3) * (del[SU2L @ S2, #1, #2] * del[SU2L @ S2, #3, #4]) + (1/3) * (del[SU2L @ S2, #1, #3] * del[SU2L @ S2, #2, #4]) + (1/3) * (del[SU2L @ S2, #1, #4] * del[SU2L @ S2, #2, #3]))`
λ_{X_2}	$S_2 (S_2^\dagger)^3$	`lambdaT3ExtraQ000013I1`; contraction: `((1/3) * (del[SU2L @ S2, #1, #2] * del[SU2L @ S2, #3, #4]) + (1/3) * (del[SU2L @ S2, #1, #3] * del[SU2L @ S2, #2, #4]) + (1/3) * (del[SU2L @ S2, #1, #4] * del[SU2L @ S2, #2, #3]))`
λ_{X_3}	definition unavailable in registry	`lambdaT3ExtraQ000112I1`; symbolic running parameter
λ_{X_4}	definition unavailable in registry	`lambdaT3ExtraQ000202I1`; symbolic running parameter
$\lambda_{X_5}^{(1)}$	$(S_1^\dagger)^2 S_2 S_2^\dagger$	`lambdaT3ExtraQ000211I1`; contraction: `del[SU2L @ S2, #3, #4]`
$\lambda_{X_5}^{(2)}$	$(S_1^\dagger)^2 S_2 S_2^\dagger$	`lambdaT3ExtraQ000211I2`; contraction: `(del[SU2L @ S2, #1, #2] * del[SU2L @ S2, #3, #4])`
λ_{X_6}	$(S_1^\dagger)^4$	`lambdaT3ExtraQ000400I1`; contraction: `1`
$\lambda_{X_7}^{(1)}$	$S_1 S_1^\dagger (S_2^\dagger)^2$	`lambdaT3ExtraQ001102I1`; contraction: `del[SU2L @ S2, #3, #4]`
$\lambda_{X_7}^{(2)}$	$S_1 S_1^\dagger (S_2^\dagger)^2$	`lambdaT3ExtraQ001102I2`; contraction: `(del[SU2L @ S2, #1, #2] * del[SU2L @ S2, #3, #4])`
λ_{X_8}	definition unavailable in registry	`lambdaT3ExtraQ001201I1`; symbolic running parameter
λ_{X_9}	$S_1 (S_1^\dagger)^3$	`lambdaT3ExtraQ001300I1`; contraction: `1`
$\lambda_{X_{10}}$	$(H^\dagger)^2 (S_2^\dagger)^2$	`lambdaT3ExtraQ020002I1`; contraction: `((1/2) * (eps[SU2L @ fund, #1, #3] * eps[SU2L @ fund, #2, #4]) + (1/2) * (eps[SU2L @ fund, #1, #4] * eps[SU2L @ fund, #2, #3]))`
$\lambda_{X_{11}}$	definition unavailable in registry	`lambdaT3ExtraQ020020I1`; symbolic running parameter
$\lambda_{X_{12}}$	$(H^\dagger)^2 S_1^\dagger S_2^\dagger$	`lambdaT3ExtraQ020101I1`; contraction: `delS2[SU2L, #4, #1, #2]`
$\lambda_{X_{13}}$	definition unavailable in registry	`lambdaT3ExtraQ020200I1`; symbolic running parameter
$\lambda_{X_{14}}$	$(H^\dagger)^2 S_1 S_2$	`lambdaT3ExtraQ021010I1`; contraction: `delS2[SU2L, #4, #1, #2]`
$\lambda_{X_{15}}$	$(H^\dagger)^2 (S_1)^2$	`lambdaT3ExtraQ022000I1`; contraction: `((1/2) * (del[SU2L @ fund, #1, #3] * del[SU2L @ fund, #2, #4]) + (1/2) * (del[SU2L @ fund, #1, #4] * del[SU2L @ fund, #2, #3]))`
$\lambda_{X_{16}}$	$H H^\dagger (S_2^\dagger)^2$	`lambdaT3ExtraQ110002I1`; contraction: `(del[SU2L @ fund, #2, #1] * del[SU2L @ S2, #3, #4])`

$\lambda_{X_{17}}^{(1)}$	$H\,H^\dagger\,S_1^\dagger\,S_2^\dagger$	`lambdaT3ExtraQ110101I1`; contraction: `(del[SU2L @ fund, #4, #1] * eps[SU2L @ fund, #2, #3])`
$\lambda_{X_{17}}^{(2)}$	$H\,H^\dagger\,S_1^\dagger\,S_2^\dagger$	`lambdaT3ExtraQ110101I2`; contraction: `(del[SU2L @ fund, #4, #1] * eps[SU2L @ fund, #2, #3])`
$\lambda_{X_{18}}$	$H\,H^\dagger\,(S_1^\dagger)^2$	`lambdaT3ExtraQ110200I1`; contraction: `del[SU2L @ fund, #2, #1]`
λ_{X_1}	$(S_2^\dagger)^4$	`lambdaT3GeneralQ000004I1`; contraction: `((1/3) * (del[SU2L @ S2, #1, #2] * del[SU2L @ S2, #3, #4]) + (1/3) * (del[SU2L @ S2, #1, #3] * del[SU2L @ S2, #2, #4]) + (1/3) * (del[SU2L @ S2, #1, #4] * del[SU2L @ S2, #2, #3]))`
λ_{X_2}	$S_2\,(S_2^\dagger)^3$	`lambdaT3GeneralQ000013I1`; contraction: `((1/3) * (del[SU2L @ S2, #1, #2] * del[SU2L @ S2, #3, #4]) + (1/3) * (del[SU2L @ S2, #1, #3] * del[SU2L @ S2, #2, #4]) + (1/3) * (del[SU2L @ S2, #1, #4] * del[SU2L @ S2, #2, #3]))`
$\lambda_{X_5}^{(1)}$	$(S_1^\dagger)^2\,S_2\,S_2^\dagger$	`lambdaT3GeneralQ000211I1`; contraction: `del[SU2L @ S2, #3, #4]`
$\lambda_{X_5}^{(2)}$	$(S_1^\dagger)^2\,S_2\,S_2^\dagger$	`lambdaT3GeneralQ000211I2`; contraction: `((1/2) * (del[SU2L @ S2, #1, #3] * del[SU2L @ S2, #2, #4]) + (1/2) * (del[SU2L @ S2, #1, #4] * del[SU2L @ S2, #2, #3]))`
λ_{X_6}	$(S_1^\dagger)^4$	`lambdaT3GeneralQ000400I1`; contraction: `1`
$\lambda_{X_7}^{(1)}$	$S_1\,S_1^\dagger\,(S_2^\dagger)^2$	`lambdaT3GeneralQ001102I1`; contraction: `del[SU2L @ S2, #3, #4]`
$\lambda_{X_7}^{(2)}$	$S_1\,S_1^\dagger\,(S_2^\dagger)^2$	`lambdaT3GeneralQ001102I2`; contraction: `((1/2) * (del[SU2L @ S2, #1, #3] * del[SU2L @ S2, #2, #4]) + (1/2) * (del[SU2L @ S2, #1, #4] * del[SU2L @ S2, #2, #3]))`
λ_{X_9}	$S_1\,(S_1^\dagger)^3$	`lambdaT3GeneralQ001300I1`; contraction: `1`
$\lambda_{X_{10}}$	$(H^\dagger)^2\,(S_2^\dagger)^2$	`lambdaT3GeneralQ020002I1`; contraction: `((1/2) * (eps[SU2L @ fund, #1, #3] * eps[SU2L @ fund, #2, #4]) + (1/2) * (eps[SU2L @ fund, #1, #4] * eps[SU2L @ fund, #2, #3]))`
$\lambda_{X_{12}}$	$(H^\dagger)^2\,S_1^\dagger\,S_2^\dagger$	`lambdaT3GeneralQ020101I1`; contraction: `delS2[SU2L, #4, #1, #2]`
$\lambda_{X_{14}}$	$(H^\dagger)^2\,S_1\,S_2$	`lambdaT3GeneralQ021010I1`; contraction: `delS2[SU2L, #4, #1, #2]`
$\lambda_{X_{15}}$	$(H^\dagger)^2\,(S_1)^2$	`lambdaT3GeneralQ022000I1`; contraction: `((1/2) * (del[SU2L @ fund, #1, #3] * del[SU2L @ fund, #2, #4]) + (1/2) * (del[SU2L @ fund, #1, #4] * del[SU2L @ fund, #2, #3]))`
$\lambda_{X_{16}}$	$H\,H^\dagger\,(S_2^\dagger)^2$	`lambdaT3GeneralQ110002I1`; contraction: `(del[SU2L @ fund, #2, #1] * del[SU2L @ S2, #3, #4])`
$\lambda_{X_{17}}^{(1)}$	$H\,H^\dagger\,S_1^\dagger\,S_2^\dagger$	`lambdaT3GeneralQ110101I1`; contraction: `(del[SU2L @ fund, #4, #1] * eps[SU2L @ fund, #2, #3])`
$\lambda_{X_{18}}$	$H\,H^\dagger\,(S_1^\dagger)^2$	`lambdaT3GeneralQ110200I1`; contraction: `del[SU2L @ fund, #2, #1]`
$\mu_{\text{T3GeneralS1S2}}$	scalar bilinear	`muT3GeneralS1S2`; see native Lagrangian for contraction

$\mu_{\mathrm{T3GeneralS1Sq}}$	scalar bilinear	`muT3GeneralS1Sq`; see native Lagrangian for contraction
$\mu_{\mathrm{T3GeneralS2Sq}}$	scalar bilinear	`muT3GeneralS2Sq`; see native Lagrangian for contraction
y_1^{conj}	$L\,F\,S_1^\dagger$	`y1Conj`; conjugate-scalar Yukawa
y_2^{conj}	$L\,F\,S_2^\dagger$	`y2Conj`; conjugate-scalar Yukawa

Field representations

run	S_1	S_2	F	α
A	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	0
A	$(1,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	-2
A	$(1,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	-4
B	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(1,1)$	1
B	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(1,0)$	-1
B	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$	-3
C	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$	1
C	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	-1
C	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$	-3
D	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$	0
D	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$	2
D	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	-2
E	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	0
E	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$	2
E	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$	-2
E	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$	-4

Complete general-interaction RGE prefactors

$\beta^{(1)}_{M_F}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$(((y_1)^T(y_1)^*M_F)_{ij}+((y_1^{\mathrm{conj}})^T(y_1^{\mathrm{conj}})^*M_F)_{ij})$	$\frac{1}{2}$	0	0
$(M_F(y_2)^\dagger y_2)_{ij}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$g_2^2(M_F)_{ij}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2(M_F)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{27}{2}$
$((y_1)^T(y_1)^*M_F)_{ij}$	0	$\frac{1}{2}$	$\frac{1}{2}$
$(M_F(y_2^{\mathrm{conj}})^\dagger y_2^{\mathrm{conj}})_{ij}$	0	$\frac{3}{4}$	0
$((M_F)^T(y_1)^\dagger y_1)_{ij}$	0	0	0
$((M_F)^T(y_2)^\dagger y_2)_{ij}$	0	0	0
$((y_2)^T(y_2)^*M_F)_{ij}$	0	0	0
$g_2^2(((M_F)^T)_{ij}+(M_F)_{ij})$	0	0	0
$((y_1^{\mathrm{conj}})^T(y_1^{\mathrm{conj}})^*M_F)_{ij}$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$(((y_1)^T(y_1)^*M_F)_{ij}+((y_1^{\mathrm{conj}})^T(y_1^{\mathrm{conj}})^*M_F)_{ij})$	0	0	0
$(M_F(y_2)^\dagger y_2)_{ij}$	1	0	1
$g_2^2(M_F)_{ij}$	0	0	0
$g_1^2(M_F)_{ij}$	-6	0	-6
$((y_1)^T(y_1)^*M_F)_{ij}$	1	1	1
$(M_F(y_2^{\mathrm{conj}})^\dagger y_2^{\mathrm{conj}})_{ij}$	0	0	0

$((M_F)^T(y_1)^\dagger y_1)_{ij}$	0	1	0
$((M_F)^T(y_2)^\dagger y_2)_{ij}$	0	1	0
$((y_2)^T(y_2)^* M_F)_{ij}$	0	1	0
$g_2^2(((M_F)^T)_{ij} + (M_F)_{ij})$	0	0	0
$((y_1^{\rm conj})^T(y_1^{\rm conj})^* M_F)_{ij}$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$(((y_1)^T(y_1)^* M_F)_{ij} + ((y_1^{\rm conj})^T(y_1^{\rm conj})^* M_F)_{ij})$	0	0	0
$(M_F(y_2)^\dagger y_2)_{ij}$	$\frac{1}{2}$	0	$\frac{1}{2}$
$g_2^2(M_F)_{ij}$	-12	0	-12
$g_1^2(M_F)_{ij}$	-6	0	-6
$((y_1)^T(y_1)^* M_F)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(M_F(y_2^{\rm conj})^\dagger y_2^{\rm conj})_{ij}$	0	0	0
$((M_F)^T(y_1)^\dagger y_1)_{ij}$	0	$\frac{1}{2}$	0
$((M_F)^T(y_2)^\dagger y_2)_{ij}$	0	$\frac{1}{2}$	0
$((y_2)^T(y_2)^* M_F)_{ij}$	0	$\frac{1}{2}$	0
$g_2^2(((M_F)^T)_{ij} + (M_F)_{ij})$	0	-6	0
$((y_1^{\rm conj})^T(y_1^{\rm conj})^* M_F)_{ij}$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$(((y_1)^T(y_1)^* M_F)_{ij} + ((y_1^{\rm conj})^T(y_1^{\rm conj})^* M_F)_{ij})$	$\frac{3}{4}$	0	0
$(M_F(y_2)^\dagger y_2)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$g_2^2(M_F)_{ij}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2(M_F)_{ij}$	$-\frac{3}{2}$	$-\frac{27}{2}$	$-\frac{3}{2}$
$((y_1)^T(y_1)^* M_F)_{ij}$	0	$\frac{3}{4}$	$\frac{3}{4}$
$(M_F(y_2^{\rm conj})^\dagger y_2^{\rm conj})_{ij}$	0	0	$\frac{1}{2}$
$((M_F)^T(y_1)^\dagger y_1)_{ij}$	0	0	0
$((M_F)^T(y_2)^\dagger y_2)_{ij}$	0	0	0
$((y_2)^T(y_2)^* M_F)_{ij}$	0	0	0
$g_2^2(((M_F)^T)_{ij} + (M_F)_{ij})$	0	0	0
$((y_1^{\rm conj})^T(y_1^{\rm conj})^* M_F)_{ij}$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$(((y_1)^T(y_1)^* M_F)_{ij} + ((y_1^{\rm conj})^T(y_1^{\rm conj})^* M_F)_{ij})$	0	0	0
$(M_F(y_2)^\dagger y_2)_{ij}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$g_2^2(M_F)_{ij}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2(M_F)_{ij}$	$-\frac{3}{2}$	$-\frac{27}{2}$	$-\frac{3}{2}$
$((y_1)^T(y_1)^* M_F)_{ij}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$(M_F(y_2^{\rm conj})^\dagger y_2^{\rm conj})_{ij}$	0	0	$\frac{3}{4}$
$((M_F)^T(y_1)^\dagger y_1)_{ij}$	0	0	0
$((M_F)^T(y_2)^\dagger y_2)_{ij}$	0	0	0
$((y_2)^T(y_2)^* M_F)_{ij}$	0	0	0
$g_2^2(((M_F)^T)_{ij} + (M_F)_{ij})$	0	0	0
$((y_1^{\rm conj})^T(y_1^{\rm conj})^* M_F)_{ij}$	$\frac{3}{4}$	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$(((y_1)^T(y_1)^* M_F)_{ij} + ((y_1^{\rm conj})^T(y_1^{\rm conj})^* M_F)_{ij})$	0
$(M_F(y_2)^\dagger y_2)_{ij}$	$\frac{3}{4}$
$g_2^2(M_F)_{ij}$	$-\frac{9}{2}$
$g_1^2(M_F)_{ij}$	$-\frac{27}{2}$

$((y_1)^T(y_1)^* M_F)_{ij}$	$\frac{3}{4}$
$(M_F(y_2^{\rm conj})^\dagger y_2^{\rm conj})_{ij}$	0
$((M_F)^T(y_1)^\dagger y_1)_{ij}$	0
$((M_F)^T(y_2)^\dagger y_2)_{ij}$	0
$((y_2)^T(y_2)^* M_F)_{ij}$	0
$g_2^2(((M_F)^T)_{ij} + (M_F)_{ij})$	0
$((y_1^{\rm conj})^T(y_1^{\rm conj})^* M_F)_{ij}$	0

$\beta_{g_2}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
g_2^3	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
g_2^3	$-\frac{17}{6}$	$-\frac{17}{6}$	$-\frac{17}{6}$

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
g_2^3	$\frac{31}{6}$	$\frac{7}{6}$	$\frac{31}{6}$

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
g_2^3	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
g_2^3	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$\mathcal{S}$	E ($\alpha = -4$)
g_2^3	$\frac{1}{6}$

$\beta_{g_3}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
g_3^3	-7	-7	-7

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
g_3^3	-7	-7	-7

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
g_3^3	-7	-7	-7

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)

g_3^3	-7	-7	-7

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
g_3^3	-7	-7	-7

$\mathcal{S}$	E ($\alpha=-4$)
g_3^3	-7

$\beta_{g_1}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
g_1^3	$\frac{59}{6}$	$\frac{55}{6}$	$\frac{163}{6}$

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
g_1^3	$\frac{25}{2}$	$\frac{43}{6}$	$\frac{25}{2}$

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
g_1^3	$\frac{41}{2}$	$\frac{43}{6}$	$\frac{41}{2}$

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
g_1^3	$\frac{55}{6}$	$\frac{163}{6}$	$\frac{59}{6}$

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
g_1^3	$\frac{59}{6}$	$\frac{179}{6}$	$\frac{59}{6}$

$\mathcal{S}$	E ($\alpha=-4$)
g_1^3	$\frac{179}{6}$

$\beta_{\lambda_{12}^{(1)}}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_{12}^{(1)}(4T_\nu^{(1)}+4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+2T_\nu^{(2)}+4\lambda_{S_2}^{(A)})$	1	0	0
$\lambda_{X_5}^{(1)}(8\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger)+4(4(\lambda_{X_5}^{(1)})^*+3(\lambda_{X_9})^*))$	1	0	0
$(\lambda_{X_5}^{(1)})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	8	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	-4	-4	-4
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger y_2(y_2)^\dagger)$	-4	0	0
$\lambda_5(\lambda_5)^*$	4	4	4
$\lambda_{X_9}(\lambda_{X_5}^{(1)})^*$	12	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	4	0	0
$g_2^2\lambda_{12}^{(1)}$	-12	-12	-12
$g_1^2\lambda_{12}^{(1)}$	-6	-6	-30

$\lambda_{12}^{(1)^2}$	4	4	4
$\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	4	4	4
$\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	8	8	8
$\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	4	4	4
$\lambda_{12}^{(1)}(4T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 4\lambda_{S_2}^{(A)})$	0	1	0
$\lambda_{X_7}^{(1)}(4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 4(5(\lambda_{X_2})^* + 4(\lambda_{X_7}^{(1)})^*))$	0	1	0
$(\lambda_{X_7}^{(1)})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	4	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0	-4	0
$\lambda_{X_2}(\lambda_{X_7}^{(1)})^*$	0	20	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	4	0
$\lambda_{12}^{(1)}(4T_\nu^{(1)} + 2T_\nu^{(2)} + 4\lambda_{S_2}^{(A)})$	0	0	1
g_1^4	0	0	48
$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$	0	0	0
g_2^4	0	0	0
$\lambda_{12}^{(A)^2}$	0	0	0
$\mathrm{Tr}(y_1(y_2)^\dagger y_2(y_1)^\dagger)$	0	0	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}(2(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}})^* + (\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$\lambda_{X_{17}}^{(1)}((\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}})^* + 2(\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$\lambda_9(\lambda_9)^*$	0	0	0
$\lambda_{10}((\lambda_{10})^* + (\lambda_{11})^*)$	0	0	0
$\lambda_{11}((\lambda_{10})^* + (\lambda_{11})^*)$	0	0	0
$\lambda_{12}^{(1)}(3T_\nu^{(1)} + 3T_\nu^{(2)})$	0	0	0
$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 4T_\nu^{(2)} + 4\lambda_{S_1}^{(A)})$	0	0	0
$\lambda_{X_5}^{(1)}(4\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 4(4(\lambda_{X_5}^{(1)})^* + 5(\lambda_{X_9})^*))$	0	0	0
$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 4T_\nu^{(2)} + 4\lambda_{S_1}^{(A)})$	0	0	0
$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 4\lambda_{S_1}^{(A)})$	0	0	0
$\lambda_{X_7}^{(1)}(8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 4(3(\lambda_{X_2})^* + 4(\lambda_{X_7}^{(1)})^*))$	0	0	0
$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)} + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_{X_9}(5(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}((\lambda_{X_5}^{(2)})^* + (\lambda_{X_9})^*)$	0	0	0
$\lambda_{S_1}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0
$\lambda_{S_2}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0
$\lambda_{12}^{(\times)^2}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	0	0	0
$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_{X_2}(5(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}((\lambda_{X_2})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_{12}^{(1)}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)} + 4\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_{X_5}^{(1)}(8\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 4(4(\lambda_{X_5}^{(1)})^* + 3(\lambda_{X_9})^*))$	0	0	0
$(\lambda_{X_5}^{(1)})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	-2	-2	-2
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger y_2(y_2)^\dagger)$	0	0	0
$\lambda_5(\lambda_5)^*$	3	3	3
$\lambda_{X_9}(\lambda_{X_5}^{(1)})^*$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$g_2^2\lambda_{12}^{(1)}$	-9	-9	-9
$g_1^2\lambda_{12}^{(1)}$	-15	-3	-15
$\lambda_{12}^{(1)^2}$	4	4	4

$\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	6	6	6
$\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	6	6	6
$\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	4	4	4
$\lambda_{12}^{(1)}(4T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 4\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_{X_7}^{(1)}(4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 4(5(\lambda_{X_2})^* + 4(\lambda_{X_7}^{(1)})^*))$	0	0	0
$(\lambda_{X_7}^{(1)})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{X_2}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0
$\lambda_{12}^{(1)}(4T_\nu^{(1)} + 2T_\nu^{(2)} + 4\lambda_{S_2}^{(A)})$	0	0	0
g_1^4	$\frac{27}{4}$	$\frac{3}{4}$	$\frac{27}{4}$
$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$	1	1	1
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
${\lambda_{12}^{(A)}}^2$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$\mathrm{Tr}(y_1(y_2)^\dagger y_2(y_1)^\dagger)$	0	-4	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}(2(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}})^* + (\lambda_{X_{17}}^{(1)})^*)$	0	1	0
$\lambda_{X_{17}}^{(1)}((\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}})^* + 2(\lambda_{X_{17}}^{(1)})^*)$	0	1	0
$\lambda_9(\lambda_9)^*$	0	24	0
$\lambda_{10}((\lambda_{10})^* + (\lambda_{11})^*)$	0	9	0
$\lambda_{11}((\lambda_{10})^* + (\lambda_{11})^*)$	0	9	0
$\lambda_{12}^{(1)}(3T_\nu^{(1)} + 3T_\nu^{(2)})$	0	0	0
$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 4T_\nu^{(2)} + 4\lambda_{S_1}^{(A)})$	0	0	0
$\lambda_{X_5}^{(1)}(4\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 4(4(\lambda_{X_5}^{(1)})^* + 5(\lambda_{X_9})^*))$	0	0	0
$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 4T_\nu^{(2)} + 4\lambda_{S_1}^{(A)})$	0	0	0
$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 4\lambda_{S_1}^{(A)})$	0	0	0
$\lambda_{X_7}^{(1)}(8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 4(3(\lambda_{X_2})^* + 4(\lambda_{X_7}^{(1)})^*))$	0	0	0
$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)} + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_{X_9}(5(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}((\lambda_{X_5}^{(2)})^* + (\lambda_{X_9})^*)$	0	0	0
$\lambda_{S_1}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0
$\lambda_{S_2}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	0	0	0
$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_{X_2}(5(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}((\lambda_{X_2})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_{12}^{(1)}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)} + 4\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_{X_5}^{(1)}(8\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 4(4(\lambda_{X_5}^{(1)})^* + 3(\lambda_{X_9})^*))$	0	0	0
$(\lambda_{X_5}^{(1)})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger y_2(y_2)^\dagger)$	0	0	0
$\lambda_5(\lambda_5)^*$	3	3	3
$\lambda_{X_9}(\lambda_{X_5}^{(1)})^*$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$g_2^2\lambda_{12}^{(1)}$	-9	-9	-9
$g_1^2\lambda_{12}^{(1)}$	-15	-3	-15
${\lambda_{12}^{(1)}}^2$	4	4	4
$\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	6	6	6

$\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	6	6	6
$\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	4	4	4
$\lambda_{12}^{(1)}(4T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 4\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_{X_7}^{(1)}(4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 4(5(\lambda_{X_2})^* + 4(\lambda_{X_7}^{(1)})^*))$	0	0	0
$(\lambda_{X_7}^{(1)})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{X_2}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0
$\lambda_{12}^{(1)}(4T_\nu^{(1)} + 2T_\nu^{(2)} + 4\lambda_{S_2}^{(A)})$	0	0	0
g_1^4	$\frac{27}{4}$	$\frac{3}{4}$	$\frac{27}{4}$
$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$	0	0	0
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
${\lambda_{12}^{(A)}}^2$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$\mathrm{Tr}(y_1(y_2)^\dagger y_2(y_1)^\dagger)$	0	-3	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + (\lambda_{X_{17}}^{(1)})^*)$	0	1	0
$\lambda_{X_{17}}^{(1)}((\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + 2(\lambda_{X_{17}}^{(1)})^*)$	0	1	0
$\lambda_9(\lambda_9)^*$	0	24	0
$\lambda_{10}((\lambda_{10})^* + (\lambda_{11})^*)$	0	9	0
$\lambda_{11}((\lambda_{10})^* + (\lambda_{11})^*)$	0	9	0
$\lambda_{12}^{(1)}(3T_\nu^{(1)} + 3T_\nu^{(2)})$	1	1	1
$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 4T_\nu^{(2)} + 4\lambda_{S_1}^{(A)})$	0	0	0
$\lambda_{X_5}^{(1)}(4\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 4(4(\lambda_{X_5}^{(1)})^* + 5(\lambda_{X_9})^*))$	0	0	0
$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 4T_\nu^{(2)} + 4\lambda_{S_1}^{(A)})$	0	0	0
$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 4\lambda_{S_1}^{(A)})$	0	0	0
$\lambda_{X_7}^{(1)}(8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 4(3(\lambda_{X_2})^* + 4(\lambda_{X_7}^{(1)})^*))$	0	0	0
$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)} + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_{X_9}(5(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}((\lambda_{X_5}^{(2)})^* + (\lambda_{X_9})^*)$	0	0	0
$\lambda_{S_1}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0
$\lambda_{S_2}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	0	0	0
$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_{X_2}(5(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}((\lambda_{X_2})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_{12}^{(1)}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)} + 4\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_{X_5}^{(1)}(8\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 4(4(\lambda_{X_5}^{(1)})^* + 3(\lambda_{X_9})^*))$	0	0	0
$(\lambda_{X_5}^{(1)})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	4	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	-4	-4	-4
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger y_2(y_2)^\dagger)$	-4	0	0
$\lambda_5(\lambda_5)^*$	4	4	4
$\lambda_{X_9}(\lambda_{X_5}^{(1)})^*$	20	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	4	0	0
$g_2^2\lambda_{12}^{(1)}$	-12	-12	-12
$g_1^2\lambda_{12}^{(1)}$	-6	-30	-6
${\lambda_{12}^{(1)}}^2$	4	4	4
$\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	8	8	8
$\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	4	4	4

$\lambda^{(1)}_{HS_1}\lambda^{(1)}_{HS_2}$	4	4	4
$\lambda^{(1)}_{12}(4T^{(1)}_\nu + 2T^{(2)}_\nu + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 4\lambda^{(A)}_{S_2})$	0	0	0
$\lambda^{(1)}_{X_7}(4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 4(5(\lambda_{X_2})^* + 4(\lambda^{(1)}_{X_7})^*))$	0	0	0
$(\lambda^{(1)}_{X_7})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	8
$\mathrm{Tr}(y_1(y_1)^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0	0	-4
$\lambda_{X_2}(\lambda^{(1)}_{X_7})^*$	0	0	12
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	4
$\lambda^{(1)}_{12}(4T^{(1)}_\nu + 2T^{(2)}_\nu + 4\lambda^{(A)}_{S_2})$	0	0	0
g_1^4	0	48	0
$\lambda^{(1)}_{12}(2T^{(1)}_\nu + 2T^{(2)}_\nu)$	0	0	0
g_2^4	0	0	0
${\lambda^{(A)}_{12}}^2$	0	0	0
$\mathrm{Tr}(y_1(y_2)^\dagger y_2(y_1)^\dagger)$	0	0	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + (\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda^{(1)}_{X_{17}}((\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + 2(\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda_9(\lambda_9)^*$	0	0	0
$\lambda_{10}((\lambda_{10})^* + (\lambda_{11})^*)$	0	0	0
$\lambda_{11}((\lambda_{10})^* + (\lambda_{11})^*)$	0	0	0
$\lambda^{(1)}_{12}(3T^{(1)}_\nu + 3T^{(2)}_\nu)$	0	0	0
$\lambda^{(1)}_{12}(2T^{(1)}_\nu + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 4T^{(2)}_\nu + 4\lambda^{(A)}_{S_1})$	1	0	0
$\lambda^{(1)}_{X_5}(4\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 4(4(\lambda^{(1)}_{X_5})^* + 5(\lambda_{X_9})^*))$	1	0	0
$\lambda^{(1)}_{12}(2T^{(1)}_\nu + 4T^{(2)}_\nu + 4\lambda^{(A)}_{S_1})$	0	1	0
$\lambda^{(1)}_{12}(2T^{(1)}_\nu + 4T^{(2)}_\nu + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 4\lambda^{(A)}_{S_1})$	0	0	1
$\lambda^{(1)}_{X_7}(8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 4(3(\lambda_{X_2})^* + 4(\lambda^{(1)}_{X_7})^*))$	0	0	1
$\lambda^{(1)}_{12}(2T^{(1)}_\nu + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T^{(2)}_\nu + 4\lambda^{(A)}_{S_1} + 4\lambda^{(A)}_{S_2})$	0	0	0
$\lambda_{X_9}(5(\lambda^{(1)}_{X_5})^* + (\lambda^{(2)}_{X_5})^*)$	0	0	0
$\lambda^{(2)}_{X_5}((\lambda^{(2)}_{X_5})^* + (\lambda_{X_9})^*)$	0	0	0
$\lambda^{(1)}_{S_1}(8\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12})$	0	0	0
$\lambda^{(1)}_{S_2}(8\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12})$	0	0	0
${\lambda^{(\times)}_{12}}^2$	0	0	0
$\lambda^{(\times)}_{12}\lambda^{(A)}_{S_1}$	0	0	0
$\lambda^{(\times)}_{12}\lambda^{(A)}_{S_2}$	0	0	0
$\lambda^{(1)}_{12}(2T^{(1)}_\nu + 2T^{(2)}_\nu + 4\lambda^{(A)}_{S_1} + 4\lambda^{(A)}_{S_2})$	0	0	0
$\lambda^{(1)}_{12}(2T^{(1)}_\nu + 2T^{(2)}_\nu + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 4\lambda^{(A)}_{S_1} + 4\lambda^{(A)}_{S_2})$	0	0	0
$\lambda_{X_2}(5(\lambda^{(1)}_{X_7})^* + (\lambda^{(2)}_{X_7})^*)$	0	0	0
$\lambda^{(2)}_{X_7}((\lambda_{X_2})^* + (\lambda^{(2)}_{X_7})^*)$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda^{(1)}_{12}(4T^{(1)}_\nu + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T^{(2)}_\nu + 4\lambda^{(A)}_{S_2})$	0	0	0
$\lambda^{(1)}_{X_5}(8\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 4(4(\lambda^{(1)}_{X_5})^* + 3(\lambda_{X_9})^*))$	0	0	0
$(\lambda^{(1)}_{X_5})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	4	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$-\frac{22}{15}$	$-\frac{22}{15}$	$-\frac{22}{15}$
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger y_2(y_2)^\dagger)$	$-\frac{22}{15}$	0	0
$\lambda_5(\lambda_5)^*$	2	2	2
$\lambda_{X_9}(\lambda^{(1)}_{X_5})^*$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	2	0	0
$g_2^2\lambda^{(1)}_{12}$	-24	-24	-24
$g_1^2\lambda^{(1)}_{12}$	-6	-30	-6
${\lambda^{(1)}_{12}}^2$	4	4	4
$\lambda^{(1)}_{12}\lambda^{(1)}_{S_1}$	0	0	0
$\lambda^{(1)}_{12}\lambda^{(1)}_{S_2}$	0	0	0
$\lambda^{(1)}_{HS_1}\lambda^{(1)}_{HS_2}$	4	4	4

$\lambda_{12}^{(1)}(4T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 4\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_{X_7}^{(1)}(4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 4(5(\lambda_{X_2})^* + 4(\lambda_{X_7}^{(1)})^*))$	0	0	1
$(\lambda_{X_7}^{(1)})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	4
$\mathrm{Tr}(y_1(y_1)^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0	0	$-\frac{22}{15}$
$\lambda_{X_2}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	2
$\lambda_{12}^{(1)}(4T_\nu^{(1)} + 2T_\nu^{(2)} + 4\lambda_{S_2}^{(A)})$	0	0	0
g_1^4	0	48	0
$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$	0	0	0
g_2^4	$\frac{63}{5}$	$\frac{63}{5}$	$\frac{63}{5}$
${\lambda_{12}^{(A)}}^2$	16	16	16
$\mathrm{Tr}(y_1(y_2)^\dagger y_2(y_1)^\dagger)$	0	0	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}(2(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}})^* + (\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$\lambda_{X_{17}}^{(1)}((\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}})^* + 2(\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$\lambda_9(\lambda_9)^*$	0	0	0
$\lambda_{10}((\lambda_{10})^* + (\lambda_{11})^*)$	0	0	0
$\lambda_{11}((\lambda_{10})^* + (\lambda_{11})^*)$	0	0	0
$\lambda_{12}^{(1)}(3T_\nu^{(1)} + 3T_\nu^{(2)})$	0	0	0
$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 4T_\nu^{(2)} + 4\lambda_{S_1}^{(A)})$	0	0	0
$\lambda_{X_5}^{(1)}(4\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 4(4(\lambda_{X_5}^{(1)})^* + 5(\lambda_{X_9})^*))$	1	0	0
$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 4T_\nu^{(2)} + 4\lambda_{S_1}^{(A)})$	0	0	0
$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 4\lambda_{S_1}^{(A)})$	0	0	0
$\lambda_{X_7}^{(1)}(8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 4(3(\lambda_{X_2})^* + 4(\lambda_{X_7}^{(1)})^*))$	0	0	0
$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)} + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)})$	1	0	0
$\lambda_{X_9}(5(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	4	0	0
$\lambda_{X_5}^{(2)}((\lambda_{X_5}^{(2)})^* + (\lambda_{X_9})^*)$	4	0	0
$\lambda_{S_1}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	1	1	1
$\lambda_{S_2}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	1	1	1
${\lambda_{12}^{(\times)}}^2$	4	4	4
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	4	4	4
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	4	4	4
$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)})$	0	1	0
$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)})$	0	0	1
$\lambda_{X_2}(5(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	4
$\lambda_{X_7}^{(2)}((\lambda_{X_2})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	4

$\mathcal{S}$	E ($\alpha = -4$)
$\lambda_{12}^{(1)}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)} + 4\lambda_{S_2}^{(A)})$	0
$\lambda_{X_5}^{(1)}(8\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 4(4(\lambda_{X_5}^{(1)})^* + 3(\lambda_{X_9})^*))$	0
$(\lambda_{X_5}^{(1)})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$-\frac{22}{15}$
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger y_2(y_2)^\dagger)$	0
$\lambda_5(\lambda_5)^*$	2
$\lambda_{X_9}(\lambda_{X_5}^{(1)})^*$	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0
$g_2^2\lambda_{12}^{(1)}$	-24
$g_1^2\lambda_{12}^{(1)}$	-30
${\lambda_{12}^{(1)}}^2$	4
$\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	0
$\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	0
$\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	4
$\lambda_{12}^{(1)}(4T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 4\lambda_{S_2}^{(A)})$	0

$\lambda_{X_7}^{(1)}(4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger)+4(5(\lambda_{X_2})^*+4(\lambda_{X_7}^{(1)})^*))$	0
$(\lambda_{X_7}^{(1)})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0
$\lambda_{X_2}(\lambda_{X_7}^{(1)})^*$	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0
$\lambda_{12}^{(1)}(4T_\nu^{(1)}+2T_\nu^{(2)}+4\lambda_{S_2}^{(A)})$	0
g_1^4	48
$\lambda_{12}^{(1)}(2T_\nu^{(1)}+2T_\nu^{(2)})$	0
g_2^4	$\frac{63}{5}$
${\lambda_{12}^{(A)}}^2$	16
$\mathrm{Tr}(y_1(y_2)^\dagger y_2(y_1)^\dagger)$	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*+(\lambda_{X_{17}}^{(1)})^*)$	0
$\lambda_{X_{17}}^{(1)}((\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*+2(\lambda_{X_{17}}^{(1)})^*)$	0
$\lambda_9(\lambda_9)^*$	0
$\lambda_{10}((\lambda_{10})^*+(\lambda_{11})^*)$	0
$\lambda_{11}((\lambda_{10})^*+(\lambda_{11})^*)$	0
$\lambda_{12}^{(1)}(3T_\nu^{(1)}+3T_\nu^{(2)})$	0
$\lambda_{12}^{(1)}(2T_\nu^{(1)}+2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+4T_\nu^{(2)}+4\lambda_{S_1}^{(A)})$	0
$\lambda_{X_5}^{(1)}(4\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger)+4(4(\lambda_{X_5}^{(1)})^*+5(\lambda_{X_9})^*))$	0
$\lambda_{12}^{(1)}(2T_\nu^{(1)}+4T_\nu^{(2)}+4\lambda_{S_1}^{(A)})$	0
$\lambda_{12}^{(1)}(2T_\nu^{(1)}+4T_\nu^{(2)}+4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+4\lambda_{S_1}^{(A)})$	0
$\lambda_{X_7}^{(1)}(8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger)+4(3(\lambda_{X_2})^*+4(\lambda_{X_7}^{(1)})^*))$	0
$\lambda_{12}^{(1)}(2T_\nu^{(1)}+2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+2T_\nu^{(2)}+4\lambda_{S_1}^{(A)}+4\lambda_{S_2}^{(A)})$	0
$\lambda_{X_9}(5(\lambda_{X_5}^{(1)})^*+(\lambda_{X_5}^{(2)})^*)$	0
$\lambda_{X_5}^{(2)}((\lambda_{X_5}^{(2)})^*+(\lambda_{X_9})^*)$	0
$\lambda_{S_1}^{(1)}(8\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$	1
$\lambda_{S_2}^{(1)}(8\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$	1
${\lambda_{12}^{(\times)}}^2$	4
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	4
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	4
$\lambda_{12}^{(1)}(2T_\nu^{(1)}+2T_\nu^{(2)}+4\lambda_{S_1}^{(A)}+4\lambda_{S_2}^{(A)})$	1
$\lambda_{12}^{(1)}(2T_\nu^{(1)}+2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+4\lambda_{S_1}^{(A)}+4\lambda_{S_2}^{(A)})$	0
$\lambda_{X_2}(5(\lambda_{X_7}^{(1)})^*+(\lambda_{X_7}^{(2)})^*)$	0
$\lambda_{X_7}^{(2)}((\lambda_{X_2})^*+(\lambda_{X_7}^{(2)})^*)$	0

$\beta^{(1)}_{\lambda_{12}^{(A)}}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_{12}^{(A)}(2T_\nu^{(1)}+2T_\nu^{(2)})$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	0	0	0
$\lambda_5(\lambda_5)^*$	0	0	0
$g_1^2(18g_2^2-15\lambda_{12}^{(A)})$	0	0	0
$g_2^2\lambda_{12}^{(A)}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	0	0	0
$\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	0	0	0
$\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	0	0	0
$\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$	0	0	0
$\lambda_{X_{17}}^{(1)}(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*$	0	0	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(\lambda_{X_{17}}^{(1)})^*$	0	0	0
$\lambda_9(\lambda_9)^*$	0	0	0
$\lambda_{10}(5(\lambda_{10})^*+(\lambda_{11})^*)$	0	0	0

$\lambda_{11}((\lambda_{10})^* + 5(\lambda_{11})^*)$	0	0	0
$g_1^2(-6g_2^2 - 3\lambda_{12}^{(A)})$	0	0	0
$\lambda_{12}^{(A)}(3T_\nu^{(1)} + 3T_\nu^{(2)})$	0	0	0
$\mathrm{Tr}(y_1(y_2)^\dagger y_2(y_1)^\dagger)$	0	0	0
$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3})$	0	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger y_2(y_2)^\dagger)$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$g_1^2\lambda_{12}^{(A)}$	0	0	0
$\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	0	0	0
$\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	0	0	0
$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3})$	0	0	0
$g_1^2(16g_2^2 - 30\lambda_{12}^{(A)})$	0	0	0
$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + \frac{16\lambda_{12}^{(\times)}}{3})$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$	1	1	1
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$-\frac{8}{3}$	$-\frac{8}{3}$	$-\frac{8}{3}$
$\lambda_5(\lambda_5)^*$	$\frac{4}{3}$	$\frac{4}{3}$	$\frac{4}{3}$
$g_1^2(18g_2^2 - 15\lambda_{12}^{(A)})$	1	0	1
$g_2^2\lambda_{12}^{(A)}$	-9	-9	-9
$\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	8	8	8
$\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	2	2	2
$\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	2	2	2
$\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$	1	1	1
$\lambda_{X_{17}}^{(1)}(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}})^*$	0	4	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}(\lambda_{X_{17}}^{(1)})^*$	0	4	0
$\lambda_9(\lambda_9)^*$	0	-64	0
$\lambda_{10}(5(\lambda_{10})^* + (\lambda_{11})^*)$	0	-4	0
$\lambda_{11}((\lambda_{10})^* + 5(\lambda_{11})^*)$	0	-4	0
$g_1^2(-6g_2^2 - 3\lambda_{12}^{(A)})$	0	1	0
$\lambda_{12}^{(A)}(3T_\nu^{(1)} + 3T_\nu^{(2)})$	0	0	0
$\mathrm{Tr}(y_1(y_2)^\dagger y_2(y_1)^\dagger)$	0	0	0
$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3})$	0	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger y_2(y_2)^\dagger)$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$g_1^2\lambda_{12}^{(A)}$	0	0	0
$\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	0	0	0
$\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	0	0	0
$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3})$	0	0	0
$g_1^2(16g_2^2 - 30\lambda_{12}^{(A)})$	0	0	0
$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + \frac{16\lambda_{12}^{(\times)}}{3})$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$-\frac{2}{3}$	$-\frac{2}{3}$	$-\frac{2}{3}$
$\lambda_5(\lambda_5)^*$	$\frac{4}{3}$	$\frac{4}{3}$	$\frac{4}{3}$

$g_1^2(18g_2^2 - 15\lambda_{12}^{(A)})$	1	0	1
$g_2^2\lambda_{12}^{(A)}$	-9	-9	-9
$\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	8	8	8
$\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	2	2	2
$\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	2	2	2
$\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$	1	1	1
$\lambda_{X_{17}}^{(1)}(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})^*$	0	4	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_{X_{17}}^{(1)})^*$	0	4	0
$\lambda_9(\lambda_9)^*$	0	-64	0
$\lambda_{10}(5(\lambda_{10})^* + (\lambda_{11})^*)$	0	-4	0
$\lambda_{11}((\lambda_{10})^* + 5(\lambda_{11})^*)$	0	-4	0
$g_1^2(-6g_2^2 - 3\lambda_{12}^{(A)})$	0	1	0
$\lambda_{12}^{(A)}(3T_\nu^{(1)} + 3T_\nu^{(2)})$	1	1	1
$\text{Tr}(y_1(y_2)^\dagger y_2(y_1)^\dagger)$	0	$-\frac{8}{3}$	0
$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2\,\text{Tr}(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger) + 2T_\nu^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3})$	0	0	0
$\text{Tr}(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger y_2(y_2)^\dagger)$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$g_1^2\lambda_{12}^{(A)}$	0	0	0
$\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	0	0	0
$\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	0	0	0
$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3})$	0	0	0
$g_1^2(16g_2^2 - 30\lambda_{12}^{(A)})$	0	0	0
$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\text{Tr}(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger) + \frac{16\lambda_{12}^{(\times)}}{3})$	0	0	0
$\text{Tr}(y_1(y_1)^\dagger y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger)$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$	0	0	0
$\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	0	0	0
$\lambda_5(\lambda_5)^*$	0	0	0
$g_1^2(18g_2^2 - 15\lambda_{12}^{(A)})$	0	0	0
$g_2^2\lambda_{12}^{(A)}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	0	0	0
$\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	0	0	0
$\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	0	0	0
$\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$	0	0	0
$\lambda_{X_{17}}^{(1)}(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})^*$	0	0	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_{X_{17}}^{(1)})^*$	0	0	0
$\lambda_9(\lambda_9)^*$	0	0	0
$\lambda_{10}(5(\lambda_{10})^* + (\lambda_{11})^*)$	0	0	0
$\lambda_{11}((\lambda_{10})^* + 5(\lambda_{11})^*)$	0	0	0
$g_1^2(-6g_2^2 - 3\lambda_{12}^{(A)})$	0	0	0
$\lambda_{12}^{(A)}(3T_\nu^{(1)} + 3T_\nu^{(2)})$	0	0	0
$\text{Tr}(y_1(y_2)^\dagger y_2(y_1)^\dagger)$	0	0	0
$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2\,\text{Tr}(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger) + 2T_\nu^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3})$	0	0	0
$\text{Tr}(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger y_2(y_2)^\dagger)$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$g_1^2\lambda_{12}^{(A)}$	0	0	0
$\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	0	0	0
$\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	0	0	0
$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3})$	0	0	0

$g_1^2(16g_2^2 - 30\lambda_{12}^{(A)})$	0	0	0
$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + \frac{16\lambda_{12}^{(\times)}}{3})$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$\frac{2}{3}$	$\frac{2}{3}$	$\frac{2}{3}$
$\lambda_5(\lambda_5)^*$	$-\frac{2}{9}$	$-\frac{2}{9}$	$-\frac{2}{9}$
$g_1^2(18g_2^2 - 15\lambda_{12}^{(A)})$	0	0	0
$g_2^2\lambda_{12}^{(A)}$	-24	-24	-24
$\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	8	8	8
$\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	2	2	2
$\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	2	2	2
$\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$	3	3	3
$\lambda_{X_{17}}^{(1)}(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}})^*$	0	0	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}(\lambda_{X_{17}}^{(1)})^*$	0	0	0
$\lambda_9(\lambda_9)^*$	0	0	0
$\lambda_{10}(5(\lambda_{10})^* + (\lambda_{11})^*)$	0	0	0
$\lambda_{11}((\lambda_{10})^* + 5(\lambda_{11})^*)$	0	0	0
$g_1^2(-6g_2^2 - 3\lambda_{12}^{(A)})$	0	0	0
$\lambda_{12}^{(A)}(3T_\nu^{(1)} + 3T_\nu^{(2)})$	0	0	0
$\mathrm{Tr}(y_1(y_2)^\dagger y_2(y_1)^\dagger)$	0	0	0
$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3})$	1	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger y_2(y_2)^\dagger)$	$-\frac{2}{3}$	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	$-\frac{1}{3}$	0	0
$g_1^2\lambda_{12}^{(A)}$	-6	0	-6
$\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	16	16	16
$\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	16	16	16
$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3})$	0	1	0
$g_1^2(16g_2^2 - 30\lambda_{12}^{(A)})$	0	1	0
$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + \frac{16\lambda_{12}^{(\times)}}{3})$	0	0	1
$\mathrm{Tr}(y_1(y_1)^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0	0	$-\frac{2}{3}$
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	$-\frac{1}{3}$

$\mathcal{S}$	E ($\alpha = -4$)
$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$\frac{2}{3}$
$\lambda_5(\lambda_5)^*$	$-\frac{2}{9}$
$g_1^2(18g_2^2 - 15\lambda_{12}^{(A)})$	0
$g_2^2\lambda_{12}^{(A)}$	-24
$\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	8
$\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	2
$\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	2
$\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$	3
$\lambda_{X_{17}}^{(1)}(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}})^*$	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}(\lambda_{X_{17}}^{(1)})^*$	0
$\lambda_9(\lambda_9)^*$	0
$\lambda_{10}(5(\lambda_{10})^* + (\lambda_{11})^*)$	0
$\lambda_{11}((\lambda_{10})^* + 5(\lambda_{11})^*)$	0

$g_1^2(-6g_2^2 - 3\lambda_{12}^{(A)})$	0
$\lambda_{12}^{(A)}(3T_\nu^{(1)} + 3T_\nu^{(2)})$	0
$\mathrm{Tr}(y_1(y_2)^\dagger y_2(y_1)^\dagger)$	0
$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3})$	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger y_2(y_2)^\dagger)$	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0
$g_1^2\lambda_{12}^{(A)}$	0
$\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	16
$\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	16
$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3})$	1
$g_1^2(16g_2^2 - 30\lambda_{12}^{(A)})$	1
$\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + \frac{16\lambda_{12}^{(\times)}}{3})$	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0

$\beta_{\lambda_{12}^{(\times)}}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda_{12}^{(\times)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)})$	0	0	0
$\lambda_{X_5}^{(2)}(2\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 8(\lambda_{X_5}^{(1)})^* + 10(\lambda_{X_5}^{(2)})^* + 4(\lambda_{X_9})^*)$	0	0	0
$(\lambda_{X_5}^{(2)})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	0	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger y_2(y_2)^\dagger)$	0	0	0
$\lambda_5(\lambda_5)^*$	0	0	0
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(2)})^*$	0	0	0
$\lambda_{X_9}(\lambda_{X_5}^{(2)})^*$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
g_2^4	0	0	0
$g_2^2\lambda_{12}^{(\times)}$	0	0	0
$g_1^2\lambda_{12}^{(\times)}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	0	0	0
$\lambda_{12}^{(\times)^2}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	0	0	0
$\lambda_{12}^{(\times)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$	0	0	0
$\lambda_{12}^{(\times)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_7}^{(2)}(2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 4(\lambda_{X_2})^* + 8(\lambda_{X_7}^{(1)})^* + 10(\lambda_{X_7}^{(2)})^*)$	0	0	0
$(\lambda_{X_7}^{(2)})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{X_2}(\lambda_{X_7}^{(2)})^*$	0	0	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(2)})^*$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_{12}^{(\times)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)})$	0	0	0
$\lambda_{X_5}^{(2)}(2\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 8(\lambda_{X_5}^{(1)})^* + 10(\lambda_{X_5}^{(2)})^* + 4(\lambda_{X_9})^*)$	0	0	0
$(\lambda_{X_5}^{(2)})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	0	0	0

$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger y_2(y_2)^\dagger)$	0	0	0
$\lambda_5(\lambda_5)^*$	0	0	0
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(2)})^*$	0	0	0
$\lambda_{X_9}(\lambda_{X_5}^{(2)})^*$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
g_2^4	0	0	0
$g_2^2\lambda_{12}^{(\times)}$	0	0	0
$g_1^2\lambda_{12}^{(\times)}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	0	0	0
$\lambda_{12}^{(\times)}(2T_\nu^{(1)}+2T_\nu^{(2)})$	0	0	0
$\lambda_{12}^{(\times)}(2T_\nu^{(1)}+2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_7}^{(2)}(2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger)+4(\lambda_{X_2})^*+8(\lambda_{X_7}^{(1)})^*+10(\lambda_{X_7}^{(2)})^*)$	0	0	0
$(\lambda_{X_7}^{(2)})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{X_2}(\lambda_{X_7}^{(2)})^*$	0	0	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(2)})^*$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda_{12}^{(\times)}(2T_\nu^{(1)}+2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+2T_\nu^{(2)})$	0	0	0
$\lambda_{X_5}^{(2)}(2\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger)+8(\lambda_{X_5}^{(1)})^*+10(\lambda_{X_5}^{(2)})^*+4(\lambda_{X_9})^*)$	0	0	0
$(\lambda_{X_5}^{(2)})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	0	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger y_2(y_2)^\dagger)$	0	0	0
$\lambda_5(\lambda_5)^*$	0	0	0
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(2)})^*$	0	0	0
$\lambda_{X_9}(\lambda_{X_5}^{(2)})^*$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
g_2^4	0	0	0
$g_2^2\lambda_{12}^{(\times)}$	0	0	0
$g_1^2\lambda_{12}^{(\times)}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	0	0	0
$\lambda_{12}^{(\times)}(2T_\nu^{(1)}+2T_\nu^{(2)})$	0	0	0
$\lambda_{12}^{(\times)}(2T_\nu^{(1)}+2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_7}^{(2)}(2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger)+4(\lambda_{X_2})^*+8(\lambda_{X_7}^{(1)})^*+10(\lambda_{X_7}^{(2)})^*)$	0	0	0
$(\lambda_{X_7}^{(2)})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{X_2}(\lambda_{X_7}^{(2)})^*$	0	0	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(2)})^*$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda_{12}^{(\times)}(2T_\nu^{(1)}+2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+2T_\nu^{(2)})$	0	0	0
$\lambda_{X_5}^{(2)}(2\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger)+8(\lambda_{X_5}^{(1)})^*+10(\lambda_{X_5}^{(2)})^*+4(\lambda_{X_9})^*)$	0	0	0
$(\lambda_{X_5}^{(2)})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	0	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger y_2(y_2)^\dagger)$	0	0	0
$\lambda_5(\lambda_5)^*$	0	0	0
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(2)})^*$	0	0	0
$\lambda_{X_9}(\lambda_{X_5}^{(2)})^*$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
g_2^4	0	0	0
$g_2^2\lambda_{12}^{(\times)}$	0	0	0
$g_1^2\lambda_{12}^{(\times)}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	0	0	0
$\lambda_{12}^{(\times)}(2T_\nu^{(1)}+2T_\nu^{(2)})$	0	0	0
$\lambda_{12}^{(\times)}(2T_\nu^{(1)}+2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_7}^{(2)}(2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger)+4(\lambda_{X_2})^*+8(\lambda_{X_7}^{(1)})^*+10(\lambda_{X_7}^{(2)})^*)$	0	0	0
$(\lambda_{X_7}^{(2)})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{X_2}(\lambda_{X_7}^{(2)})^*$	0	0	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(2)})^*$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda_{12}^{(\times)}(2T_\nu^{(1)}+2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+2T_\nu^{(2)})$	1	0	0
$\lambda_{X_5}^{(2)}(2\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger)+8(\lambda_{X_5}^{(1)})^*+10(\lambda_{X_5}^{(2)})^*+4(\lambda_{X_9})^*)$	1	0	0
$(\lambda_{X_5}^{(2)})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	2	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$-\frac{4}{5}$	$-\frac{4}{5}$	$-\frac{4}{5}$
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger y_2(y_2)^\dagger)$	$-\frac{4}{5}$	0	0
$\lambda_5(\lambda_5)^*$	-1	-1	-1
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(2)})^*$	8	0	0
$\lambda_{X_9}(\lambda_{X_5}^{(2)})^*$	4	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	-1	0	0
g_2^4	$\frac{51}{10}$	$\frac{51}{10}$	$\frac{51}{10}$
$g_2^2\lambda_{12}^{(\times)}$	-24	-24	-24
$g_1^2\lambda_{12}^{(\times)}$	-6	-30	-6
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	8	8	8
${\lambda_{12}^{(\times)}}^2$	10	10	10
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)}$	2	2	2
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	-2	-2	-2
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)}$	2	2	2
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	-2	-2	-2
$\lambda_{12}^{(\times)}(2T_\nu^{(1)}+2T_\nu^{(2)})$	0	1	0
$\lambda_{12}^{(\times)}(2T_\nu^{(1)}+2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	1
$\lambda_{X_7}^{(2)}(2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger)+4(\lambda_{X_2})^*+8(\lambda_{X_7}^{(1)})^*+10(\lambda_{X_7}^{(2)})^*)$	0	0	1
$(\lambda_{X_7}^{(2)})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	2

$\mathrm{Tr}(y_1(y_1)^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0	0	$-\frac{4}{5}$
$\lambda_{X_2}(\lambda_{X_7}^{(2)})^*$	0	0	4
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(2)})^*$	0	0	8
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	-1

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_{12}^{(\times)}(2T_\nu^{(1)}+2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+2T_\nu^{(2)})$	0
$\lambda_{X_5}^{(2)}(2\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger)+8(\lambda_{X_5}^{(1)})^*+10(\lambda_{X_5}^{(2)})^*+4(\lambda_{X_9})^*)$	0
$(\lambda_{X_5}^{(2)})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$-\frac{4}{5}$
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger y_2(y_2)^\dagger)$	0
$\lambda_5(\lambda_5)^*$	-1
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(2)})^*$	0
$\lambda_{X_9}(\lambda_{X_5}^{(2)})^*$	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0
g_2^4	$\frac{51}{10}$
$g_2^2\lambda_{12}^{(\times)}$	-24
$g_1^2\lambda_{12}^{(\times)}$	-30
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	8
${\lambda_{12}^{(\times)}}^2$	10
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)}$	2
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	-2
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)}$	2
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	-2
$\lambda_{12}^{(\times)}(2T_\nu^{(1)}+2T_\nu^{(2)})$	1
$\lambda_{12}^{(\times)}(2T_\nu^{(1)}+2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0
$\lambda_{X_7}^{(2)}(2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger)+4(\lambda_{X_2})^*+8(\lambda_{X_7}^{(1)})^*+10(\lambda_{X_7}^{(2)})^*)$	0
$(\lambda_{X_7}^{(2)})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0
$\lambda_{X_2}(\lambda_{X_7}^{(2)})^*$	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(2)})^*$	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0

$\beta_{\lambda_1}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_1(12\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+4\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+12\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	1	1
$\lambda_5(\lambda_5)^*$	4	4	4
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	4	0	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	8	0	0
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
$g_1^2(\frac{3g_2^2}{2}-3\lambda_1)$	1	1	1
$g_2^2\lambda_1$	-9	-9	-9
g_1^4	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
${\lambda_1}^2$	12	12	12
${\lambda_{HS_1}^{(1)}}^2$	2	2	2
${\lambda_{HS_2}^{(1)}}^2$	6	6	6
${\lambda_{HS_2}^{(A)}}^2$	3	3	3
T	-4	-4	-4
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	4	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	24	0

$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0	0	0
${\lambda_{HS_1}^{(A)}}^2$	0	0	0
$\lambda_7(\lambda_7)^*$	0	0	0
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*$	0	0	0
$\lambda_8(\lambda_8)^*$	0	0	0
$\lambda^{(1)}_{X_{17}}(\lambda^{(1)}_{X_{17}})^*$	0	0	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_1(12\,\text{Tr}(Y_d(Y_d)^\dagger) + 4\,\text{Tr}(Y_e(Y_e)^\dagger) + 12\,\text{Tr}(Y_u(Y_u)^\dagger))$	1	1	1
$\lambda_5(\lambda_5)^*$	4	4	4
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	0	0	0
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
$g_1^2(\frac{3g_2^2}{2} - 3\lambda_1)$	1	1	1
$g_2^2\lambda_1$	-9	-9	-9
g_1^4	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
${\lambda_1}^2$	12	12	12
${\lambda_{HS_1}^{(1)}}^2$	4	4	4
${\lambda_{HS_2}^{(1)}}^2$	4	4	4
${\lambda_{HS_2}^{(A)}}^2$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
T	-4	-4	-4
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	0	0
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	8	0	0
${\lambda_{HS_1}^{(A)}}^2$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
$\lambda_7(\lambda_7)^*$	0	8	0
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*$	0	4	0
$\lambda_8(\lambda_8)^*$	0	8	0
$\lambda^{(1)}_{X_{17}}(\lambda^{(1)}_{X_{17}})^*$	0	4	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	8

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_1(12\,\text{Tr}(Y_d(Y_d)^\dagger) + 4\,\text{Tr}(Y_e(Y_e)^\dagger) + 12\,\text{Tr}(Y_u(Y_u)^\dagger))$	1	1	1
$\lambda_5(\lambda_5)^*$	4	4	4
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	0	0	0
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
$g_1^2(\frac{3g_2^2}{2} - 3\lambda_1)$	1	1	1
$g_2^2\lambda_1$	-9	-9	-9
g_1^4	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
${\lambda_1}^2$	12	12	12
${\lambda_{HS_1}^{(1)}}^2$	4	4	4
${\lambda_{HS_2}^{(1)}}^2$	4	4	4
${\lambda_{HS_2}^{(A)}}^2$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
T	-4	-4	-4
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	0	0
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	8	0	0
${\lambda_{HS_1}^{(A)}}^2$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

$\lambda_7(\lambda_7)^*$	0	8	0
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*$	0	4	0
$\lambda_8(\lambda_8)^*$	0	8	0
$\lambda^{(1)}_{X_{17}}(\lambda^{(1)}_{X_{17}})^*$	0	4	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	8

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_1(12\,\text{Tr}(Y_d(Y_d)^\dagger) + 4\,\text{Tr}(Y_e(Y_e)^\dagger) + 12\,\text{Tr}(Y_u(Y_u)^\dagger))$	1	1	1
$\lambda_5(\lambda_5)^*$	4	4	4
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	4	0	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	24	0	0
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
$g_1^2(\frac{3g_2^2}{2} - 3\lambda_1)$	1	1	1
$g_2^2\lambda_1$	-9	-9	-9
g_1^4	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
${\lambda_1}^2$	12	12	12
${\lambda^{(1)}_{HS_1}}^2$	6	6	6
${\lambda^{(1)}_{HS_2}}^2$	2	2	2
${\lambda^{(A)}_{HS_2}}^2$	0	0	0
T	-4	-4	-4
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	4
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	0	8
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0	0	0
${\lambda^{(A)}_{HS_1}}^2$	3	3	3
$\lambda_7(\lambda_7)^*$	0	0	0
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*$	0	0	0
$\lambda_8(\lambda_8)^*$	0	0	0
$\lambda^{(1)}_{X_{17}}(\lambda^{(1)}_{X_{17}})^*$	0	0	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda_1(12\,\text{Tr}(Y_d(Y_d)^\dagger) + 4\,\text{Tr}(Y_e(Y_e)^\dagger) + 12\,\text{Tr}(Y_u(Y_u)^\dagger))$	1	1	1
$\lambda_5(\lambda_5)^*$	4	4	4
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	4	0	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	24	0	0
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
$g_1^2(\frac{3g_2^2}{2} - 3\lambda_1)$	1	1	1
$g_2^2\lambda_1$	-9	-9	-9
g_1^4	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
${\lambda_1}^2$	12	12	12
${\lambda^{(1)}_{HS_1}}^2$	6	6	6
${\lambda^{(1)}_{HS_2}}^2$	6	6	6
${\lambda^{(A)}_{HS_2}}^2$	3	3	3
T	-4	-4	-4
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	4
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	0	24
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0	0	0
${\lambda^{(A)}_{HS_1}}^2$	3	3	3
$\lambda_7(\lambda_7)^*$	0	0	0
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*$	0	0	0

$\lambda_8(\lambda_8)^*$	0	0	0
$\lambda^{(1)}_{X_{17}}(\lambda^{(1)}_{X_{17}})^*$	0	0	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	0

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_1(12\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+4\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+12\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1
$\lambda_5(\lambda_5)^*$	4
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	0
g_2^4	$\frac{9}{4}$
$g_1^2(\frac{3g_2^2}{2}-3\lambda_1)$	1
$g_2^2\lambda_1$	-9
g_1^4	$\frac{3}{4}$
${\lambda_1}^2$	12
${\lambda^{(1)}_{HS_1}}^2$	6
${\lambda^{(1)}_{HS_2}}^2$	6
${\lambda^{(A)}_{HS_2}}^2$	3
T	-4
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0
${\lambda^{(A)}_{HS_1}}^2$	3
$\lambda_7(\lambda_7)^*$	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*$	0
$\lambda_8(\lambda_8)^*$	0
$\lambda^{(1)}_{X_{17}}(\lambda^{(1)}_{X_{17}})^*$	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0

$\beta^{(1)}_{\lambda^{(1)}_{HS_1}}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda^{(1)}_{HS_1}(4T^{(1)}_\nu+4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	0	0
$\lambda_{X_{18}}(8\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger)+4(3(\lambda_{X_9})^*+4(\lambda_{X_{18}})^*))$	1	0	0
$(\lambda_{X_{18}})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	8	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger Y_e(Y_e)^\dagger)$	-4	-4	-4
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger Y_e(Y_e)^\dagger)$	-4	0	0
$\lambda_5(\lambda_5)^*$	12	12	12
$\lambda_{X_9}(\lambda_{X_{18}})^*$	12	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	12	0	0
$g_2^2\lambda^{(1)}_{HS_1}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2\lambda^{(1)}_{HS_1}$	$-\frac{3}{2}$	$-\frac{15}{2}$	$-\frac{51}{2}$
$\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$	6	6	6
$\lambda_1\lambda^{(1)}_{HS_1}$	6	6	6
${\lambda^{(1)}_{HS_1}}^2$	4	4	4
$\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	4	4	4
$\lambda^{(1)}_{HS_1}(4T^{(1)}_\nu+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	1	1
$\lambda_{X_{16}}(\lambda^{(1)}_{X_7})^*$	0	12	0
$\lambda^{(1)}_{X_7}(\lambda_{X_{16}})^*$	0	12	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	12	0
g_1^4	0	3	12

$\lambda^{(1)}_{HS_1}(2T^{(1)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0	0	0
g_2^4	0	0	0
$\lambda^{(A)\ 2}_{HS_1}$	0	0	0
$\lambda_7(\lambda_7)^*$	0	0	0
$\lambda_{11}((\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - (\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + 3(\lambda_{11})^* + 2(\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda^{(1)}_{X_{17}}(2(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - 3(\lambda_{11})^* + 4(\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda^{(1)}_{HS_1}(3T^{(1)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda^{(1)}_{HS_1}(2T^{(1)}_\nu + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda^{(A)}_{S_1})$	0	0	0
$\lambda_{X_{18}}(4\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 4(5(\lambda_{X_9})^* + 4(\lambda_{X_{18}})^*))$	0	0	0
$\lambda^{(1)}_{HS_1}(2T^{(1)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda^{(A)}_{S_1})$	0	0	0
$\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12})$	0	0	0
$\lambda_{X_{16}}(3(\lambda^{(1)}_{X_7})^* + (\lambda^{(2)}_{X_7})^*)$	0	0	0
$\lambda^{(2)}_{X_7}(\lambda_{X_{16}})^*$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda^{(1)}_{HS_1}(4T^{(1)}_\nu + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{X_{18}}(8\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 4(3(\lambda_{X_9})^* + 4(\lambda_{X_{18}})^*))$	0	0	0
$(\lambda_{X_{18}})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger Y_e(Y_e)^\dagger)$	-2	-2	-2
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger Y_e(Y_e)^\dagger)$	0	0	0
$\lambda_5(\lambda_5)^*$	6	6	6
$\lambda_{X_9}(\lambda_{X_{18}})^*$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$g_2^2\lambda^{(1)}_{HS_1}$	-9	-9	-9
$g_1^2\lambda^{(1)}_{HS_1}$	-3	-3	-15
$\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$	4	4	4
$\lambda_1\lambda^{(1)}_{HS_1}$	6	6	6
$\lambda^{(1)\ 2}_{HS_1}$	4	4	4
$\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	6	6	6
$\lambda^{(1)}_{HS_1}(4T^{(1)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{X_{16}}(\lambda^{(1)}_{X_7})^*$	0	0	0
$\lambda^{(1)}_{X_7}(\lambda_{X_{16}})^*$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0
g_1^4	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{27}{4}$
$\lambda^{(1)}_{HS_1}(2T^{(1)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	1	1
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	24	0	0
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
$\lambda^{(A)\ 2}_{HS_1}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$\lambda_7(\lambda_7)^*$	0	24	0
$\lambda_{11}((\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - (\lambda^{(1)}_{X_{17}})^*)$	0	3	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + 3(\lambda_{11})^* + 2(\lambda^{(1)}_{X_{17}})^*)$	0	1	0
$\lambda^{(1)}_{X_{17}}(2(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - 3(\lambda_{11})^* + 4(\lambda^{(1)}_{X_{17}})^*)$	0	1	0
$\lambda^{(1)}_{HS_1}(3T^{(1)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda^{(1)}_{HS_1}(2T^{(1)}_\nu + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda^{(A)}_{S_1})$	0	0	0
$\lambda_{X_{18}}(4\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 4(5(\lambda_{X_9})^* + 4(\lambda_{X_{18}})^*))$	0	0	0
$\lambda^{(1)}_{HS_1}(2T^{(1)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda^{(A)}_{S_1})$	0	0	0
$\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12})$	0	0	0
$\lambda_{X_{16}}(3(\lambda^{(1)}_{X_7})^* + (\lambda^{(2)}_{X_7})^*)$	0	0	0
$\lambda^{(2)}_{X_7}(\lambda_{X_{16}})^*$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda^{(1)}_{HS_1}(4T^{(1)}_\nu + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{X_{18}}(8\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 4(3(\lambda_{X_9})^* + 4(\lambda_{X_{18}})^*))$	0	0	0
$(\lambda_{X_{18}})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger Y_e(Y_e)^\dagger)$	-3	-3	-3
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger Y_e(Y_e)^\dagger)$	0	0	0
$\lambda_5(\lambda_5)^*$	6	6	6
$\lambda_{X_9}(\lambda_{X_{18}})^*$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$g_2^2\lambda^{(1)}_{HS_1}$	-9	-9	-9
$g_1^2\lambda^{(1)}_{HS_1}$	-3	-3	-15
$\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$	4	4	4
$\lambda_1\lambda^{(1)}_{HS_1}$	6	6	6
${\lambda^{(1)}_{HS_1}}^2$	4	4	4
$\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	6	6	6
$\lambda^{(1)}_{HS_1}(4T^{(1)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{X_{16}}(\lambda^{(1)}_{X_7})^*$	0	0	0
$\lambda^{(1)}_{X_7}(\lambda_{X_{16}})^*$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0
g_1^4	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{27}{4}$
$\lambda^{(1)}_{HS_1}(2T^{(1)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	24	0	0
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
${\lambda^{(A)}_{HS_1}}^2$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$\lambda_7(\lambda_7)^*$	0	24	0
$\lambda_{11}((\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - (\lambda^{(1)}_{X_{17}})^*)$	0	3	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + 3(\lambda_{11})^* + 2(\lambda^{(1)}_{X_{17}})^*)$	0	1	0
$\lambda^{(1)}_{X_{17}}(2(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - 3(\lambda_{11})^* + 4(\lambda^{(1)}_{X_{17}})^*)$	0	1	0
$\lambda^{(1)}_{HS_1}(3T^{(1)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	1	1
$\lambda^{(1)}_{HS_1}(2T^{(1)}_\nu + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda^{(A)}_{S_1})$	0	0	0
$\lambda_{X_{18}}(4\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 4(5(\lambda_{X_9})^* + 4(\lambda_{X_{18}})^*))$	0	0	0
$\lambda^{(1)}_{HS_1}(2T^{(1)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda^{(A)}_{S_1})$	0	0	0
$\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12})$	0	0	0
$\lambda_{X_{16}}(3(\lambda^{(1)}_{X_7})^* + (\lambda^{(2)}_{X_7})^*)$	0	0	0
$\lambda^{(2)}_{X_7}(\lambda_{X_{16}})^*$	0	0	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda^{(1)}_{HS_1}(4T^{(1)}_\nu + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{X_{18}}(8\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 4(3(\lambda_{X_9})^* + 4(\lambda_{X_{18}})^*))$	0	0	0
$(\lambda_{X_{18}})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	4	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger Y_e(Y_e)^\dagger)$	-2	-2	-2
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger Y_e(Y_e)^\dagger)$	-2	0	0
$\lambda_5(\lambda_5)^*$	4	4	4
$\lambda_{X_9}(\lambda_{X_{18}})^*$	20	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	4	0	0
$g_2^2\lambda^{(1)}_{HS_1}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$g_1^2\lambda^{(1)}_{HS_1}$	$-\frac{3}{2}$	$-\frac{15}{2}$	$-\frac{15}{2}$
$\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$	2	2	2
$\lambda_1\lambda^{(1)}_{HS_1}$	6	6	6
${\lambda^{(1)}_{HS_1}}^2$	4	4	4
$\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	8	8	8

$\lambda^{(1)}_{HS_1}(4T^{(1)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{X_{16}}(\lambda^{(1)}_{X_7})^*$	0	0	4
$\lambda^{(1)}_{X_7}(\lambda_{X_{16}})^*$	0	0	4
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	4
g_1^4	0	3	3
$\lambda^{(1)}_{HS_1}(2T^{(1)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0	0	0
g_2^4	6	6	6
${\lambda^{(A)}_{HS_1}}^2$	6	6	6
$\lambda_7(\lambda_7)^*$	0	0	0
$\lambda_{11}((\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - (\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + 3(\lambda_{11})^* + 2(\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda^{(1)}_{X_{17}}(2(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - 3(\lambda_{11})^* + 4(\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda^{(1)}_{HS_1}(3T^{(1)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda^{(1)}_{HS_1}(2T^{(1)}_\nu + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda^{(A)}_{S_1})$	1	0	0
$\lambda_{X_{18}}(4\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 4(5(\lambda_{X_9})^* + 4(\lambda_{X_{18}})^*))$	1	0	0
$\lambda^{(1)}_{HS_1}(2T^{(1)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda^{(A)}_{S_1})$	0	1	1
$\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12})$	0	0	0
$\lambda_{X_{16}}(3(\lambda^{(1)}_{X_7})^* + (\lambda^{(2)}_{X_7})^*)$	0	0	0
$\lambda^{(2)}_{X_7}(\lambda_{X_{16}})^*$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda^{(1)}_{HS_1}(4T^{(1)}_\nu + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{X_{18}}(8\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 4(3(\lambda_{X_9})^* + 4(\lambda_{X_{18}})^*))$	0	0	0
$(\lambda_{X_{18}})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	4	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger Y_e(Y_e)^\dagger)$	-2	-2	-2
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger Y_e(Y_e)^\dagger)$	-2	0	0
$\lambda_5(\lambda_5)^*$	4	4	4
$\lambda_{X_9}(\lambda_{X_{18}})^*$	20	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	4	0	0
$g_2^2\lambda^{(1)}_{HS_1}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$g_1^2\lambda^{(1)}_{HS_1}$	$-\frac{3}{2}$	$-\frac{15}{2}$	$-\frac{15}{2}$
$\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$	0	0	0
$\lambda_1\lambda^{(1)}_{HS_1}$	6	6	6
${\lambda^{(1)}_{HS_1}}^2$	4	4	4
$\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	8	8	8
$\lambda^{(1)}_{HS_1}(4T^{(1)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{X_{16}}(\lambda^{(1)}_{X_7})^*$	0	0	0
$\lambda^{(1)}_{X_7}(\lambda_{X_{16}})^*$	0	0	12
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	4
g_1^4	0	3	3
$\lambda^{(1)}_{HS_1}(2T^{(1)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0	0	0
g_2^4	6	6	6
${\lambda^{(A)}_{HS_1}}^2$	6	6	6
$\lambda_7(\lambda_7)^*$	0	0	0
$\lambda_{11}((\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - (\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + 3(\lambda_{11})^* + 2(\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda^{(1)}_{X_{17}}(2(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - 3(\lambda_{11})^* + 4(\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda^{(1)}_{HS_1}(3T^{(1)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda^{(1)}_{HS_1}(2T^{(1)}_\nu + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda^{(A)}_{S_1})$	1	0	0
$\lambda_{X_{18}}(4\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 4(5(\lambda_{X_9})^* + 4(\lambda_{X_{18}})^*))$	1	0	0

$\lambda^{(1)}_{HS_1}(2T^{(1)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda^{(A)}_{S_1})$	0	1	1
$\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12})$	1	1	1
$\lambda_{X_{16}}(3(\lambda^{(1)}_{X_7})^* + (\lambda^{(2)}_{X_7})^*)$	0	0	4
$\lambda^{(2)}_{X_7}(\lambda_{X_{16}})^*$	0	0	4

$\mathcal{S}$	E ($\alpha = -4$)
$\lambda^{(1)}_{HS_1}(4T^{(1)}_\nu + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$\lambda_{X_{18}}(8\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 4(3(\lambda_{X_9})^* + 4(\lambda_{X_{18}})^*))$	0
$(\lambda_{X_{18}})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	0
$\mathrm{Tr}(y_1(y_1)^\dagger Y_e(Y_e)^\dagger)$	-2
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger Y_e(Y_e)^\dagger)$	0
$\lambda_5(\lambda_5)^*$	4
$\lambda_{X_9}(\lambda_{X_{18}})^*$	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0
$g_2^2\lambda^{(1)}_{HS_1}$	$-\frac{33}{2}$
$g_1^2\lambda^{(1)}_{HS_1}$	$-\frac{51}{2}$
$\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$	0
$\lambda_1\lambda^{(1)}_{HS_1}$	6
$\lambda^{(1)\ 2}_{HS_1}$	4
$\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	8
$\lambda^{(1)}_{HS_1}(4T^{(1)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$\lambda_{X_{16}}(\lambda^{(1)}_{X_7})^*$	0
$\lambda^{(1)}_{X_7}(\lambda_{X_{16}})^*$	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0
g_1^4	12
$\lambda^{(1)}_{HS_1}(2T^{(1)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0
g_2^4	6
$\lambda^{(A)\ 2}_{HS_1}$	6
$\lambda_7(\lambda_7)^*$	0
$\lambda_{11}((\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - (\lambda^{(1)}_{X_{17}})^*)$	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + 3(\lambda_{11})^* + 2(\lambda^{(1)}_{X_{17}})^*)$	0
$\lambda^{(1)}_{X_{17}}(2(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - 3(\lambda_{11})^* + 4(\lambda^{(1)}_{X_{17}})^*)$	0
$\lambda^{(1)}_{HS_1}(3T^{(1)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$\lambda^{(1)}_{HS_1}(2T^{(1)}_\nu + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda^{(A)}_{S_1})$	0
$\lambda_{X_{18}}(4\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 4(5(\lambda_{X_9})^* + 4(\lambda_{X_{18}})^*))$	0
$\lambda^{(1)}_{HS_1}(2T^{(1)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda^{(A)}_{S_1})$	1
$\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12})$	1
$\lambda_{X_{16}}(3(\lambda^{(1)}_{X_7})^* + (\lambda^{(2)}_{X_7})^*)$	0
$\lambda^{(2)}_{X_7}(\lambda_{X_{16}})^*$	0

$\beta^{(1)}_{\lambda^{(A)}_{HS_1}}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda^{(A)}_{HS_1}(2T^{(1)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger Y_e(Y_e)^\dagger)$	0	0	0
$\lambda_5(\lambda_5)^*$	0	0	0
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0	0	0
$g_1^2(6g_2^2 - 3\lambda^{(A)}_{HS_1})$	0	0	0
$g_2^2\lambda^{(A)}_{HS_1}$	0	0	0

$\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	0	0	0
$\lambda_1\lambda_{HS_1}^{(A)}$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	0	0	0
$\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$	0	0	0
$\lambda_7(\lambda_7)^*$	0	0	0
$\lambda_{11}((\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + (\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + (\lambda_{11})^*)$	0	0	0
$\lambda_{X_{17}}^{(1)}((\lambda_{11})^* - 2(\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$g_1^2(-6g_2^2 - 3\lambda_{HS_1}^{(A)})$	0	0	0
$g_1^2(-18g_2^2 - 15\lambda_{HS_1}^{(A)})$	0	0	0
$\lambda_{HS_1}^{(A)}(3T_\nu^{(1)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)} + 2\,\text{Tr}(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger) + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger) + 16\lambda_{S_1}^{(A)})$	0	0	0
$\text{Tr}(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger Y_e(Y_e)^\dagger)$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$g_1^2\lambda_{HS_1}^{(A)}$	0	0	0
$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger) + 16\lambda_{S_1}^{(A)})$	0	0	0
$g_1^2(4g_2^2 - \frac{15\lambda_{HS_1}^{(A)}}{2})$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0
$g_1^2(-4g_2^2 - \frac{15\lambda_{HS_1}^{(A)}}{2})$	0	0	0
$g_1^2(-8g_2^2 - \frac{51\lambda_{HS_1}^{(A)}}{2})$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	1	1	1
$\text{Tr}(y_1(y_1)^\dagger Y_e(Y_e)^\dagger)$	$\frac{8}{3}$	$\frac{8}{3}$	$\frac{8}{3}$
$\lambda_5(\lambda_5)^*$	$\frac{16}{3}$	$\frac{16}{3}$	$\frac{16}{3}$
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	64	0	0
$g_1^2(6g_2^2 - 3\lambda_{HS_1}^{(A)})$	1	0	0
$g_2^2\lambda_{HS_1}^{(A)}$	-9	-9	-9
$\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	1	1	1
$\lambda_1\lambda_{HS_1}^{(A)}$	2	2	2
$\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	8	8	8
$\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$	2	2	2
$\lambda_7(\lambda_7)^*$	0	-64	0
$\lambda_{11}((\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + (\lambda_{X_{17}}^{(1)})^*)$	0	4	0
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + (\lambda_{11})^*)$	0	4	0
$\lambda_{X_{17}}^{(1)}((\lambda_{11})^* - 2(\lambda_{X_{17}}^{(1)})^*)$	0	4	0
$g_1^2(-6g_2^2 - 3\lambda_{HS_1}^{(A)})$	0	1	0
$g_1^2(-18g_2^2 - 15\lambda_{HS_1}^{(A)})$	0	0	1
$\lambda_{HS_1}^{(A)}(3T_\nu^{(1)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)} + 2\,\text{Tr}(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger) + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger) + 16\lambda_{S_1}^{(A)})$	0	0	0
$\text{Tr}(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger Y_e(Y_e)^\dagger)$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$g_1^2\lambda_{HS_1}^{(A)}$	0	0	0
$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger) + 16\lambda_{S_1}^{(A)})$	0	0	0
$g_1^2(4g_2^2 - \frac{15\lambda_{HS_1}^{(A)}}{2})$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0
$g_1^2(-4g_2^2 - \frac{15\lambda_{HS_1}^{(A)}}{2})$	0	0	0
$g_1^2(-8g_2^2 - \frac{51\lambda_{HS_1}^{(A)}}{2})$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger Y_e(Y_e)^\dagger)$	$-\frac{4}{3}$	$-\frac{4}{3}$	$-\frac{4}{3}$
$\lambda_5(\lambda_5)^*$	$\frac{16}{3}$	$\frac{16}{3}$	$\frac{16}{3}$
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	64	0	0
$g_1^2(6g_2^2-3\lambda_{HS_1}^{(A)})$	1	0	0
$g_2^2\lambda_{HS_1}^{(A)}$	-9	-9	-9
$\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	1	1	1
$\lambda_1\lambda_{HS_1}^{(A)}$	2	2	2
$\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	8	8	8
$\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$	2	2	2
$\lambda_7(\lambda_7)^*$	0	-64	0
$\lambda_{11}((\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*+(\lambda_{X_{17}}^{(1)})^*)$	0	4	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*+(\lambda_{11})^*)$	0	4	0
$\lambda_{X_{17}}^{(1)}((\lambda_{11})^*-2(\lambda_{X_{17}}^{(1)})^*)$	0	4	0
$g_1^2(-6g_2^2-3\lambda_{HS_1}^{(A)})$	0	1	0
$g_1^2(-18g_2^2-15\lambda_{HS_1}^{(A)})$	0	0	1
$\lambda_{HS_1}^{(A)}(3T_\nu^{(1)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	1	1
$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)}+2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+16\lambda_{S_1}^{(A)})$	0	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger Y_e(Y_e)^\dagger)$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$g_1^2\lambda_{HS_1}^{(A)}$	0	0	0
$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+16\lambda_{S_1}^{(A)})$	0	0	0
$g_1^2(4g_2^2-\frac{15\lambda_{HS_1}^{(A)}}{2})$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0
$g_1^2(-4g_2^2-\frac{15\lambda_{HS_1}^{(A)}}{2})$	0	0	0
$g_1^2(-8g_2^2-\frac{51\lambda_{HS_1}^{(A)}}{2})$	0	0	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger Y_e(Y_e)^\dagger)$	$-\frac{4}{3}$	$-\frac{4}{3}$	$-\frac{4}{3}$
$\lambda_5(\lambda_5)^*$	$-\frac{8}{3}$	$-\frac{8}{3}$	$-\frac{8}{3}$
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0	0	0
$g_1^2(6g_2^2-3\lambda_{HS_1}^{(A)})$	0	0	0
$g_2^2\lambda_{HS_1}^{(A)}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	0	0	0
$\lambda_1\lambda_{HS_1}^{(A)}$	2	2	2
$\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	8	8	8
$\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$	2	2	2
$\lambda_7(\lambda_7)^*$	0	0	0
$\lambda_{11}((\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*+(\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*+(\lambda_{11})^*)$	0	0	0
$\lambda_{X_{17}}^{(1)}((\lambda_{11})^*-2(\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$g_1^2(-6g_2^2-3\lambda_{HS_1}^{(A)})$	0	0	0
$g_1^2(-18g_2^2-15\lambda_{HS_1}^{(A)})$	0	0	0
$\lambda_{HS_1}^{(A)}(3T_\nu^{(1)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)}+2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+16\lambda_{S_1}^{(A)})$	1	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger Y_e(Y_e)^\dagger)$	$\frac{4}{3}$	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	$\frac{8}{3}$	0	0
$g_1^2\lambda_{HS_1}^{(A)}$	$-\frac{3}{2}$	0	0

$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 16\lambda_{S_1}^{(A)})$	0	1	1
$g_1^2(4g_2^2 - \frac{15\lambda_{HS_1}^{(A)}}{2})$	0	1	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	$-\frac{8}{3}$
$g_1^2(-4g_2^2 - \frac{15\lambda_{HS_1}^{(A)}}{2})$	0	0	1
$g_1^2(-8g_2^2 - \frac{51\lambda_{HS_1}^{(A)}}{2})$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger Y_e(Y_e)^\dagger)$	$-\frac{4}{3}$	$-\frac{4}{3}$	$-\frac{4}{3}$
$\lambda_5(\lambda_5)^*$	$-\frac{4}{3}$	$-\frac{4}{3}$	$-\frac{4}{3}$
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0	0	0
$g_1^2(6g_2^2 - 3\lambda_{HS_1}^{(A)})$	0	0	0
$g_2^2\lambda_{HS_1}^{(A)}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	12	12	12
$\lambda_1\lambda_{HS_1}^{(A)}$	2	2	2
$\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	8	8	8
$\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$	2	2	2
$\lambda_7(\lambda_7)^*$	0	0	0
$\lambda_{11}((\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}})^* + (\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}(2(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}})^* + (\lambda_{11})^*)$	0	0	0
$\lambda_{X_{17}}^{(1)}((\lambda_{11})^* - 2(\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$g_1^2(-6g_2^2 - 3\lambda_{HS_1}^{(A)})$	0	0	0
$g_1^2(-18g_2^2 - 15\lambda_{HS_1}^{(A)})$	0	0	0
$\lambda_{HS_1}^{(A)}(3T_\nu^{(1)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 16\lambda_{S_1}^{(A)})$	1	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger Y_e(Y_e)^\dagger)$	$\frac{4}{3}$	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	$-\frac{8}{9}$	0	0
$g_1^2\lambda_{HS_1}^{(A)}$	$-\frac{3}{2}$	0	0
$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 16\lambda_{S_1}^{(A)})$	0	1	1
$g_1^2(4g_2^2 - \frac{15\lambda_{HS_1}^{(A)}}{2})$	0	1	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	$-\frac{4}{3}$
$g_1^2(-4g_2^2 - \frac{15\lambda_{HS_1}^{(A)}}{2})$	0	0	1
$g_1^2(-8g_2^2 - \frac{51\lambda_{HS_1}^{(A)}}{2})$	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$\mathrm{Tr}(y_1(y_1)^\dagger Y_e(Y_e)^\dagger)$	$-\frac{4}{3}$
$\lambda_5(\lambda_5)^*$	$-\frac{4}{3}$
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0
$g_1^2(6g_2^2 - 3\lambda_{HS_1}^{(A)})$	0
$g_2^2\lambda_{HS_1}^{(A)}$	$-\frac{33}{2}$
$\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	12
$\lambda_1\lambda_{HS_1}^{(A)}$	2
$\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	8
$\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$	2
$\lambda_7(\lambda_7)^*$	0
$\lambda_{11}((\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}})^* + (\lambda_{X_{17}}^{(1)})^*)$	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}(2(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}})^* + (\lambda_{11})^*)$	0
$\lambda_{X_{17}}^{(1)}((\lambda_{11})^* - 2(\lambda_{X_{17}}^{(1)})^*)$	0

$g_1^2(-6g_2^2 - 3\lambda_{HS_1}^{(A)})$	0
$g_1^2(-18g_2^2 - 15\lambda_{HS_1}^{(A)})$	0
$\lambda_{HS_1}^{(A)}(3T_\nu^{(1)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 16\lambda_{S_1}^{(A)})$	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger Y_e(Y_e)^\dagger)$	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0
$g_1^2\lambda_{HS_1}^{(A)}$	0
$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 16\lambda_{S_1}^{(A)})$	1
$g_1^2(4g_2^2 - \frac{15\lambda_{HS_1}^{(A)}}{2})$	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0
$g_1^2(-4g_2^2 - \frac{15\lambda_{HS_1}^{(A)}}{2})$	0
$g_1^2(-8g_2^2 - \frac{51\lambda_{HS_1}^{(A)}}{2})$	1

$\beta_{\lambda_{HS_2}^{(1)}}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_{S_2}^{(A)})$	1	0	1
$\mathrm{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	-2	-2	-2
$\lambda_5(\lambda_5)^*$	4	4	4
$\lambda_{X_{18}}(\lambda_{X_5}^{(1)})^*$	4	0	0
$\lambda_{X_5}^{(1)}(\lambda_{X_{18}})^*$	4	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	4	0	0
g_2^4	6	6	6
$g_2^2\lambda_{HS_2}^{(1)}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
g_1^4	3	0	3
$g_1^2\lambda_{HS_2}^{(1)}$	$-\frac{15}{2}$	$-\frac{3}{2}$	$-\frac{15}{2}$
$\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	2	2	2
$\lambda_1\lambda_{HS_2}^{(1)}$	6	6	6
$\lambda_{HS_2}^{(1)\,2}$	4	4	4
$\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	8	8	8
$\lambda_{HS_2}^{(A)\,2}$	6	6	6
$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_{S_2}^{(A)})$	0	1	0
$\lambda_{X_{16}}(4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 4(5(\lambda_{X_2})^* + 4(\lambda_{X_{16}})^*))$	0	1	0
$(\lambda_{X_{16}})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	4	0
$\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger Y_e(Y_e)^\dagger)$	0	-2	0
$\lambda_{X_2}(\lambda_{X_{16}})^*$	0	20	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	4	0
$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{10}((\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}})^* - (\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}(4(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}})^* + 3(\lambda_{10})^* + 2(\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$\lambda_{X_{17}}^{(1)}(2(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}})^* - 3(\lambda_{10})^* + 4(\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$\lambda_8(\lambda_8)^*$	0	0	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	0
$\lambda_{HS_2}^{(1)}(3T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{HS_2}^{(1)}(4T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{HS_2}^{(1)}(4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{X_{16}}(8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 4(3(\lambda_{X_2})^* + 4(\lambda_{X_{16}})^*))$	0	0	0
$\lambda_{X_{18}}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}(\lambda_{X_{18}})^*$	0	0	0
$\lambda_{HS_1}^{(1)}(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$\lambda^{(1)}_{HS_2}(2T^{(2)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda^{(A)}_{S_2})$	0	0	0
$\mathrm{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	-2	-2	-2
$\lambda_5(\lambda_5)^*$	6	6	6
$\lambda_{X_{18}}(\lambda^{(1)}_{X_5})^*$	0	0	0
$\lambda^{(1)}_{X_5}(\lambda_{X_{18}})^*$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
$g_2^2\lambda^{(1)}_{HS_2}$	-9	-9	-9
g_1^4	$\frac{27}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$g_1^2\lambda^{(1)}_{HS_2}$	-15	-3	-3
$\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$	4	4	4
$\lambda_1\lambda^{(1)}_{HS_2}$	6	6	6
$\lambda^{(1)}_{HS_2}{}^2$	4	4	4
$\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	6	6	6
$\lambda^{(A)}_{HS_2}{}^2$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$\lambda^{(1)}_{HS_2}(2T^{(2)}_\nu + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda^{(A)}_{S_2})$	0	0	0
$\lambda_{X_{16}}(4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 4(5(\lambda_{X_2})^* + 4(\lambda_{X_{16}})^*))$	0	0	0
$(\lambda_{X_{16}})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger Y_e(Y_e)^\dagger)$	0	0	0
$\lambda_{X_2}(\lambda_{X_{16}})^*$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0
$\lambda^{(1)}_{HS_2}(2T^{(2)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	1	1
$\lambda_{10}((\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - (\lambda^{(1)}_{X_{17}})^*)$	0	3	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + 3(\lambda_{10})^* + 2(\lambda^{(1)}_{X_{17}})^*)$	0	1	0
$\lambda^{(1)}_{X_{17}}(2(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - 3(\lambda_{10})^* + 4(\lambda^{(1)}_{X_{17}})^*)$	0	1	0
$\lambda_8(\lambda_8)^*$	0	24	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	24
$\lambda^{(1)}_{HS_2}(3T^{(2)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda^{(1)}_{HS_2}(4T^{(2)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda^{(1)}_{HS_2}(4T^{(2)}_\nu + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{X_{16}}(8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 4(3(\lambda_{X_2})^* + 4(\lambda_{X_{16}})^*))$	0	0	0
$\lambda_{X_{18}}(3(\lambda^{(1)}_{X_5})^* + (\lambda^{(2)}_{X_5})^*)$	0	0	0
$\lambda^{(2)}_{X_5}(\lambda_{X_{18}})^*$	0	0	0
$\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12})$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda^{(1)}_{HS_2}(2T^{(2)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda^{(A)}_{S_2})$	0	0	0
$\mathrm{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	-3	-3	-3
$\lambda_5(\lambda_5)^*$	6	6	6
$\lambda_{X_{18}}(\lambda^{(1)}_{X_5})^*$	0	0	0
$\lambda^{(1)}_{X_5}(\lambda_{X_{18}})^*$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
$g_2^2\lambda^{(1)}_{HS_2}$	-9	-9	-9
g_1^4	$\frac{27}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$g_1^2\lambda^{(1)}_{HS_2}$	-15	-3	-3
$\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$	4	4	4
$\lambda_1\lambda^{(1)}_{HS_2}$	6	6	6
$\lambda^{(1)}_{HS_2}{}^2$	4	4	4
$\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	6	6	6

$\lambda_{HS_2}^{(A)\ 2}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_{X_{16}}(4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 4(5(\lambda_{X_2})^* + 4(\lambda_{X_{16}})^*))$	0	0	0
$(\lambda_{X_{16}})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger Y_e(Y_e)^\dagger)$	0	0	0
$\lambda_{X_2}(\lambda_{X_{16}})^*$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0
$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{10}((\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - (\lambda_{X_{17}}^{(1)})^*)$	0	3	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + 3(\lambda_{10})^* + 2(\lambda_{X_{17}}^{(1)})^*)$	0	1	0
$\lambda_{X_{17}}^{(1)}(2(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - 3(\lambda_{10})^* + 4(\lambda_{X_{17}}^{(1)})^*)$	0	1	0
$\lambda_8(\lambda_8)^*$	0	24	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	24
$\lambda_{HS_2}^{(1)}(3T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	1	1
$\lambda_{HS_2}^{(1)}(4T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{HS_2}^{(1)}(4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{X_{16}}(8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 4(3(\lambda_{X_2})^* + 4(\lambda_{X_{16}})^*))$	0	0	0
$\lambda_{X_{18}}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}(\lambda_{X_{18}})^*$	0	0	0
$\lambda_{HS_1}^{(1)}(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_{S_2}^{(A)})$	0	0	0
$\mathrm{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	-4	-4	-4
$\lambda_5(\lambda_5)^*$	12	12	12
$\lambda_{X_{18}}(\lambda_{X_5}^{(1)})^*$	12	0	0
$\lambda_{X_5}^{(1)}(\lambda_{X_{18}})^*$	12	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	12	0	0
g_2^4	0	0	0
$g_2^2\lambda_{HS_2}^{(1)}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
g_1^4	3	12	0
$g_1^2\lambda_{HS_2}^{(1)}$	$-\frac{15}{2}$	$-\frac{51}{2}$	$-\frac{3}{2}$
$\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	6	6	6
$\lambda_1\lambda_{HS_2}^{(1)}$	6	6	6
$\lambda_{HS_2}^{(1)\ 2}$	4	4	4
$\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	4	4	4
$\lambda_{HS_2}^{(A)\ 2}$	0	0	0
$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_{X_{16}}(4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 4(5(\lambda_{X_2})^* + 4(\lambda_{X_{16}})^*))$	0	0	0
$(\lambda_{X_{16}})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	8
$\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger Y_e(Y_e)^\dagger)$	0	0	-4
$\lambda_{X_2}(\lambda_{X_{16}})^*$	0	0	12
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	12
$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{10}((\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - (\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + 3(\lambda_{10})^* + 2(\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$\lambda_{X_{17}}^{(1)}(2(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - 3(\lambda_{10})^* + 4(\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$\lambda_8(\lambda_8)^*$	0	0	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	0
$\lambda_{HS_2}^{(1)}(3T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{HS_2}^{(1)}(4T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	1	0
$\lambda_{HS_2}^{(1)}(4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	1

$\lambda_{X_{16}}(8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger)+4(3(\lambda_{X_2})^*+4(\lambda_{X_{16}})^*))$	0	0	1
$\lambda_{X_{18}}(3(\lambda^{(1)}_{X_5})^*+(\lambda^{(2)}_{X_5})^*)$	0	0	0
$\lambda^{(2)}_{X_5}(\lambda_{X_{18}})^*$	0	0	0
$\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12})$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda^{(1)}_{HS_2}(2T_\nu^{(2)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+4\lambda^{(A)}_{S_2})$	1	1	0
$\mathrm{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	-2	-2	-2
$\lambda_5(\lambda_5)^*$	4	4	4
$\lambda_{X_{18}}(\lambda^{(1)}_{X_5})^*$	0	0	0
$\lambda^{(1)}_{X_5}(\lambda_{X_{18}})^*$	12	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	4	0	0
g_2^4	6	6	6
$g_2^2\lambda^{(1)}_{HS_2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
g_1^4	3	12	0
$g_1^2\lambda^{(1)}_{HS_2}$	$-\frac{15}{2}$	$-\frac{51}{2}$	$-\frac{3}{2}$
$\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$	0	0	0
$\lambda_1\lambda^{(1)}_{HS_2}$	6	6	6
${\lambda^{(1)}_{HS_2}}^2$	4	4	4
$\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	8	8	8
${\lambda^{(A)}_{HS_2}}^2$	6	6	6
$\lambda^{(1)}_{HS_2}(2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+4\lambda^{(A)}_{S_2})$	0	0	1
$\lambda_{X_{16}}(4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger)+4(5(\lambda_{X_2})^*+4(\lambda_{X_{16}})^*))$	0	0	1
$(\lambda_{X_{16}})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	4
$\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger Y_e(Y_e)^\dagger)$	0	0	-2
$\lambda_{X_2}(\lambda_{X_{16}})^*$	0	0	20
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	4
$\lambda^{(1)}_{HS_2}(2T_\nu^{(2)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{10}((\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*-(\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*+3(\lambda_{10})^*+2(\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda^{(1)}_{X_{17}}(2(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*-3(\lambda_{10})^*+4(\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda_8(\lambda_8)^*$	0	0	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	0
$\lambda^{(1)}_{HS_2}(3T_\nu^{(2)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda^{(1)}_{HS_2}(4T_\nu^{(2)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda^{(1)}_{HS_2}(4T_\nu^{(2)}+4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{X_{16}}(8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger)+4(3(\lambda_{X_2})^*+4(\lambda_{X_{16}})^*))$	0	0	0
$\lambda_{X_{18}}(3(\lambda^{(1)}_{X_5})^*+(\lambda^{(2)}_{X_5})^*)$	4	0	0
$\lambda^{(2)}_{X_5}(\lambda_{X_{18}})^*$	4	0	0
$\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12})$	1	1	1

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda^{(1)}_{HS_2}(2T_\nu^{(2)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+4\lambda^{(A)}_{S_2})$	1
$\mathrm{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	-2
$\lambda_5(\lambda_5)^*$	4
$\lambda_{X_{18}}(\lambda^{(1)}_{X_5})^*$	0
$\lambda^{(1)}_{X_5}(\lambda_{X_{18}})^*$	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0
g_2^4	6
$g_2^2\lambda^{(1)}_{HS_2}$	$-\frac{33}{2}$
g_1^4	3

$g_1^2\lambda_{HS_2}^{(1)}$	$-\frac{15}{2}$
$\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	0
$\lambda_1\lambda_{HS_2}^{(1)}$	6
${\lambda_{HS_2}^{(1)}}^2$	4
$\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	8
${\lambda_{HS_2}^{(A)}}^2$	6
$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_{S_2}^{(A)})$	0
$\lambda_{X_{16}}(4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 4(5(\lambda_{X_2})^* + 4(\lambda_{X_{16}})^*))$	0
$(\lambda_{X_{16}})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0
$\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger Y_e(Y_e)^\dagger)$	0
$\lambda_{X_2}(\lambda_{X_{16}})^*$	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0
$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$\lambda_{10}((\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - (\lambda_{X_{17}}^{(1)})^*)$	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + 3(\lambda_{10})^* + 2(\lambda_{X_{17}}^{(1)})^*)$	0
$\lambda_{X_{17}}^{(1)}(2(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - 3(\lambda_{10})^* + 4(\lambda_{X_{17}}^{(1)})^*)$	0
$\lambda_8(\lambda_8)^*$	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0
$\lambda_{HS_2}^{(1)}(3T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$\lambda_{HS_2}^{(1)}(4T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$\lambda_{HS_2}^{(1)}(4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$\lambda_{X_{16}}(8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 4(3(\lambda_{X_2})^* + 4(\lambda_{X_{16}})^*))$	0
$\lambda_{X_{18}}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0
$\lambda_{X_5}^{(2)}(\lambda_{X_{18}})^*$	0
$\lambda_{HS_1}^{(1)}(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	1

$\beta^{(1)}_{\lambda^{(A)}_{HS_2}}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 16\lambda_{S_2}^{(A)})$	1	0	1
$\mathrm{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$\frac{4}{3}$	$\frac{4}{3}$	$\frac{4}{3}$
$\lambda_5(\lambda_5)^*$	$\frac{8}{3}$	$\frac{8}{3}$	$\frac{8}{3}$
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	$\frac{8}{3}$	0	0
$g_1^2(4g_2^2 - \frac{15\lambda_{HS_2}^{(A)}}{2})$	1	0	0
$g_2^2\lambda_{HS_2}^{(A)}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$\lambda_1\lambda_{HS_2}^{(A)}$	2	2	2
$\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	8	8	8
$\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	2	2	2
$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 16\lambda_{S_2}^{(A)})$	0	1	0
$\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger Y_e(Y_e)^\dagger)$	0	$-\frac{4}{3}$	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	$-\frac{8}{3}$	0
$g_1^2\lambda_{HS_2}^{(A)}$	0	$-\frac{3}{2}$	0
$g_1^2(-4g_2^2 - \frac{15\lambda_{HS_2}^{(A)}}{2})$	0	0	1
$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$g_1^2(18g_2^2 - 15\lambda_{HS_2}^{(A)})$	0	0	0
$\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$	0	0	0
$\lambda_{10}((\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + (\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + (\lambda_{10})^*)$	0	0	0
$\lambda_8(\lambda_8)^*$	0	0	0
$\lambda_{X_{17}}^{(1)}(8(\lambda_{X_{17}}^{(1)})^* - 4(\lambda_{10})^*)$	0	0	0
$g_1^2(6g_2^2 - 3\lambda_{HS_2}^{(A)})$	0	0	0

$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	0
$g_1^2(-6g_2^2 - 3\lambda_{HS_2}^{(A)})$	0	0	0
$\lambda_{HS_2}^{(A)}(3T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$g_1^2(8g_2^2 - \frac{51\lambda_{HS_2}^{(A)}}{2})$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 16\lambda_{S_2}^{(A)})$	0	0	0
$\mathrm{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	8	8	8
$\lambda_5(\lambda_5)^*$	16	16	16
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$g_1^2(4g_2^2 - \frac{15\lambda_{HS_2}^{(A)}}{2})$	0	0	0
$g_2^2\lambda_{HS_2}^{(A)}$	-9	-9	-9
$\lambda_1\lambda_{HS_2}^{(A)}$	2	2	2
$\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	8	8	8
$\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	2	2	2
$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 16\lambda_{S_2}^{(A)})$	0	0	0
$\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger Y_e(Y_e)^\dagger)$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0
$g_1^2\lambda_{HS_2}^{(A)}$	0	0	0
$g_1^2(-4g_2^2 - \frac{15\lambda_{HS_2}^{(A)}}{2})$	0	0	0
$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	1	1
$g_1^2(18g_2^2 - 15\lambda_{HS_2}^{(A)})$	1	0	0
$\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$	1	1	1
$\lambda_{10}((\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}})^* + (\lambda_{X_{17}}^{(1)})^*)$	0	-4	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}(2(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}})^* + (\lambda_{10})^*)$	0	-4	0
$\lambda_8(\lambda_8)^*$	0	64	0
$\lambda_{X_{17}}^{(1)}(8(\lambda_{X_{17}}^{(1)})^* - 4(\lambda_{10})^*)$	0	1	0
$g_1^2(6g_2^2 - 3\lambda_{HS_2}^{(A)})$	0	1	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	-64
$g_1^2(-6g_2^2 - 3\lambda_{HS_2}^{(A)})$	0	0	1
$\lambda_{HS_2}^{(A)}(3T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$g_1^2(8g_2^2 - \frac{51\lambda_{HS_2}^{(A)}}{2})$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 16\lambda_{S_2}^{(A)})$	0	0	0
$\mathrm{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	-4	-4	-4
$\lambda_5(\lambda_5)^*$	16	16	16
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$g_1^2(4g_2^2 - \frac{15\lambda_{HS_2}^{(A)}}{2})$	0	0	0
$g_2^2\lambda_{HS_2}^{(A)}$	-9	-9	-9
$\lambda_1\lambda_{HS_2}^{(A)}$	2	2	2
$\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	8	8	8
$\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	2	2	2
$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 16\lambda_{S_2}^{(A)})$	0	0	0
$\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger Y_e(Y_e)^\dagger)$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0
$g_1^2\lambda_{HS_2}^{(A)}$	0	0	0
$g_1^2(-4g_2^2 - \frac{15\lambda_{HS_2}^{(A)}}{2})$	0	0	0
$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0

$g_1^2(18g_2^2 - 15\lambda_{HS_2}^{(A)})$	1	0	0
$\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$	1	1	1
$\lambda_{10}((\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})^* + (\lambda_{X_{17}}^{(1)})^*)$	0	-4	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})^* + (\lambda_{10})^*)$	0	-4	0
$\lambda_8(\lambda_8)^*$	0	64	0
$\lambda_{X_{17}}^{(1)}(8(\lambda_{X_{17}}^{(1)})^* - 4(\lambda_{10})^*)$	0	1	0
$g_1^2(6g_2^2 - 3\lambda_{HS_2}^{(A)})$	0	1	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	-64
$g_1^2(-6g_2^2 - 3\lambda_{HS_2}^{(A)})$	0	0	1
$\lambda_{HS_2}^{(A)}(3T_\nu^{(2)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	1	1	1
$g_1^2(8g_2^2 - \frac{51\lambda_{HS_2}^{(A)}}{2})$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger) + 16\lambda_{S_2}^{(A)})$	0	0	0
$\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	0	0	0
$\lambda_5(\lambda_5)^*$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$g_1^2(4g_2^2 - \frac{15\lambda_{HS_2}^{(A)}}{2})$	0	0	0
$g_2^2\lambda_{HS_2}^{(A)}$	0	0	0
$\lambda_1\lambda_{HS_2}^{(A)}$	0	0	0
$\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	0	0	0
$\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	0	0	0
$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)} + 2\,\text{Tr}(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger) + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger) + 16\lambda_{S_2}^{(A)})$	0	0	0
$\text{Tr}(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger Y_e(Y_e)^\dagger)$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0
$g_1^2\lambda_{HS_2}^{(A)}$	0	0	0
$g_1^2(-4g_2^2 - \frac{15\lambda_{HS_2}^{(A)}}{2})$	0	0	0
$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$g_1^2(18g_2^2 - 15\lambda_{HS_2}^{(A)})$	0	0	0
$\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$	0	0	0
$\lambda_{10}((\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})^* + (\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})^* + (\lambda_{10})^*)$	0	0	0
$\lambda_8(\lambda_8)^*$	0	0	0
$\lambda_{X_{17}}^{(1)}(8(\lambda_{X_{17}}^{(1)})^* - 4(\lambda_{10})^*)$	0	0	0
$g_1^2(6g_2^2 - 3\lambda_{HS_2}^{(A)})$	0	0	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	0
$g_1^2(-6g_2^2 - 3\lambda_{HS_2}^{(A)})$	0	0	0
$\lambda_{HS_2}^{(A)}(3T_\nu^{(2)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$g_1^2(8g_2^2 - \frac{51\lambda_{HS_2}^{(A)}}{2})$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger) + 16\lambda_{S_2}^{(A)})$	1	1	0
$\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$\frac{4}{3}$	$\frac{4}{3}$	$\frac{4}{3}$
$\lambda_5(\lambda_5)^*$	$-\frac{8}{9}$	$-\frac{8}{9}$	$-\frac{8}{9}$
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	$-\frac{8}{9}$	0	0
$g_1^2(4g_2^2 - \frac{15\lambda_{HS_2}^{(A)}}{2})$	1	0	0
$g_2^2\lambda_{HS_2}^{(A)}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$\lambda_1\lambda_{HS_2}^{(A)}$	2	2	2
$\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	8	8	8

$\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	2	2	2
$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 16\lambda_{S_2}^{(A)})$	0	0	1
$\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger Y_e(Y_e)^\dagger)$	0	0	$-\frac{4}{3}$
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	$-\frac{4}{3}$
$g_1^2\lambda_{HS_2}^{(A)}$	0	0	$-\frac{3}{2}$
$g_1^2(-4g_2^2 - \frac{15\lambda_{HS_2}^{(A)}}{2})$	0	0	0
$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$g_1^2(18g_2^2 - 15\lambda_{HS_2}^{(A)})$	0	0	0
$\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$	12	12	12
$\lambda_{10}((\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}})^* + (\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}(2(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}})^* + (\lambda_{10})^*)$	0	0	0
$\lambda_8(\lambda_8)^*$	0	0	0
$\lambda_{X_{17}}^{(1)}(8(\lambda_{X_{17}}^{(1)})^* - 4(\lambda_{10})^*)$	0	0	0
$g_1^2(6g_2^2 - 3\lambda_{HS_2}^{(A)})$	0	0	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	0
$g_1^2(-6g_2^2 - 3\lambda_{HS_2}^{(A)})$	0	0	0
$\lambda_{HS_2}^{(A)}(3T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$g_1^2(8g_2^2 - \frac{51\lambda_{HS_2}^{(A)}}{2})$	0	1	0

$\mathcal{S}$	E ($\alpha = -4$)
$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 16\lambda_{S_2}^{(A)})$	1
$\mathrm{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$\frac{4}{3}$
$\lambda_5(\lambda_5)^*$	$-\frac{8}{9}$
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0
$g_1^2(4g_2^2 - \frac{15\lambda_{HS_2}^{(A)}}{2})$	0
$g_2^2\lambda_{HS_2}^{(A)}$	$-\frac{33}{2}$
$\lambda_1\lambda_{HS_2}^{(A)}$	2
$\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	8
$\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	2
$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 16\lambda_{S_2}^{(A)})$	0
$\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger Y_e(Y_e)^\dagger)$	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0
$g_1^2\lambda_{HS_2}^{(A)}$	0
$g_1^2(-4g_2^2 - \frac{15\lambda_{HS_2}^{(A)}}{2})$	1
$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$g_1^2(18g_2^2 - 15\lambda_{HS_2}^{(A)})$	0
$\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$	12
$\lambda_{10}((\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}})^* + (\lambda_{X_{17}}^{(1)})^*)$	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}(2(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}})^* + (\lambda_{10})^*)$	0
$\lambda_8(\lambda_8)^*$	0
$\lambda_{X_{17}}^{(1)}(8(\lambda_{X_{17}}^{(1)})^* - 4(\lambda_{10})^*)$	0
$g_1^2(6g_2^2 - 3\lambda_{HS_2}^{(A)})$	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0
$g_1^2(-6g_2^2 - 3\lambda_{HS_2}^{(A)})$	0
$\lambda_{HS_2}^{(A)}(3T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$g_1^2(8g_2^2 - \frac{51\lambda_{HS_2}^{(A)}}{2})$	0

$\beta_{\lambda_7}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda_7(2T_\nu^{(1)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$(\lambda_5)^*\,\mathrm{Tr}(y_2(y_1)^\dagger)$	0	0	0
$g_2^2\lambda_7$	0	0	0
$g_1^2\lambda_7$	0	0	0
$\lambda_1\lambda_7$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_7$	0	0	0
$\lambda_{HS_1}^{(A)}\lambda_7$	0	0	0
$\lambda_7\lambda_{S_1}^{(1)}$	0	0	0
$\lambda_8\lambda_9$	0	0	0
$\lambda_7(3T_\nu^{(1)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_7(2T_\nu^{(1)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	1	0
$(\lambda_5)^*\,\mathrm{Tr}(y_2(y_1)^\dagger)$	0	$-\frac{1}{3}$	0
$g_2^2\lambda_7$	0	-9	0
$g_1^2\lambda_7$	0	-3	0
$\lambda_1\lambda_7$	0	2	0
$\lambda_{HS_1}^{(1)}\lambda_7$	0	8	0
$\lambda_{HS_1}^{(A)}\lambda_7$	0	-4	0
$\lambda_7\lambda_{S_1}^{(1)}$	0	2	0
$\lambda_8\lambda_9$	0	4	0
$\lambda_7(3T_\nu^{(1)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_7(2T_\nu^{(1)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$(\lambda_5)^*\,\mathrm{Tr}(y_2(y_1)^\dagger)$	0	$-\frac{1}{2}$	0
$g_2^2\lambda_7$	0	-9	0
$g_1^2\lambda_7$	0	-3	0
$\lambda_1\lambda_7$	0	2	0
$\lambda_{HS_1}^{(1)}\lambda_7$	0	8	0
$\lambda_{HS_1}^{(A)}\lambda_7$	0	-4	0
$\lambda_7\lambda_{S_1}^{(1)}$	0	2	0
$\lambda_8\lambda_9$	0	4	0
$\lambda_7(3T_\nu^{(1)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	1	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_7(2T_\nu^{(1)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$(\lambda_5)^*\,\mathrm{Tr}(y_2(y_1)^\dagger)$	0	0	0
$g_2^2\lambda_7$	0	0	0
$g_1^2\lambda_7$	0	0	0
$\lambda_1\lambda_7$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_7$	0	0	0
$\lambda_{HS_1}^{(A)}\lambda_7$	0	0	0
$\lambda_7\lambda_{S_1}^{(1)}$	0	0	0
$\lambda_8\lambda_9$	0	0	0
$\lambda_7(3T_\nu^{(1)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda_7(2T_\nu^{(1)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$(\lambda_5)^*\,\mathrm{Tr}(y_2(y_1)^\dagger)$	0	0	0
$g_2^2\lambda_7$	0	0	0
$g_1^2\lambda_7$	0	0	0
$\lambda_1\lambda_7$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_7$	0	0	0
$\lambda_{HS_1}^{(A)}\lambda_7$	0	0	0
$\lambda_7\lambda_{S_1}^{(1)}$	0	0	0
$\lambda_8\lambda_9$	0	0	0
$\lambda_7(3T_\nu^{(1)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_7(2T_\nu^{(1)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$(\lambda_5)^*\,\mathrm{Tr}(y_2(y_1)^\dagger)$	0
$g_2^2\lambda_7$	0
$g_1^2\lambda_7$	0
$\lambda_1\lambda_7$	0
$\lambda_{HS_1}^{(1)}\lambda_7$	0
$\lambda_{HS_1}^{(A)}\lambda_7$	0
$\lambda_7\lambda_{S_1}^{(1)}$	0
$\lambda_8\lambda_9$	0
$\lambda_7(3T_\nu^{(1)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0

$\beta_{\lambda_{12}}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}(T_\nu^{(1)}+T_\nu^{(2)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_9(8(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}})^*-4(\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$g_2^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}$	0	0	0
$g_1^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}$	0	0	0
$\lambda_{12}^{(A)}(\lambda_{X_{17}}^{(1)}-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}}{2})$	0	0	0
$\lambda_1(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}-2\lambda_{X_{17}}^{(1)})$	0	0	0
$\lambda_{HS_1}^{(1)}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}+6\lambda_{11})$	0	0	0
$\lambda_{HS_1}^{(A)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}+\frac{\lambda_{11}}{2}+\lambda_{X_{17}}^{(1)})$	0	0	0
$\lambda_{HS_2}^{(1)}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}+6\lambda_{10})$	0	0	0
$\lambda_{HS_2}^{(A)}(-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}-\frac{\lambda_{10}}{2}-\lambda_{X_{17}}^{(1)})$	0	0	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}(\frac{3}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}(T_\nu^{(1)}+T_\nu^{(2)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	1	0
$\lambda_9(8(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}})^*-4(\lambda_{X_{17}}^{(1)})^*)$	0	1	0
$g_2^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}$	0	-9	0
$g_1^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}$	0	-3	0
$\lambda_{12}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}$	0	2	0

$\lambda_{12}^{(A)}(\lambda_{X_{17}}^{(1)} - \frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2})$	0	1	0
$\lambda_1(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} - 2\lambda_{X_{17}}^{(1)})$	0	1	0
$\lambda_{HS_1}^{(1)}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} + 6\lambda_{11})$	0	1	0
$\lambda_{HS_1}^{(A)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} + \frac{\lambda_{11}}{2} + \lambda_{X_{17}}^{(1)})$	0	1	0
$\lambda_{HS_2}^{(1)}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} + 6\lambda_{10})$	0	1	0
$\lambda_{HS_2}^{(A)}(-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} - \frac{\lambda_{10}}{2} - \lambda_{X_{17}}^{(1)})$	0	1	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\frac{3}{2}T_\nu^{(1)} + \frac{3}{2}T_\nu^{(2)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(T_\nu^{(1)} + T_\nu^{(2)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_9(8(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})^* - 4(\lambda_{X_{17}}^{(1)})^*)$	0	1	0
$g_2^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	0	-9	0
$g_1^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	0	-3	0
$\lambda_{12}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	0	2	0
$\lambda_{12}^{(A)}(\lambda_{X_{17}}^{(1)} - \frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2})$	0	1	0
$\lambda_1(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} - 2\lambda_{X_{17}}^{(1)})$	0	1	0
$\lambda_{HS_1}^{(1)}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} + 6\lambda_{11})$	0	1	0
$\lambda_{HS_1}^{(A)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} + \frac{\lambda_{11}}{2} + \lambda_{X_{17}}^{(1)})$	0	1	0
$\lambda_{HS_2}^{(1)}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} + 6\lambda_{10})$	0	1	0
$\lambda_{HS_2}^{(A)}(-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} - \frac{\lambda_{10}}{2} - \lambda_{X_{17}}^{(1)})$	0	1	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\frac{3}{2}T_\nu^{(1)} + \frac{3}{2}T_\nu^{(2)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	1	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(T_\nu^{(1)} + T_\nu^{(2)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_9(8(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})^* - 4(\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$g_2^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	0	0	0
$g_1^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	0	0	0
$\lambda_{12}^{(A)}(\lambda_{X_{17}}^{(1)} - \frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2})$	0	0	0
$\lambda_1(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} - 2\lambda_{X_{17}}^{(1)})$	0	0	0
$\lambda_{HS_1}^{(1)}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} + 6\lambda_{11})$	0	0	0
$\lambda_{HS_1}^{(A)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} + \frac{\lambda_{11}}{2} + \lambda_{X_{17}}^{(1)})$	0	0	0
$\lambda_{HS_2}^{(1)}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} + 6\lambda_{10})$	0	0	0
$\lambda_{HS_2}^{(A)}(-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} - \frac{\lambda_{10}}{2} - \lambda_{X_{17}}^{(1)})$	0	0	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\frac{3}{2}T_\nu^{(1)} + \frac{3}{2}T_\nu^{(2)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(T_\nu^{(1)} + T_\nu^{(2)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_9(8(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})^* - 4(\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$g_2^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	0	0	0
$g_1^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	0	0	0
$\lambda_{12}^{(A)}(\lambda_{X_{17}}^{(1)} - \frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2})$	0	0	0

$\lambda_1(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - 2\lambda^{(1)}_{X_{17}})$	0	0	0
$\lambda^{(1)}_{HS_1}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + 6\lambda_{11})$	0	0	0
$\lambda^{(A)}_{HS_1}(2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + \frac{\lambda_{11}}{2} + \lambda^{(1)}_{X_{17}})$	0	0	0
$\lambda^{(1)}_{HS_2}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + 6\lambda_{10})$	0	0	0
$\lambda^{(A)}_{HS_2}(-2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - \frac{\lambda_{10}}{2} - \lambda^{(1)}_{X_{17}})$	0	0	0
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(\frac{3}{2}T_\nu^{(1)} + \frac{3}{2}T_\nu^{(2)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(T_\nu^{(1)} + T_\nu^{(2)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0
$\lambda_9(8(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - 4(\lambda^{(1)}_{X_{17}})^*)$	0
$g_2^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	0
$g_1^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	0
$\lambda^{(1)}_{12}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	0
$\lambda^{(A)}_{12}(\lambda^{(1)}_{X_{17}} - \frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2})$	0
$\lambda_1(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - 2\lambda^{(1)}_{X_{17}})$	0
$\lambda^{(1)}_{HS_1}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + 6\lambda_{11})$	0
$\lambda^{(A)}_{HS_1}(2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + \frac{\lambda_{11}}{2} + \lambda^{(1)}_{X_{17}})$	0
$\lambda^{(1)}_{HS_2}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + 6\lambda_{10})$	0
$\lambda^{(A)}_{HS_2}(-2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - \frac{\lambda_{10}}{2} - \lambda^{(1)}_{X_{17}})$	0
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(\frac{3}{2}T_\nu^{(1)} + \frac{3}{2}T_\nu^{(2)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0

$\beta^{(1)}_{\lambda_8}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda_8(2T_\nu^{(2)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$(\lambda_5)^*\,\text{Tr}(y_1(y_2)^\dagger)$	0	0	0
$\lambda_7(\lambda_9)^*$	0	0	0
$g_2^2\lambda_8$	0	0	0
$g_1^2\lambda_8$	0	0	0
$\lambda_1\lambda_8$	0	0	0
$\lambda^{(1)}_{HS_2}\lambda_8$	0	0	0
$\lambda^{(A)}_{HS_2}\lambda_8$	0	0	0
$\lambda_8\lambda^{(1)}_{S_2}$	0	0	0
$\lambda_8(3T_\nu^{(2)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_8(2T_\nu^{(2)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	1	0
$(\lambda_5)^*\,\text{Tr}(y_1(y_2)^\dagger)$	0	1	0
$\lambda_7(\lambda_9)^*$	0	4	0
$g_2^2\lambda_8$	0	-9	0
$g_1^2\lambda_8$	0	-3	0
$\lambda_1\lambda_8$	0	2	0
$\lambda^{(1)}_{HS_2}\lambda_8$	0	8	0
$\lambda^{(A)}_{HS_2}\lambda_8$	0	4	0
$\lambda_8\lambda^{(1)}_{S_2}$	0	2	0
$\lambda_8(3T_\nu^{(2)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_8(2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$(\lambda_5)^*\,\mathrm{Tr}(y_1(y_2)^\dagger)$	0	$\frac{3}{2}$	0
$\lambda_7(\lambda_9)^*$	0	4	0
$g_2^2\lambda_8$	0	-9	0
$g_1^2\lambda_8$	0	-3	0
$\lambda_1\lambda_8$	0	2	0
$\lambda_{HS_2}^{(1)}\lambda_8$	0	8	0
$\lambda_{HS_2}^{(A)}\lambda_8$	0	4	0
$\lambda_8\lambda_{S_2}^{(1)}$	0	2	0
$\lambda_8(3T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	1	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_8(2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$(\lambda_5)^*\,\mathrm{Tr}(y_1(y_2)^\dagger)$	0	0	0
$\lambda_7(\lambda_9)^*$	0	0	0
$g_2^2\lambda_8$	0	0	0
$g_1^2\lambda_8$	0	0	0
$\lambda_1\lambda_8$	0	0	0
$\lambda_{HS_2}^{(1)}\lambda_8$	0	0	0
$\lambda_{HS_2}^{(A)}\lambda_8$	0	0	0
$\lambda_8\lambda_{S_2}^{(1)}$	0	0	0
$\lambda_8(3T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda_8(2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$(\lambda_5)^*\,\mathrm{Tr}(y_1(y_2)^\dagger)$	0	0	0
$\lambda_7(\lambda_9)^*$	0	0	0
$g_2^2\lambda_8$	0	0	0
$g_1^2\lambda_8$	0	0	0
$\lambda_1\lambda_8$	0	0	0
$\lambda_{HS_2}^{(1)}\lambda_8$	0	0	0
$\lambda_{HS_2}^{(A)}\lambda_8$	0	0	0
$\lambda_8\lambda_{S_2}^{(1)}$	0	0	0
$\lambda_8(3T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$\lambda_8(2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$(\lambda_5)^*\,\mathrm{Tr}(y_1(y_2)^\dagger)$	0
$\lambda_7(\lambda_9)^*$	0
$g_2^2\lambda_8$	0
$g_1^2\lambda_8$	0
$\lambda_1\lambda_8$	0
$\lambda_{HS_2}^{(1)}\lambda_8$	0
$\lambda_{HS_2}^{(A)}\lambda_8$	0
$\lambda_8\lambda_{S_2}^{(1)}$	0
$\lambda_8(3T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0

$\beta_{\lambda_{S_1}^{(1)}}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda_{S_1}^{(1)}(8T_\nu^{(1)} + 8\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	1	0	0
$\lambda_{X_9}(24\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 108(\lambda_{X_9})^*)$	1	0	0
$(\lambda_{X_9})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	24	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)$	-16	0	0
$\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger y_1^{\mathrm{conj}}(y_1)^\dagger)$	-16	0	0
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	12	0	0
$\lambda_{X_6}(\lambda_{X_6})^*$	288	0	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	8	0	0
$\lambda_{12}^{(1)\,2}$	6	6	6
$\lambda_{HS_1}^{(1)\,2}$	4	4	4
$\lambda_{S_1}^{(1)\,2}$	10	10	10
$T_\nu^{(1)}$	-8	-8	-8
T_ν	-8	0	0
$\lambda_{S_1}^{(1)}T_\nu^{(1)}$	0	8	8
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0	24	0
g_1^4	0	12	192
$g_1^2\lambda_{S_1}^{(1)}$	0	-12	-48
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0	0	0
g_2^4	0	0	0
$g_1^2(\frac{3g_2^2}{2} - 3\lambda_{S_1}^{(1)})$	0	0	0
$g_2^2\lambda_{S_1}^{(1)}$	0	0	0
$\lambda_{12}^{(A)\,2}$	0	0	0
$\lambda_{HS_1}^{(A)\,2}$	0	0	0
$\lambda_7(\lambda_7)^*$	0	0	0
$\lambda_9(\lambda_9)^*$	0	0	0
$\lambda_{11}(\lambda_{11})^*$	0	0	0
$g_1^2(\frac{27g_2^2}{2} - 27\lambda_{S_1}^{(1)})$	0	0	0
$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 8\lambda_{S_1}^{(A)})$	0	0	0
$\lambda_{X_9}(\frac{32}{3}\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + \frac{352(\lambda_{X_9})^*}{3})$	0	0	0
$\lambda_{S_1}^{(A)\,2}$	0	0	0
$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 8\lambda_{S_1}^{(A)})$	0	0	0
$g_1^2(8g_2^2 - 12\lambda_{S_1}^{(1)})$	0	0	0
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}(\frac{8(\lambda_{X_5}^{(1)})^*}{3} + 4(\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	0	0	0
$\lambda_{12}^{(\times)\,2}$	0	0	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}(8(\lambda_{X_7}^{(1)})^* + \frac{20(\lambda_{X_7}^{(2)})^*}{3})$	0	0	0
$g_1^2(32g_2^2 - 48\lambda_{S_1}^{(1)})$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_{S_1}^{(1)}(8T_\nu^{(1)} + 8\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_9}(24\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 108(\lambda_{X_9})^*)$	0	0	0
$(\lambda_{X_9})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)$	0	0	0
$\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger y_1^{\mathrm{conj}}(y_1)^\dagger)$	0	0	0
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	0	0	0
$\lambda_{X_6}(\lambda_{X_6})^*$	0	0	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	0	0	0

$\lambda_{12}^{(1)^2}$	4	4	4
$\lambda_{HS_1}^{(1)^2}$	4	4	4
$\lambda_{S_1}^{(1)^2}$	12	12	12
$T_\nu^{(1)}$	-4	-4	-4
T_ν	0	0	0
$\lambda_{S_1}^{(1)} T_\nu^{(1)}$	4	4	4
$\lambda_{X_7}^{(1)} (\lambda_{X_7}^{(1)})^*$	0	0	0
g_1^4	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{243}{4}$
$g_1^2 \lambda_{S_1}^{(1)}$	0	0	0
$\lambda_{X_{15}} (\lambda_{X_{15}})^*$	8	0	0
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
$g_1^2 (\frac{3g_2^2}{2} - 3\lambda_{S_1}^{(1)})$	1	1	0
$g_2^2 \lambda_{S_1}^{(1)}$	-9	-9	-9
$\lambda_{12}^{(A)^2}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
$\lambda_{HS_1}^{(A)^2}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
$\lambda_7 (\lambda_7)^*$	0	8	0
$\lambda_9 (\lambda_9)^*$	0	8	0
$\lambda_{11} (\lambda_{11})^*$	0	24	0
$g_1^2 (\frac{27g_2^2}{2} - 27\lambda_{S_1}^{(1)})$	0	0	1
$\lambda_{S_1}^{(1)} (4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}} (y_1^{\mathrm{conj}})^\dagger) + 8\lambda_{S_1}^{(A)})$	0	0	0
$\lambda_{X_9} (\frac{32}{3}\,\mathrm{Tr}(y_1 (y_1^{\mathrm{conj}})^\dagger) + \frac{352(\lambda_{X_9})^*}{3})$	0	0	0
$\lambda_{S_1}^{(A)^2}$	0	0	0
$\lambda_{S_1}^{(1)} (4T_\nu^{(1)} + 8\lambda_{S_1}^{(A)})$	0	0	0
$g_1^2 (8g_2^2 - 12\lambda_{S_1}^{(1)})$	0	0	0
$\lambda_{X_5}^{(1)} (3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)} (\frac{8(\lambda_{X_5}^{(1)})^*}{3} + 4(\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{12}^{(1)} \lambda_{12}^{(\times)}$	0	0	0
$\lambda_{12}^{(\times)^2}$	0	0	0
$\lambda_{X_7}^{(1)} (3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)} (8(\lambda_{X_7}^{(1)})^* + \frac{20(\lambda_{X_7}^{(2)})^*}{3})$	0	0	0
$g_1^2 (32g_2^2 - 48\lambda_{S_1}^{(1)})$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_{S_1}^{(1)} (8T_\nu^{(1)} + 8\,\mathrm{Tr}(y_1^{\mathrm{conj}} (y_1^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_9} (24\,\mathrm{Tr}(y_1 (y_1^{\mathrm{conj}})^\dagger) + 108(\lambda_{X_9})^*)$	0	0	0
$(\lambda_{X_9})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}} (y_1)^\dagger)$	0	0	0
$\mathrm{Tr}(y_1 (y_1)^\dagger y_1^{\mathrm{conj}} (y_1^{\mathrm{conj}})^\dagger)$	0	0	0
$\mathrm{Tr}(y_1 (y_1^{\mathrm{conj}})^\dagger y_1^{\mathrm{conj}} (y_1)^\dagger)$	0	0	0
$\lambda_{X_5}^{(1)} (\lambda_{X_5}^{(1)})^*$	0	0	0
$\lambda_{X_6} (\lambda_{X_6})^*$	0	0	0
$\lambda_{X_{18}} (\lambda_{X_{18}})^*$	0	0	0
$\lambda_{12}^{(1)^2}$	4	4	4
$\lambda_{HS_1}^{(1)^2}$	4	4	4
$\lambda_{S_1}^{(1)^2}$	12	12	12
$T_\nu^{(1)}$	-5	-5	-5
T_ν	0	0	0
$\lambda_{S_1}^{(1)} T_\nu^{(1)}$	6	6	6
$\lambda_{X_7}^{(1)} (\lambda_{X_7}^{(1)})^*$	0	0	0
g_1^4	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{243}{4}$
$g_1^2 \lambda_{S_1}^{(1)}$	0	0	0

$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	8	0	0
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
$g_1^2(\frac{3g_2^2}{2} - 3\lambda_{S_1}^{(1)})$	1	1	0
$g_2^2\lambda_{S_1}^{(1)}$	-9	-9	-9
${\lambda_{12}^{(A)}}^2$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
${\lambda_{HS_1}^{(A)}}^2$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
$\lambda_7(\lambda_7)^*$	0	8	0
$\lambda_9(\lambda_9)^*$	0	8	0
$\lambda_{11}(\lambda_{11})^*$	0	24	0
$g_1^2(\frac{27g_2^2}{2} - 27\lambda_{S_1}^{(1)})$	0	0	1
$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 8\lambda_{S_1}^{(A)})$	0	0	0
$\lambda_{X_9}(\frac{32}{3}\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + \frac{352(\lambda_{X_9})^*}{3})$	0	0	0
${\lambda_{S_1}^{(A)}}^2$	0	0	0
$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 8\lambda_{S_1}^{(A)})$	0	0	0
$g_1^2(8g_2^2 - 12\lambda_{S_1}^{(1)})$	0	0	0
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}(\frac{8(\lambda_{X_5}^{(1)})^*}{3} + 4(\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}(8(\lambda_{X_7}^{(1)})^* + \frac{20(\lambda_{X_7}^{(2)})^*}{3})$	0	0	0
$g_1^2(32g_2^2 - 48\lambda_{S_1}^{(1)})$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_{S_1}^{(1)}(8T_\nu^{(1)} + 8\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_9}(24\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 108(\lambda_{X_9})^*)$	0	0	0
$(\lambda_{X_9})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	$\frac{32}{3}$	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)$	$-\frac{248}{45}$	0	0
$\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger y_1^{\mathrm{conj}}(y_1)^\dagger)$	$-\frac{248}{45}$	0	0
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	$\frac{8}{3}$	0	0
$\lambda_{X_6}(\lambda_{X_6})^*$	$\frac{832}{3}$	0	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	$\frac{16}{3}$	0	0
${\lambda_{12}^{(1)}}^2$	2	2	2
${\lambda_{HS_1}^{(1)}}^2$	4	4	4
${\lambda_{S_1}^{(1)}}^2$	14	14	14
$T_\nu^{(1)}$	$-\frac{8}{3}$	$-\frac{8}{3}$	$-\frac{8}{3}$
T_ν	$-\frac{8}{3}$	0	0
$\lambda_{S_1}^{(1)}T_\nu^{(1)}$	0	0	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0	0	8
g_1^4	0	12	12
$g_1^2\lambda_{S_1}^{(1)}$	0	0	0
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0	0	0
g_2^4	22	22	22
$g_1^2(\frac{3g_2^2}{2} - 3\lambda_{S_1}^{(1)})$	0	0	0
$g_2^2\lambda_{S_1}^{(1)}$	-24	-24	-24
${\lambda_{12}^{(A)}}^2$	0	0	0
${\lambda_{HS_1}^{(A)}}^2$	0	0	0
$\lambda_7(\lambda_7)^*$	0	0	0
$\lambda_9(\lambda_9)^*$	0	0	0
$\lambda_{11}(\lambda_{11})^*$	0	0	0
$g_1^2(\frac{27g_2^2}{2} - 27\lambda_{S_1}^{(1)})$	0	0	0

$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 8\lambda_{S_1}^{(A)})$	1	0	0
$\lambda_{X_9}(\frac{32}{3}\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + \frac{352(\lambda_{X_9})^*}{3})$	1	0	0
${\lambda_{S_1}^{(A)}}^2$	16	16	16
$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 8\lambda_{S_1}^{(A)})$	0	1	1
$g_1^2(8g_2^2 - 12\lambda_{S_1}^{(1)})$	0	1	1
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}(\frac{8(\lambda_{X_5}^{(1)})^*}{3} + 4(\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}(8(\lambda_{X_7}^{(1)})^* + \frac{20(\lambda_{X_7}^{(2)})^*}{3})$	0	0	0
$g_1^2(32g_2^2 - 48\lambda_{S_1}^{(1)})$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda_{S_1}^{(1)}(8T_\nu^{(1)} + 8\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_9}(24\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 108(\lambda_{X_9})^*)$	0	0	0
$(\lambda_{X_9})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	$\frac{32}{3}$	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)$	$-\frac{248}{45}$	0	0
$\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger y_1^{\mathrm{conj}}(y_1)^\dagger)$	$-\frac{248}{45}$	0	0
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	0	0	0
$\lambda_{X_6}(\lambda_{X_6})^*$	$\frac{832}{3}$	0	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	$\frac{16}{3}$	0	0
${\lambda_{12}^{(1)}}^2$	6	6	6
${\lambda_{HS_1}^{(1)}}^2$	4	4	4
${\lambda_{S_1}^{(1)}}^2$	14	14	14
$T_\nu^{(1)}$	$-\frac{8}{3}$	$-\frac{8}{3}$	$-\frac{8}{3}$
T_ν	$-\frac{8}{3}$	0	0
$\lambda_{S_1}^{(1)}T_\nu^{(1)}$	0	0	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0	0	0
g_1^4	0	12	12
$g_1^2\lambda_{S_1}^{(1)}$	0	0	0
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0	0	0
g_2^4	22	22	22
$g_1^2(\frac{3g_2^2}{2} - 3\lambda_{S_1}^{(1)})$	0	0	0
$g_2^2\lambda_{S_1}^{(1)}$	-24	-24	-24
${\lambda_{12}^{(A)}}^2$	0	0	0
${\lambda_{HS_1}^{(A)}}^2$	0	0	0
$\lambda_7(\lambda_7)^*$	0	0	0
$\lambda_9(\lambda_9)^*$	0	0	0
$\lambda_{11}(\lambda_{11})^*$	0	0	0
$g_1^2(\frac{27g_2^2}{2} - 27\lambda_{S_1}^{(1)})$	0	0	0
$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 8\lambda_{S_1}^{(A)})$	1	0	0
$\lambda_{X_9}(\frac{32}{3}\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + \frac{352(\lambda_{X_9})^*}{3})$	1	0	0
${\lambda_{S_1}^{(A)}}^2$	16	16	16
$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 8\lambda_{S_1}^{(A)})$	0	1	1
$g_1^2(8g_2^2 - 12\lambda_{S_1}^{(1)})$	0	1	1
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	$\frac{8}{3}$	0	0
$\lambda_{X_5}^{(2)}(\frac{8(\lambda_{X_5}^{(1)})^*}{3} + 4(\lambda_{X_5}^{(2)})^*)$	1	0	0
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	8	8	8
${\lambda_{12}^{(\times)}}^2$	$\frac{20}{3}$	$\frac{20}{3}$	$\frac{20}{3}$

$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	8
$\lambda_{X_7}^{(2)}(8(\lambda_{X_7}^{(1)})^* + \frac{20(\lambda_{X_7}^{(2)})^*}{3})$	0	0	1
$g_1^2(32g_2^2 - 48\lambda_{S_1}^{(1)})$	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$\lambda_{S_1}^{(1)}(8T_\nu^{(1)} + 8\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0
$\lambda_{X_9}(24\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 108(\lambda_{X_9})^*)$	0
$(\lambda_{X_9})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)$	0
$\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger y_1^{\mathrm{conj}}(y_1)^\dagger)$	0
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	0
$\lambda_{X_6}(\lambda_{X_6})^*$	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	0
$\lambda_{12}^{(1)^2}$	6
$\lambda_{HS_1}^{(1)^2}$	4
$\lambda_{S_1}^{(1)^2}$	14
$T_\nu^{(1)}$	$-\frac{8}{3}$
T_ν	0
$\lambda_{S_1}^{(1)}T_\nu^{(1)}$	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0
g_1^4	192
$g_1^2\lambda_{S_1}^{(1)}$	0
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0
g_2^4	22
$g_1^2(\frac{3g_2^2}{2} - 3\lambda_{S_1}^{(1)})$	0
$g_2^2\lambda_{S_1}^{(1)}$	-24
$\lambda_{12}^{(A)^2}$	0
$\lambda_{HS_1}^{(A)^2}$	0
$\lambda_7(\lambda_7)^*$	0
$\lambda_9(\lambda_9)^*$	0
$\lambda_{11}(\lambda_{11})^*$	0
$g_1^2(\frac{27g_2^2}{2} - 27\lambda_{S_1}^{(1)})$	0
$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 8\lambda_{S_1}^{(A)})$	0
$\lambda_{X_9}(\frac{32}{3}\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + \frac{352(\lambda_{X_9})^*}{3})$	0
$\lambda_{S_1}^{(A)^2}$	16
$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 8\lambda_{S_1}^{(A)})$	1
$g_1^2(8g_2^2 - 12\lambda_{S_1}^{(1)})$	0
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0
$\lambda_{X_5}^{(2)}(\frac{8(\lambda_{X_5}^{(1)})^*}{3} + 4(\lambda_{X_5}^{(2)})^*)$	0
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	8
$\lambda_{12}^{(\times)^2}$	$\frac{20}{3}$
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0
$\lambda_{X_7}^{(2)}(8(\lambda_{X_7}^{(1)})^* + \frac{20(\lambda_{X_7}^{(2)})^*}{3})$	0
$g_1^2(32g_2^2 - 48\lambda_{S_1}^{(1)})$	1

$\beta_{\lambda_{S_1}^{(A)}}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda_{S_1}^{(A)}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0	0	0

$\lambda_{X_9}(-\frac{4}{3}\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) - \frac{44(\lambda_{X_9})^*}{3})$	0	0	0
$(\lambda_{X_9})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)$	0	0	0
$\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger y_1^{\mathrm{conj}}(y_1)^\dagger)$	0	0	0
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	0	0	0
$\lambda_{X_6}(\lambda_{X_6})^*$	0	0	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	0	0	0
g_2^4	0	0	0
$g_2^2\lambda_{S_1}^{(A)}$	0	0	0
${\lambda_{HS_1}^{(A)}}^2$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	0	0	0
${\lambda_{S_1}^{(A)}}^2$	0	0	0
$T_\nu^{(1)}$	0	0	0
T_ν	0	0	0
$\lambda_{S_1}^{(A)}T_\nu^{(1)}$	0	0	0
$g_1^2(8g_2^2 - 12\lambda_{S_1}^{(A)})$	0	0	0
$\lambda_{X_5}^{(2)}(\lambda_{X_5}^{(1)})^*$	0	0	0
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
${\lambda_{12}^{(A)}}^2$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$\lambda_{X_7}^{(2)}(\lambda_{X_7}^{(2)})^*$	0	0	0
$g_1^2(32g_2^2 - 48\lambda_{S_1}^{(A)})$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_{S_1}^{(A)}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_9}(-\frac{4}{3}\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) - \frac{44(\lambda_{X_9})^*}{3})$	0	0	0
$(\lambda_{X_9})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)$	0	0	0
$\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger y_1^{\mathrm{conj}}(y_1)^\dagger)$	0	0	0
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	0	0	0
$\lambda_{X_6}(\lambda_{X_6})^*$	0	0	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	0	0	0
g_2^4	0	0	0
$g_2^2\lambda_{S_1}^{(A)}$	0	0	0
${\lambda_{HS_1}^{(A)}}^2$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	0	0	0
${\lambda_{S_1}^{(A)}}^2$	0	0	0
$T_\nu^{(1)}$	0	0	0
T_ν	0	0	0
$\lambda_{S_1}^{(A)}T_\nu^{(1)}$	0	0	0
$g_1^2(8g_2^2 - 12\lambda_{S_1}^{(A)})$	0	0	0
$\lambda_{X_5}^{(2)}(\lambda_{X_5}^{(1)})^*$	0	0	0
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
${\lambda_{12}^{(A)}}^2$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$\lambda_{X_7}^{(2)}(\lambda_{X_7}^{(2)})^*$	0	0	0
$g_1^2(32g_2^2 - 48\lambda_{S_1}^{(A)})$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_{S_1}^{(A)}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0	0	0

$\lambda_{X_9}(-\frac{4}{3}\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) - \frac{44(\lambda_{X_9})^*}{3})$	0	0	0
$(\lambda_{X_9})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)$	0	0	0
$\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger y_1^{\mathrm{conj}}(y_1)^\dagger)$	0	0	0
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	0	0	0
$\lambda_{X_6}(\lambda_{X_6})^*$	0	0	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	0	0	0
g_2^4	0	0	0
$g_2^2\lambda_{S_1}^{(A)}$	0	0	0
${\lambda_{HS_1}^{(A)}}^2$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	0	0	0
${\lambda_{S_1}^{(A)}}^2$	0	0	0
$T_\nu^{(1)}$	0	0	0
T_ν	0	0	0
$\lambda_{S_1}^{(A)}T_\nu^{(1)}$	0	0	0
$g_1^2(8g_2^2 - 12\lambda_{S_1}^{(A)})$	0	0	0
$\lambda_{X_5}^{(2)}(\lambda_{X_5}^{(1)})^*$	0	0	0
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
${\lambda_{12}^{(A)}}^2$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$\lambda_{X_7}^{(2)}(\lambda_{X_7}^{(2)})^*$	0	0	0
$g_1^2(32g_2^2 - 48\lambda_{S_1}^{(A)})$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_{S_1}^{(A)}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	1	0	0
$\lambda_{X_9}(-\frac{4}{3}\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) - \frac{44(\lambda_{X_9})^*}{3})$	1	0	0
$(\lambda_{X_9})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	$-\frac{4}{3}$	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)$	$\frac{76}{45}$	0	0
$\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger y_1^{\mathrm{conj}}(y_1)^\dagger)$	$\frac{76}{45}$	0	0
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	$-\frac{4}{3}$	0	0
$\lambda_{X_6}(\lambda_{X_6})^*$	$-\frac{224}{3}$	0	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	$-\frac{8}{3}$	0	0
g_2^4	-2	-2	-2
$g_2^2\lambda_{S_1}^{(A)}$	-24	-24	-24
${\lambda_{HS_1}^{(A)}}^2$	3	3	3
$\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	12	12	12
${\lambda_{S_1}^{(A)}}^2$	20	20	20
$T_\nu^{(1)}$	$-\frac{2}{3}$	$-\frac{2}{3}$	$-\frac{2}{3}$
T_ν	$-\frac{2}{3}$	0	0
$\lambda_{S_1}^{(A)}T_\nu^{(1)}$	0	4	4
$g_1^2(8g_2^2 - 12\lambda_{S_1}^{(A)})$	0	1	1
$\lambda_{X_5}^{(2)}(\lambda_{X_5}^{(1)})^*$	0	0	0
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
${\lambda_{12}^{(A)}}^2$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$\lambda_{X_7}^{(2)}(\lambda_{X_7}^{(2)})^*$	0	0	0
$g_1^2(32g_2^2 - 48\lambda_{S_1}^{(A)})$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda_{S_1}^{(A)}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	1	0	0

$\lambda_{X_9}(-\frac{4}{3}\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) - \frac{44(\lambda_{X_9})^*}{3})$	1	0	0
$(\lambda_{X_9})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	$-\frac{4}{3}$	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)$	$\frac{76}{45}$	0	0
$\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger y_1^{\mathrm{conj}}(y_1)^\dagger)$	$\frac{76}{45}$	0	0
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	0	0	0
$\lambda_{X_6}(\lambda_{X_6})^*$	$-\frac{224}{3}$	0	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	$-\frac{8}{3}$	0	0
g_2^4	-2	-2	-2
$g_2^2\lambda_{S_1}^{(A)}$	-24	-24	-24
${\lambda_{HS_1}^{(A)}}^2$	3	3	3
$\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	12	12	12
${\lambda_{S_1}^{(A)}}^2$	20	20	20
$T_\nu^{(1)}$	$-\frac{2}{3}$	$-\frac{2}{3}$	$-\frac{2}{3}$
T_ν	$-\frac{2}{3}$	0	0
$\lambda_{S_1}^{(A)}T_\nu^{(1)}$	0	4	4
$g_1^2(8g_2^2 - 12\lambda_{S_1}^{(A)})$	0	1	1
$\lambda_{X_5}^{(2)}(\lambda_{X_5}^{(1)})^*$	$-\frac{4}{3}$	0	0
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	$-\frac{4}{3}$	0	0
${\lambda_{12}^{(A)}}^2$	12	12	12
${\lambda_{12}^{(\times)}}^2$	$-\frac{4}{3}$	$-\frac{4}{3}$	$-\frac{4}{3}$
$\lambda_{X_7}^{(2)}(\lambda_{X_7}^{(2)})^*$	0	0	$-\frac{4}{3}$
$g_1^2(32g_2^2 - 48\lambda_{S_1}^{(A)})$	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$\lambda_{S_1}^{(A)}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0
$\lambda_{X_9}(-\frac{4}{3}\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) - \frac{44(\lambda_{X_9})^*}{3})$	0
$(\lambda_{X_9})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)$	0
$\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger y_1^{\mathrm{conj}}(y_1)^\dagger)$	0
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	0
$\lambda_{X_6}(\lambda_{X_6})^*$	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	0
g_2^4	-2
$g_2^2\lambda_{S_1}^{(A)}$	-24
${\lambda_{HS_1}^{(A)}}^2$	3
$\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	12
${\lambda_{S_1}^{(A)}}^2$	20
$T_\nu^{(1)}$	$-\frac{2}{3}$
T_ν	0
$\lambda_{S_1}^{(A)}T_\nu^{(1)}$	4
$g_1^2(8g_2^2 - 12\lambda_{S_1}^{(A)})$	0
$\lambda_{X_5}^{(2)}(\lambda_{X_5}^{(1)})^*$	0
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0
${\lambda_{12}^{(A)}}^2$	12
${\lambda_{12}^{(\times)}}^2$	$-\frac{4}{3}$
$\lambda_{X_7}^{(2)}(\lambda_{X_7}^{(2)})^*$	0
$g_1^2(32g_2^2 - 48\lambda_{S_1}^{(A)})$	1

$\beta_{\lambda_{11}}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_{11}(3T_\nu^{(1)}+T_\nu^{(2)})$	0	0	0
$\lambda_9((\lambda_{10})^*+5(\lambda_{11})^*)$	0	0	0
$g_2^2\lambda_{11}$	0	0	0
$g_1^2\lambda_{11}$	0	0	0
$\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	0	0	0
$\lambda_{12}^{(A)}(-\frac{\lambda_{10}}{2}-\frac{5\lambda_{11}}{2})$	0	0	0
$\lambda_{HS_1}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}-2\lambda_{X_{17}}^{(1)})$	0	0	0
$\lambda_{HS_1}^{(A)}(\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}+\frac{\lambda_{X_{17}}^{(1)}}{2})$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{11}$	0	0	0
$\lambda_{11}(\frac{9}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)})$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$\lambda_{11}(3T_\nu^{(1)}+T_\nu^{(2)})$	0	1	0
$\lambda_9((\lambda_{10})^*+5(\lambda_{11})^*)$	0	4	0
$g_2^2\lambda_{11}$	0	-9	0
$g_1^2\lambda_{11}$	0	-3	0
$\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	0	1	0
$\lambda_{12}^{(A)}(-\frac{\lambda_{10}}{2}-\frac{5\lambda_{11}}{2})$	0	1	0
$\lambda_{HS_1}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}-2\lambda_{X_{17}}^{(1)})$	0	1	0
$\lambda_{HS_1}^{(A)}(\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}+\frac{\lambda_{X_{17}}^{(1)}}{2})$	0	1	0
$\lambda_{S_1}^{(1)}\lambda_{11}$	0	12	0
$\lambda_{11}(\frac{9}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)})$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda_{11}(3T_\nu^{(1)}+T_\nu^{(2)})$	0	0	0
$\lambda_9((\lambda_{10})^*+5(\lambda_{11})^*)$	0	4	0
$g_2^2\lambda_{11}$	0	-9	0
$g_1^2\lambda_{11}$	0	-3	0
$\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	0	1	0
$\lambda_{12}^{(A)}(-\frac{\lambda_{10}}{2}-\frac{5\lambda_{11}}{2})$	0	1	0
$\lambda_{HS_1}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}-2\lambda_{X_{17}}^{(1)})$	0	1	0
$\lambda_{HS_1}^{(A)}(\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}+\frac{\lambda_{X_{17}}^{(1)}}{2})$	0	1	0
$\lambda_{S_1}^{(1)}\lambda_{11}$	0	12	0
$\lambda_{11}(\frac{9}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)})$	0	1	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda_{11}(3T_\nu^{(1)}+T_\nu^{(2)})$	0	0	0
$\lambda_9((\lambda_{10})^*+5(\lambda_{11})^*)$	0	0	0
$g_2^2\lambda_{11}$	0	0	0
$g_1^2\lambda_{11}$	0	0	0
$\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	0	0	0
$\lambda_{12}^{(A)}(-\frac{\lambda_{10}}{2}-\frac{5\lambda_{11}}{2})$	0	0	0
$\lambda_{HS_1}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}-2\lambda_{X_{17}}^{(1)})$	0	0	0
$\lambda_{HS_1}^{(A)}(\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}+\frac{\lambda_{X_{17}}^{(1)}}{2})$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{11}$	0	0	0

$\lambda_{11}(\frac{9}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)})$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda_{11}(3T_\nu^{(1)}+T_\nu^{(2)})$	0	0	0
$\lambda_9((\lambda_{10})^*+5(\lambda_{11})^*)$	0	0	0
$g_2^2\lambda_{11}$	0	0	0
$g_1^2\lambda_{11}$	0	0	0
$\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	0	0	0
$\lambda_{12}^{(A)}(-\frac{\lambda_{10}}{2}-\frac{5\lambda_{11}}{2})$	0	0	0
$\lambda_{HS_1}^{(1)}(2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-2\lambda_{X_{17}}^{(1)})$	0	0	0
$\lambda_{HS_1}^{(A)}(\frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}+\frac{\lambda_{X_{17}}^{(1)}}{2})$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{11}$	0	0	0
$\lambda_{11}(\frac{9}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)})$	0	0	0

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_{11}(3T_\nu^{(1)}+T_\nu^{(2)})$	0
$\lambda_9((\lambda_{10})^*+5(\lambda_{11})^*)$	0
$g_2^2\lambda_{11}$	0
$g_1^2\lambda_{11}$	0
$\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	0
$\lambda_{12}^{(A)}(-\frac{\lambda_{10}}{2}-\frac{5\lambda_{11}}{2})$	0
$\lambda_{HS_1}^{(1)}(2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-2\lambda_{X_{17}}^{(1)})$	0
$\lambda_{HS_1}^{(A)}(\frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}+\frac{\lambda_{X_{17}}^{(1)}}{2})$	0
$\lambda_{S_1}^{(1)}\lambda_{11}$	0
$\lambda_{11}(\frac{9}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)})$	0

$\beta_{\lambda_9}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_9(2T_\nu^{(1)}+2T_\nu^{(2)})$	0	0	0
$\text{Tr}(y_2(y_1)^\dagger y_2(y_1)^\dagger)$	0	0	0
$\lambda_7(\lambda_8)^*$	0	0	0
$g_2^2\lambda_9$	0	0	0
$g_1^2\lambda_9$	0	0	0
$\lambda_{12}^{(1)}\lambda_9$	0	0	0
$\lambda_{12}^{(A)}\lambda_9$	0	0	0
${\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}^2$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_9$	0	0	0
$\lambda_9\lambda_{S_2}^{(1)}$	0	0	0
${\lambda_{10}}^2$	0	0	0
$\lambda_{10}\lambda_{11}$	0	0	0
${\lambda_{11}}^2$	0	0	0
${\lambda_{X_{17}}^{(1)}}^2$	0	0	0
$\lambda_9(3T_\nu^{(1)}+3T_\nu^{(2)})$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$\lambda_9(2T_\nu^{(1)}+2T_\nu^{(2)})$	0	1	0

$\mathrm{Tr}(y_2(y_1)^\dagger y_2(y_1)^\dagger)$	0	$\frac{2}{3}$	0
$\lambda_7(\lambda_8)^*$	0	4	0
$g_2^2\lambda_9$	0	-9	0
$g_1^2\lambda_9$	0	-3	0
$\lambda_{12}^{(1)}\lambda_9$	0	8	0
$\lambda_{12}^{(A)}\lambda_9$	0	-4	0
${\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}}^2$	0	1	0
$\lambda_{S_1}^{(1)}\lambda_9$	0	2	0
$\lambda_9\lambda_{S_2}^{(1)}$	0	2	0
${\lambda_{10}}^2$	0	5	0
$\lambda_{10}\lambda_{11}$	0	2	0
${\lambda_{11}}^2$	0	5	0
${\lambda_{X_{17}}^{(1)}}^2$	0	1	0
$\lambda_9(3T_\nu^{(1)}+3T_\nu^{(2)})$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda_9(2T_\nu^{(1)}+2T_\nu^{(2)})$	0	0	0
$\mathrm{Tr}(y_2(y_1)^\dagger y_2(y_1)^\dagger)$	0	$\frac{5}{6}$	0
$\lambda_7(\lambda_8)^*$	0	4	0
$g_2^2\lambda_9$	0	-9	0
$g_1^2\lambda_9$	0	-3	0
$\lambda_{12}^{(1)}\lambda_9$	0	8	0
$\lambda_{12}^{(A)}\lambda_9$	0	-4	0
${\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}}^2$	0	1	0
$\lambda_{S_1}^{(1)}\lambda_9$	0	2	0
$\lambda_9\lambda_{S_2}^{(1)}$	0	2	0
${\lambda_{10}}^2$	0	5	0
$\lambda_{10}\lambda_{11}$	0	2	0
${\lambda_{11}}^2$	0	5	0
${\lambda_{X_{17}}^{(1)}}^2$	0	1	0
$\lambda_9(3T_\nu^{(1)}+3T_\nu^{(2)})$	0	1	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda_9(2T_\nu^{(1)}+2T_\nu^{(2)})$	0	0	0
$\mathrm{Tr}(y_2(y_1)^\dagger y_2(y_1)^\dagger)$	0	0	0
$\lambda_7(\lambda_8)^*$	0	0	0
$g_2^2\lambda_9$	0	0	0
$g_1^2\lambda_9$	0	0	0
$\lambda_{12}^{(1)}\lambda_9$	0	0	0
$\lambda_{12}^{(A)}\lambda_9$	0	0	0
${\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}}^2$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_9$	0	0	0
$\lambda_9\lambda_{S_2}^{(1)}$	0	0	0
${\lambda_{10}}^2$	0	0	0
$\lambda_{10}\lambda_{11}$	0	0	0
${\lambda_{11}}^2$	0	0	0
${\lambda_{X_{17}}^{(1)}}^2$	0	0	0
$\lambda_9(3T_\nu^{(1)}+3T_\nu^{(2)})$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda_9(2T_\nu^{(1)}+2T_\nu^{(2)})$	0	0	0
$\mathrm{Tr}(y_2(y_1)^\dagger y_2(y_1)^\dagger)$	0	0	0
$\lambda_7(\lambda_8)^*$	0	0	0
$g_2^2\lambda_9$	0	0	0
$g_1^2\lambda_9$	0	0	0
$\lambda_{12}^{(1)}\lambda_9$	0	0	0
$\lambda_{12}^{(A)}\lambda_9$	0	0	0
${\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}^2$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_9$	0	0	0
$\lambda_9\lambda_{S_2}^{(1)}$	0	0	0
${\lambda_{10}}^2$	0	0	0
$\lambda_{10}\lambda_{11}$	0	0	0
${\lambda_{11}}^2$	0	0	0
${\lambda_{X_{17}}^{(1)}}^2$	0	0	0
$\lambda_9(3T_\nu^{(1)}+3T_\nu^{(2)})$	0	0	0

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_9(2T_\nu^{(1)}+2T_\nu^{(2)})$	0
$\mathrm{Tr}(y_2(y_1)^\dagger y_2(y_1)^\dagger)$	0
$\lambda_7(\lambda_8)^*$	0
$g_2^2\lambda_9$	0
$g_1^2\lambda_9$	0
$\lambda_{12}^{(1)}\lambda_9$	0
$\lambda_{12}^{(A)}\lambda_9$	0
${\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}^2$	0
$\lambda_{S_1}^{(1)}\lambda_9$	0
$\lambda_9\lambda_{S_2}^{(1)}$	0
${\lambda_{10}}^2$	0
$\lambda_{10}\lambda_{11}$	0
${\lambda_{11}}^2$	0
${\lambda_{X_{17}}^{(1)}}^2$	0
$\lambda_9(3T_\nu^{(1)}+3T_\nu^{(2)})$	0

$\beta_{\lambda_{10}}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_{10}(T_\nu^{(1)}+3T_\nu^{(2)})$	0	0	0
$\lambda_9(5(\lambda_{10})^*+(\lambda_{11})^*)$	0	0	0
$g_2^2\lambda_{10}$	0	0	0
$g_1^2\lambda_{10}$	0	0	0
$\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	0	0	0
$\lambda_{12}^{(A)}(-\frac{5\lambda_{10}}{2}-\frac{\lambda_{11}}{2})$	0	0	0
$\lambda_{HS_2}^{(1)}(2\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-2\lambda_{X_{17}}^{(1)})$	0	0	0
$\lambda_{HS_2}^{(A)}(-\frac{\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}-\frac{\lambda_{X_{17}}^{(1)}}{2})$	0	0	0
$\lambda_{10}\lambda_{S_2}^{(1)}$	0	0	0
$\lambda_{10}(\frac{3}{2}T_\nu^{(1)}+\frac{9}{2}T_\nu^{(2)})$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_{10}(T_\nu^{(1)} + 3T_\nu^{(2)})$	0	1	0
$\lambda_9(5(\lambda_{10})^* + (\lambda_{11})^*)$	0	4	0
$g_2^2\lambda_{10}$	0	-9	0
$g_1^2\lambda_{10}$	0	-3	0
$\lambda_{12}^{(1)}(6\lambda_{10} + 6\lambda_{11})$	0	1	0
$\lambda_{12}^{(A)}(-\frac{5\lambda_{10}}{2} - \frac{\lambda_{11}}{2})$	0	1	0
$\lambda_{HS_2}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} - 2\lambda_{X_{17}}^{(1)})$	0	1	0
$\lambda_{HS_2}^{(A)}(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} - \frac{\lambda_{X_{17}}^{(1)}}{2})$	0	1	0
$\lambda_{10}\lambda_{S_2}^{(1)}$	0	12	0
$\lambda_{10}(\frac{3}{2}T_\nu^{(1)} + \frac{9}{2}T_\nu^{(2)})$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_{10}(T_\nu^{(1)} + 3T_\nu^{(2)})$	0	0	0
$\lambda_9(5(\lambda_{10})^* + (\lambda_{11})^*)$	0	4	0
$g_2^2\lambda_{10}$	0	-9	0
$g_1^2\lambda_{10}$	0	-3	0
$\lambda_{12}^{(1)}(6\lambda_{10} + 6\lambda_{11})$	0	1	0
$\lambda_{12}^{(A)}(-\frac{5\lambda_{10}}{2} - \frac{\lambda_{11}}{2})$	0	1	0
$\lambda_{HS_2}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} - 2\lambda_{X_{17}}^{(1)})$	0	1	0
$\lambda_{HS_2}^{(A)}(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} - \frac{\lambda_{X_{17}}^{(1)}}{2})$	0	1	0
$\lambda_{10}\lambda_{S_2}^{(1)}$	0	12	0
$\lambda_{10}(\frac{3}{2}T_\nu^{(1)} + \frac{9}{2}T_\nu^{(2)})$	0	1	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_{10}(T_\nu^{(1)} + 3T_\nu^{(2)})$	0	0	0
$\lambda_9(5(\lambda_{10})^* + (\lambda_{11})^*)$	0	0	0
$g_2^2\lambda_{10}$	0	0	0
$g_1^2\lambda_{10}$	0	0	0
$\lambda_{12}^{(1)}(6\lambda_{10} + 6\lambda_{11})$	0	0	0
$\lambda_{12}^{(A)}(-\frac{5\lambda_{10}}{2} - \frac{\lambda_{11}}{2})$	0	0	0
$\lambda_{HS_2}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} - 2\lambda_{X_{17}}^{(1)})$	0	0	0
$\lambda_{HS_2}^{(A)}(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} - \frac{\lambda_{X_{17}}^{(1)}}{2})$	0	0	0
$\lambda_{10}\lambda_{S_2}^{(1)}$	0	0	0
$\lambda_{10}(\frac{3}{2}T_\nu^{(1)} + \frac{9}{2}T_\nu^{(2)})$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda_{10}(T_\nu^{(1)} + 3T_\nu^{(2)})$	0	0	0
$\lambda_9(5(\lambda_{10})^* + (\lambda_{11})^*)$	0	0	0
$g_2^2\lambda_{10}$	0	0	0
$g_1^2\lambda_{10}$	0	0	0
$\lambda_{12}^{(1)}(6\lambda_{10} + 6\lambda_{11})$	0	0	0
$\lambda_{12}^{(A)}(-\frac{5\lambda_{10}}{2} - \frac{\lambda_{11}}{2})$	0	0	0
$\lambda_{HS_2}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} - 2\lambda_{X_{17}}^{(1)})$	0	0	0
$\lambda_{HS_2}^{(A)}(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} - \frac{\lambda_{X_{17}}^{(1)}}{2})$	0	0	0
$\lambda_{10}\lambda_{S_2}^{(1)}$	0	0	0

$\lambda_{10}(\frac{3}{2}T_\nu^{(1)} + \frac{9}{2}T_\nu^{(2)})$	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$\lambda_{10}(T_\nu^{(1)} + 3T_\nu^{(2)})$	0
$\lambda_9(5(\lambda_{10})^* + (\lambda_{11})^*)$	0
$g_2^2\lambda_{10}$	0
$g_1^2\lambda_{10}$	0
$\lambda_{12}^{(1)}(6\lambda_{10} + 6\lambda_{11})$	0
$\lambda_{12}^{(A)}(-\frac{5\lambda_{10}}{2} - \frac{\lambda_{11}}{2})$	0
$\lambda_{HS_2}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} - 2\lambda_{X_{17}}^{(1)})$	0
$\lambda_{HS_2}^{(A)}(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} - \frac{\lambda_{X_{17}}^{(1)}}{2})$	0
$\lambda_{10}\lambda_{S_2}^{(1)}$	0
$\lambda_{10}(\frac{3}{2}T_\nu^{(1)} + \frac{9}{2}T_\nu^{(2)})$	0

$\beta_{\lambda_{S_2}^{(1)}}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda_{S_2}^{(1)}(4T_\nu^{(2)} + 8\lambda_{S_2}^{(A)})$	1	0	1
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	8	0	0
g_2^4	22	22	22
$g_1^2(8g_2^2 - 12\lambda_{S_2}^{(1)})$	1	0	1
$g_2^2\lambda_{S_2}^{(1)}$	-24	-24	-24
g_1^4	12	0	12
${\lambda_{12}^{(1)}}^2$	2	2	2
${\lambda_{HS_2}^{(1)}}^2$	4	4	4
${\lambda_{S_2}^{(1)}}^2$	14	14	14
${\lambda_{S_2}^{(A)}}^2$	16	16	16
$T_\nu^{(2)}$	$-\frac{8}{3}$	$-\frac{8}{3}$	$-\frac{8}{3}$
$\lambda_{S_2}^{(1)}(4T_\nu^{(2)} + 4\,\text{Tr}(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger) + 8\lambda_{S_2}^{(A)})$	0	1	0
$(\lambda_{X_2})^*\,\text{Tr}(y_2(y_2^{\text{conj}})^\dagger)$	0	$\frac{32}{3}$	0
$\lambda_{X_2}(\frac{32}{3}\,\text{Tr}(y_2^{\text{conj}}(y_2)^\dagger) + \frac{352(\lambda_{X_2})^*}{3})$	0	1	0
$\text{Tr}(y_2(y_2)^\dagger y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger)$	0	$-\frac{248}{45}$	0
$\text{Tr}(y_2(y_2^{\text{conj}})^\dagger y_2^{\text{conj}}(y_2)^\dagger)$	0	$-\frac{248}{45}$	0
$\lambda_{X_1}(\lambda_{X_1})^*$	0	$\frac{832}{3}$	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0	$\frac{8}{3}$	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	$\frac{16}{3}$	0
T_ν	0	$-\frac{8}{3}$	0
$\lambda_{S_2}^{(1)}T_\nu^{(2)}$	0	0	0
$g_1^2(\frac{27g_2^2}{2} - 27\lambda_{S_2}^{(1)})$	0	0	0
${\lambda_{12}^{(A)}}^2$	0	0	0
${\lambda_{HS_2}^{(A)}}^2$	0	0	0
$\lambda_8(\lambda_8)^*$	0	0	0
$\lambda_9(\lambda_9)^*$	0	0	0
$\lambda_{10}(\lambda_{10})^*$	0	0	0
$g_1^2(\frac{3g_2^2}{2} - 3\lambda_{S_2}^{(1)})$	0	0	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	0
$g_1^2\lambda_{S_2}^{(1)}$	0	0	0
$\lambda_{S_2}^{(1)}(8T_\nu^{(2)} + 8\,\text{Tr}(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger))$	0	0	0
$\lambda_{X_2}(24\,\text{Tr}(y_2^{\text{conj}}(y_2)^\dagger) + 108(\lambda_{X_2})^*)$	0	0	0

$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}(8(\lambda_{X_5}^{(1)})^* + \frac{20(\lambda_{X_5}^{(2)})^*}{3})$	0	0	0
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$g_1^2(32g_2^2 - 48\lambda_{S_2}^{(1)})$	0	0	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}(\frac{8(\lambda_{X_7}^{(1)})^*}{3} + 4(\lambda_{X_7}^{(2)})^*)$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_{S_2}^{(1)}(4T_\nu^{(2)} + 8\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	0	0	0
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
$g_1^2(8g_2^2 - 12\lambda_{S_2}^{(1)})$	0	0	0
$g_2^2\lambda_{S_2}^{(1)}$	-9	-9	-9
g_1^4	$\frac{243}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
${\lambda_{12}^{(1)}}^2$	4	4	4
${\lambda_{HS_2}^{(1)}}^2$	4	4	4
${\lambda_{S_2}^{(1)}}^2$	12	12	12
${\lambda_{S_2}^{(A)}}^2$	0	0	0
$T_\nu^{(2)}$	-4	-4	-4
$\lambda_{S_2}^{(1)}(4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 8\lambda_{S_2}^{(A)})$	0	0	0
$(\lambda_{X_2})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{X_2}(\frac{32}{3}\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + \frac{352(\lambda_{X_2})^*}{3})$	0	0	0
$\mathrm{Tr}(y_2(y_2)^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger y_2^{\mathrm{conj}}(y_2)^\dagger)$	0	0	0
$\lambda_{X_1}(\lambda_{X_1})^*$	0	0	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	0	0
T_ν	0	0	0
$\lambda_{S_2}^{(1)}T_\nu^{(2)}$	4	4	4
$g_1^2(\frac{27g_2^2}{2} - 27\lambda_{S_2}^{(1)})$	1	0	0
${\lambda_{12}^{(A)}}^2$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
${\lambda_{HS_2}^{(A)}}^2$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
$\lambda_8(\lambda_8)^*$	0	8	0
$\lambda_9(\lambda_9)^*$	0	8	0
$\lambda_{10}(\lambda_{10})^*$	0	24	0
$g_1^2(\frac{3g_2^2}{2} - 3\lambda_{S_2}^{(1)})$	0	1	1
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	8
$g_1^2\lambda_{S_2}^{(1)}$	0	0	0
$\lambda_{S_2}^{(1)}(8T_\nu^{(2)} + 8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_2}(24\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 108(\lambda_{X_2})^*)$	0	0	0
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}(8(\lambda_{X_5}^{(1)})^* + \frac{20(\lambda_{X_5}^{(2)})^*}{3})$	0	0	0
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$g_1^2(32g_2^2 - 48\lambda_{S_2}^{(1)})$	0	0	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}(\frac{8(\lambda_{X_7}^{(1)})^*}{3} + 4(\lambda_{X_7}^{(2)})^*)$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_{S_2}^{(1)}(4T_\nu^{(2)} + 8\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	0	0	0
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
$g_1^2(8g_2^2 - 12\lambda_{S_2}^{(1)})$	0	0	0
$g_2^2\lambda_{S_2}^{(1)}$	-9	-9	-9
g_1^4	$\frac{243}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
${\lambda_{12}^{(1)}}^2$	4	4	4
${\lambda_{HS_2}^{(1)}}^2$	4	4	4
${\lambda_{S_2}^{(1)}}^2$	12	12	12
${\lambda_{S_2}^{(A)}}^2$	0	0	0
$T_\nu^{(2)}$	-5	-5	-5
$\lambda_{S_2}^{(1)}(4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 8\lambda_{S_2}^{(A)})$	0	0	0
$(\lambda_{X_2})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{X_2}(\frac{32}{3}\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + \frac{352(\lambda_{X_2})^*}{3})$	0	0	0
$\mathrm{Tr}(y_2(y_2)^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger y_2^{\mathrm{conj}}(y_2)^\dagger)$	0	0	0
$\lambda_{X_1}(\lambda_{X_1})^*$	0	0	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	0	0
T_ν	0	0	0
$\lambda_{S_2}^{(1)}T_\nu^{(2)}$	6	6	6
$g_1^2(\frac{27g_2^2}{2} - 27\lambda_{S_2}^{(1)})$	1	0	0
${\lambda_{12}^{(A)}}^2$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
${\lambda_{HS_2}^{(A)}}^2$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
$\lambda_8(\lambda_8)^*$	0	8	0
$\lambda_9(\lambda_9)^*$	0	8	0
$\lambda_{10}(\lambda_{10})^*$	0	24	0
$g_1^2(\frac{3g_2^2}{2} - 3\lambda_{S_2}^{(1)})$	0	1	1
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	8
$g_1^2\lambda_{S_2}^{(1)}$	0	0	0
$\lambda_{S_2}^{(1)}(8T_\nu^{(2)} + 8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_2}(24\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 108(\lambda_{X_2})^*)$	0	0	0
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}(8(\lambda_{X_5}^{(1)})^* + \frac{20(\lambda_{X_5}^{(2)})^*}{3})$	0	0	0
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$g_1^2(32g_2^2 - 48\lambda_{S_2}^{(1)})$	0	0	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}(\frac{8(\lambda_{X_7}^{(1)})^*}{3} + 4(\lambda_{X_7}^{(2)})^*)$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_{S_2}^{(1)}(4T_\nu^{(2)} + 8\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	24	0	0
g_2^4	0	0	0
$g_1^2(8g_2^2 - 12\lambda_{S_2}^{(1)})$	0	0	0
$g_2^2\lambda_{S_2}^{(1)}$	0	0	0
g_1^4	12	192	0
${\lambda_{12}^{(1)}}^2$	6	6	6

$\lambda_{HS_2}^{(1)\ 2}$	4	4	4
$\lambda_{S_2}^{(1)2}$	10	10	10
$\lambda_{S_2}^{(A)2}$	0	0	0
$T_\nu^{(2)}$	-8	-8	-8
$\lambda_{S_2}^{(1)}(4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 8\lambda_{S_2}^{(A)})$	0	0	0
$(\lambda_{X_2})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	24
$\lambda_{X_2}(\frac{32}{3}\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + \frac{352(\lambda_{X_2})^*}{3})$	0	0	0
$\mathrm{Tr}(y_2(y_2)^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0	0	-16
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger y_2^{\mathrm{conj}}(y_2)^\dagger)$	0	0	-16
$\lambda_{X_1}(\lambda_{X_1})^*$	0	0	288
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0	0	12
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	0	8
T_ν	0	0	-8
$\lambda_{S_2}^{(1)}T_\nu^{(2)}$	8	8	0
$g_1^2(\frac{27g_2^2}{2} - 27\lambda_{S_2}^{(1)})$	0	0	0
$\lambda_{12}^{(A)2}$	0	0	0
$\lambda_{HS_2}^{(A)\ 2}$	0	0	0
$\lambda_8(\lambda_8)^*$	0	0	0
$\lambda_9(\lambda_9)^*$	0	0	0
$\lambda_{10}(\lambda_{10})^*$	0	0	0
$g_1^2(\frac{3g_2^2}{2} - 3\lambda_{S_2}^{(1)})$	0	0	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	0
$g_1^2\lambda_{S_2}^{(1)}$	-12	-48	0
$\lambda_{S_2}^{(1)}(8T_\nu^{(2)} + 8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	1
$\lambda_{X_2}(24\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 108(\lambda_{X_2})^*)$	0	0	1
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}(8(\lambda_{X_5}^{(1)})^* + \frac{20(\lambda_{X_5}^{(2)})^*}{3})$	0	0	0
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	0	0	0
$\lambda_{12}^{(\times)2}$	0	0	0
$g_1^2(32g_2^2 - 48\lambda_{S_2}^{(1)})$	0	0	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}(\frac{8(\lambda_{X_7}^{(1)})^*}{3} + 4(\lambda_{X_7}^{(2)})^*)$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda_{S_2}^{(1)}(4T_\nu^{(2)} + 8\lambda_{S_2}^{(A)})$	1	1	0
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	0	0	0
g_2^4	22	22	22
$g_1^2(8g_2^2 - 12\lambda_{S_2}^{(1)})$	1	0	0
$g_2^2\lambda_{S_2}^{(1)}$	-24	-24	-24
g_1^4	12	192	0
$\lambda_{12}^{(1)2}$	6	6	6
$\lambda_{HS_2}^{(1)\ 2}$	4	4	4
$\lambda_{S_2}^{(1)2}$	14	14	14
$\lambda_{S_2}^{(A)2}$	16	16	16
$T_\nu^{(2)}$	$-\frac{8}{3}$	$-\frac{8}{3}$	$-\frac{8}{3}$
$\lambda_{S_2}^{(1)}(4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 8\lambda_{S_2}^{(A)})$	0	0	1
$(\lambda_{X_2})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	$\frac{32}{3}$
$\lambda_{X_2}(\frac{32}{3}\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + \frac{352(\lambda_{X_2})^*}{3})$	0	0	1
$\mathrm{Tr}(y_2(y_2)^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0	0	$-\frac{248}{45}$
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger y_2^{\mathrm{conj}}(y_2)^\dagger)$	0	0	$-\frac{248}{45}$

$\lambda_{X_1}(\lambda_{X_1})^*$	0	0	$\frac{832}{3}$
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	0	$\frac{16}{3}$
T_ν	0	0	$-\frac{8}{3}$
$\lambda_{S_2}^{(1)}T_\nu^{(2)}$	0	0	0
$g_1^2(\frac{27g_2^2}{2}-27\lambda_{S_2}^{(1)})$	0	0	0
${\lambda_{12}^{(A)}}^2$	0	0	0
$\lambda_{HS_2}^{(A)\ 2}$	0	0	0
$\lambda_8(\lambda_8)^*$	0	0	0
$\lambda_9(\lambda_9)^*$	0	0	0
$\lambda_{10}(\lambda_{10})^*$	0	0	0
$g_1^2(\frac{3g_2^2}{2}-3\lambda_{S_2}^{(1)})$	0	0	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	0
$g_1^2\lambda_{S_2}^{(1)}$	0	0	0
$\lambda_{S_2}^{(1)}(8T_\nu^{(2)}+8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_2}(24\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger)+108(\lambda_{X_2})^*)$	0	0	0
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^*+(\lambda_{X_5}^{(2)})^*)$	8	0	0
$\lambda_{X_5}^{(2)}(8(\lambda_{X_5}^{(1)})^*+\frac{20(\lambda_{X_5}^{(2)})^*}{3})$	1	0	0
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	8	8	8
${\lambda_{12}^{(\times)}}^2$	$\frac{20}{3}$	$\frac{20}{3}$	$\frac{20}{3}$
$g_1^2(32g_2^2-48\lambda_{S_2}^{(1)})$	0	1	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^*+(\lambda_{X_7}^{(2)})^*)$	0	0	$\frac{8}{3}$
$\lambda_{X_7}^{(2)}(\frac{8(\lambda_{X_7}^{(1)})^*}{3}+4(\lambda_{X_7}^{(2)})^*)$	0	0	1

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_{S_2}^{(1)}(4T_\nu^{(2)}+8\lambda_{S_2}^{(A)})$	1
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	0
g_2^4	22
$g_1^2(8g_2^2-12\lambda_{S_2}^{(1)})$	1
$g_2^2\lambda_{S_2}^{(1)}$	-24
g_1^4	12
${\lambda_{12}^{(1)}}^2$	6
$\lambda_{HS_2}^{(1)\ 2}$	4
${\lambda_{S_2}^{(1)}}^2$	14
${\lambda_{S_2}^{(A)}}^2$	16
$T_\nu^{(2)}$	$-\frac{8}{3}$
$\lambda_{S_2}^{(1)}(4T_\nu^{(2)}+4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+8\lambda_{S_2}^{(A)})$	0
$(\lambda_{X_2})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0
$\lambda_{X_2}(\frac{32}{3}\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger)+\frac{352(\lambda_{X_2})^*}{3})$	0
$\mathrm{Tr}(y_2(y_2)^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger y_2^{\mathrm{conj}}(y_2)^\dagger)$	0
$\lambda_{X_1}(\lambda_{X_1})^*$	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0
T_ν	0
$\lambda_{S_2}^{(1)}T_\nu^{(2)}$	0
$g_1^2(\frac{27g_2^2}{2}-27\lambda_{S_2}^{(1)})$	0
${\lambda_{12}^{(A)}}^2$	0
$\lambda_{HS_2}^{(A)\ 2}$	0
$\lambda_8(\lambda_8)^*$	0
$\lambda_9(\lambda_9)^*$	0

$\lambda_{10}(\lambda_{10})^*$	0
$g_1^2(\frac{3g_2^2}{2} - 3\lambda_{S_2}^{(1)})$	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0
$g_1^2\lambda_{S_2}^{(1)}$	0
$\lambda_{S_2}^{(1)}(8T_\nu^{(2)} + 8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0
$\lambda_{X_2}(24\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 108(\lambda_{X_2})^*)$	0
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0
$\lambda_{X_5}^{(2)}(8(\lambda_{X_5}^{(1)})^* + \frac{20(\lambda_{X_5}^{(2)})^*}{3})$	0
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	8
${\lambda_{12}^{(\times)}}^2$	$\frac{20}{3}$
$g_1^2(32g_2^2 - 48\lambda_{S_2}^{(1)})$	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0
$\lambda_{X_7}^{(2)}(\frac{8(\lambda_{X_7}^{(1)})^*}{3} + 4(\lambda_{X_7}^{(2)})^*)$	0

$\beta_{\lambda_{S_2}^{(A)}}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda_{S_2}^{(A)}T_\nu^{(2)}$	4	0	4
g_2^4	-2	-2	-2
$g_1^2(8g_2^2 - 12\lambda_{S_2}^{(A)})$	1	0	1
$g_2^2\lambda_{S_2}^{(A)}$	-24	-24	-24
${\lambda_{HS_2}^{(A)}}^2$	3	3	3
$\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	12	12	12
${\lambda_{S_2}^{(A)}}^2$	20	20	20
$T_\nu^{(2)}$	$-\frac{2}{3}$	$-\frac{2}{3}$	$-\frac{2}{3}$
$\lambda_{S_2}^{(A)}(4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	1	0
$(\lambda_{X_2})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	$-\frac{4}{3}$	0
$\lambda_{X_2}(-\frac{4}{3}\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) - \frac{44(\lambda_{X_2})^*}{3})$	0	1	0
$\mathrm{Tr}(y_2(y_2)^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0	$\frac{76}{45}$	0
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger y_2^{\mathrm{conj}}(y_2)^\dagger)$	0	$\frac{76}{45}$	0
$\lambda_{X_1}(\lambda_{X_1})^*$	0	$-\frac{224}{3}$	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0	$-\frac{4}{3}$	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	$-\frac{8}{3}$	0
T_ν	0	$-\frac{2}{3}$	0
$\lambda_{X_5}^{(2)}(\lambda_{X_5}^{(2)})^*$	0	0	0
${\lambda_{12}^{(A)}}^2$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$g_1^2(32g_2^2 - 48\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_{X_7}^{(2)}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_{S_2}^{(A)}T_\nu^{(2)}$	0	0	0
g_2^4	0	0	0
$g_1^2(8g_2^2 - 12\lambda_{S_2}^{(A)})$	0	0	0
$g_2^2\lambda_{S_2}^{(A)}$	0	0	0
${\lambda_{HS_2}^{(A)}}^2$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	0	0	0
${\lambda_{S_2}^{(A)}}^2$	0	0	0

$T_\nu^{(2)}$	0	0	0
$\lambda_{S_2}^{(A)}(4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$(\lambda_{X_2})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{X_2}(-\frac{4}{3}\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) - \frac{44(\lambda_{X_2})^*}{3})$	0	0	0
$\mathrm{Tr}(y_2(y_2)^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger y_2^{\mathrm{conj}}(y_2)^\dagger)$	0	0	0
$\lambda_{X_1}(\lambda_{X_1})^*$	0	0	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	0	0
T_ν	0	0	0
$\lambda_{X_5}^{(2)}(\lambda_{X_5}^{(2)})^*$	0	0	0
${\lambda_{12}^{(A)}}^2$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$g_1^2(32g_2^2 - 48\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_{X_7}^{(2)}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_{S_2}^{(A)}T_\nu^{(2)}$	0	0	0
g_2^4	0	0	0
$g_1^2(8g_2^2 - 12\lambda_{S_2}^{(A)})$	0	0	0
$g_2^2{\lambda_{S_2}^{(A)}}^2$	0	0	0
${\lambda_{HS_2}^{(A)}}^2$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	0	0	0
${\lambda_{S_2}^{(A)}}^2$	0	0	0
$T_\nu^{(2)}$	0	0	0
$\lambda_{S_2}^{(A)}(4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$(\lambda_{X_2})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{X_2}(-\frac{4}{3}\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) - \frac{44(\lambda_{X_2})^*}{3})$	0	0	0
$\mathrm{Tr}(y_2(y_2)^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger y_2^{\mathrm{conj}}(y_2)^\dagger)$	0	0	0
$\lambda_{X_1}(\lambda_{X_1})^*$	0	0	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	0	0
T_ν	0	0	0
$\lambda_{X_5}^{(2)}(\lambda_{X_5}^{(2)})^*$	0	0	0
${\lambda_{12}^{(A)}}^2$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$g_1^2(32g_2^2 - 48\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_{X_7}^{(2)}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_{S_2}^{(A)}T_\nu^{(2)}$	0	0	0
g_2^4	0	0	0
$g_1^2(8g_2^2 - 12\lambda_{S_2}^{(A)})$	0	0	0
$g_2^2{\lambda_{S_2}^{(A)}}^2$	0	0	0
${\lambda_{HS_2}^{(A)}}^2$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	0	0	0
${\lambda_{S_2}^{(A)}}^2$	0	0	0

$T_\nu^{(2)}$	0	0	0
$\lambda_{S_2}^{(A)}(4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$(\lambda_{X_2})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{X_2}(-\frac{4}{3}\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) - \frac{44(\lambda_{X_2})^*}{3})$	0	0	0
$\mathrm{Tr}(y_2(y_2)^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger y_2^{\mathrm{conj}}(y_2)^\dagger)$	0	0	0
$\lambda_{X_1}(\lambda_{X_1})^*$	0	0	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	0	0
T_ν	0	0	0
$\lambda_{X_5}^{(2)}(\lambda_{X_5}^{(2)})^*$	0	0	0
${\lambda_{12}^{(A)}}^2$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$g_1^2(32g_2^2 - 48\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_{X_7}^{(2)}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda_{S_2}^{(A)}T_\nu^{(2)}$	4	4	0
g_2^4	-2	-2	-2
$g_1^2(8g_2^2 - 12\lambda_{S_2}^{(A)})$	1	0	0
$g_2^2\lambda_{S_2}^{(A)}$	-24	-24	-24
${\lambda_{HS_2}^{(A)}}^2$	3	3	3
$\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	12	12	12
${\lambda_{S_2}^{(A)}}^2$	20	20	20
$T_\nu^{(2)}$	$-\frac{2}{3}$	$-\frac{2}{3}$	$-\frac{2}{3}$
$\lambda_{S_2}^{(A)}(4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	1
$(\lambda_{X_2})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	$-\frac{4}{3}$
$\lambda_{X_2}(-\frac{4}{3}\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) - \frac{44(\lambda_{X_2})^*}{3})$	0	0	1
$\mathrm{Tr}(y_2(y_2)^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0	0	$\frac{76}{45}$
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger y_2^{\mathrm{conj}}(y_2)^\dagger)$	0	0	$\frac{76}{45}$
$\lambda_{X_1}(\lambda_{X_1})^*$	0	0	$-\frac{224}{3}$
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	0	$-\frac{8}{3}$
T_ν	0	0	$-\frac{2}{3}$
$\lambda_{X_5}^{(2)}(\lambda_{X_5}^{(2)})^*$	$-\frac{4}{3}$	0	0
${\lambda_{12}^{(A)}}^2$	12	12	12
${\lambda_{12}^{(\times)}}^2$	$-\frac{4}{3}$	$-\frac{4}{3}$	$-\frac{4}{3}$
$g_1^2(32g_2^2 - 48\lambda_{S_2}^{(A)})$	0	1	0
$\lambda_{X_7}^{(2)}(\lambda_{X_7}^{(1)})^*$	0	0	$-\frac{4}{3}$
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	$-\frac{4}{3}$

$\mathcal{S}$	E ($\alpha = -4$)
$\lambda_{S_2}^{(A)}T_\nu^{(2)}$	4
g_2^4	-2
$g_1^2(8g_2^2 - 12\lambda_{S_2}^{(A)})$	1
$g_2^2\lambda_{S_2}^{(A)}$	-24
${\lambda_{HS_2}^{(A)}}^2$	3
$\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	12
${\lambda_{S_2}^{(A)}}^2$	20

$T_\nu^{(2)}$	$-\frac{2}{3}$
$\lambda_{S_2}^{(A)}(4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0
$(\lambda_{X_2})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0
$\lambda_{X_2}(-\frac{4}{3}\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) - \frac{44(\lambda_{X_2})^*}{3})$	0
$\mathrm{Tr}(y_2(y_2)^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger y_2^{\mathrm{conj}}(y_2)^\dagger)$	0
$\lambda_{X_1}(\lambda_{X_1})^*$	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0
$\lambda_{X16}(\lambda_{X16})^*$	0
T_ν	0
$\lambda_{X_5}^{(2)}(\lambda_{X_5}^{(2)})^*$	0
${\lambda_{12}^{(A)}}^2$	12
${\lambda_{12}^{(\times)}}^2$	$-\frac{4}{3}$
$g_1^2(32g_2^2 - 48\lambda_{S_2}^{(A)})$	0
$\lambda_{X_7}^{(2)}(\lambda_{X_7}^{(1)})^*$	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0

$\beta_{\lambda_5}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda_5(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_{HS_2}^{(A)})$	1	0	0
$(\lambda_{X_{14}})^*\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger)$	4	0	0
$(\lambda_{X_{14}})^*((\lambda_{X_5}^{(1)})^* + 2(\lambda_{X_{18}})^*)$	4	0	0
$g_2^2\lambda_5$	$-\frac{21}{2}$	$-\frac{21}{2}$	$-\frac{21}{2}$
$g_1^2\lambda_5$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{33}{2}$
$\lambda_{12}^{(1)}\lambda_5$	2	2	2
$\lambda_1\lambda_5$	2	2	2
$\lambda_{HS_1}^{(1)}\lambda_5$	4	4	4
$\lambda_{HS_2}^{(1)}\lambda_5$	4	4	4
$\lambda_5(2T_\nu^{(1)} + T_\nu^{(2)} + \mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_{HS_2}^{(A)})$	0	1	0
$(\lambda_{X_{12}})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	2	0
$\lambda_{X_7}^{(1)}(\lambda_{X_{12}})^*$	0	4	0
$\lambda_{X_{16}}(\lambda_{X_{12}})^*$	0	8	0
$\lambda_5(2T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_{HS_2}^{(A)})$	0	0	1
$\lambda_5(T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + \frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)})$	0	0	0
$(\lambda_7)^*\,\mathrm{Tr}(y_2(y_1)^\dagger)$	0	0	0
$(\lambda_8)^*\,\mathrm{Tr}(y_1(y_2)^\dagger)$	0	0	0
$\lambda_5(\frac{3}{2}T_\nu^{(1)} + \frac{3}{2}T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + \frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)})$	0	0	0
$\lambda_5(T_\nu^{(1)} + \mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) - 4\lambda_{HS_1}^{(A)})$	0	0	0
$\lambda_5(T_\nu^{(1)} + 2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) - 4\lambda_{HS_1}^{(A)})$	0	0	0
$\lambda_5(T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) - 4\lambda_{HS_1}^{(A)})$	0	0	0
$\lambda_5(T_\nu^{(1)} + \mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3})$	0	0	0
$(\lambda_{X_{14}})^*(2(\lambda_{X_5}^{(1)})^* - (\lambda_{X_5}^{(2)})^* + 4(\lambda_{X_{18}})^*)$	0	0	0
$\lambda_5(T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3})$	0	0	0
$\lambda_5(T_\nu^{(1)} + T_\nu^{(2)} + \mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3})$	0	0	0
$\lambda_{X_7}^{(2)}(\lambda_{X_{12}})^*$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_5(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_{HS_2}^{(A)})$	0	0	0
$(\lambda_{X_{14}})^*\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger)$	0	0	0
$(\lambda_{X_{14}})^*((\lambda_{X_5}^{(1)})^* + 2(\lambda_{X_{18}})^*)$	0	0	0

$g_2^2\lambda_5$	-9	-9	-9
$g_1^2\lambda_5$	-9	-3	-9
$\lambda_{12}^{(1)}\lambda_5$	2	2	2
$\lambda_1\lambda_5$	2	2	2
$\lambda_{HS_1}^{(1)}\lambda_5$	4	4	4
$\lambda_{HS_2}^{(1)}\lambda_5$	4	4	4
$\lambda_5(2T_\nu^{(1)}+T_\nu^{(2)}+\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+4\lambda_{HS_2}^{(A)})$	0	0	0
$(\lambda_{X_{12}})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{X_7}^{(1)}(\lambda_{X_{12}})^*$	0	0	0
$\lambda_{X_{16}}(\lambda_{X_{12}})^*$	0	0	0
$\lambda_5(2T_\nu^{(1)}+T_\nu^{(2)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+4\lambda_{HS_2}^{(A)})$	0	0	0
$\lambda_5(T_\nu^{(1)}+T_\nu^{(2)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)})$	1	1	1
$(\lambda_7)^*\,\mathrm{Tr}(y_2(y_1)^\dagger)$	0	$-\frac{2}{3}$	0
$(\lambda_8)^*\,\mathrm{Tr}(y_1(y_2)^\dagger)$	0	2	0
$\lambda_5(\frac{3}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)})$	0	0	0
$\lambda_5(T_\nu^{(1)}+\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+2T_\nu^{(2)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)-4\lambda_{HS_1}^{(A)})$	0	0	0
$\lambda_5(T_\nu^{(1)}+2T_\nu^{(2)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)-4\lambda_{HS_1}^{(A)})$	0	0	0
$\lambda_5(T_\nu^{(1)}+2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)-4\lambda_{HS_1}^{(A)})$	0	0	0
$\lambda_5(T_\nu^{(1)}+\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+T_\nu^{(2)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)-\frac{4\lambda_{12}^{(A)}}{3}-2\lambda_{12}^{(\times)}-2\lambda_{HS_1}^{(A)}-\frac{4\lambda_{HS_2}^{(A)}}{3})$	0	0	0
$(\lambda_{X_{14}})^*(2(\lambda_{X_5}^{(1)})^*-(\lambda_{X_5}^{(2)})^*+4(\lambda_{X_{18}})^*)$	0	0	0
$\lambda_5(T_\nu^{(1)}+T_\nu^{(2)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)-\frac{4\lambda_{12}^{(A)}}{3}-2\lambda_{12}^{(\times)}-2\lambda_{HS_1}^{(A)}-\frac{4\lambda_{HS_2}^{(A)}}{3})$	0	0	0
$\lambda_5(T_\nu^{(1)}+T_\nu^{(2)}+\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)-\frac{4\lambda_{12}^{(A)}}{3}-2\lambda_{12}^{(\times)}-2\lambda_{HS_1}^{(A)}-\frac{4\lambda_{HS_2}^{(A)}}{3})$	0	0	0
$\lambda_{X_7}^{(2)}(\lambda_{X_{12}})^*$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda_5(2T_\nu^{(1)}+2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+T_\nu^{(2)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+4\lambda_{HS_2}^{(A)})$	0	0	0
$(\lambda_{X_{14}})^*\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger)$	0	0	0
$(\lambda_{X_{14}})^*((\lambda_{X_5}^{(1)})^*+2(\lambda_{X_{18}})^*)$	0	0	0
$g_2^2\lambda_5$	-9	-9	-9
$g_1^2\lambda_5$	-9	-3	-9
$\lambda_{12}^{(1)}\lambda_5$	2	2	2
$\lambda_1\lambda_5$	2	2	2
$\lambda_{HS_1}^{(1)}\lambda_5$	4	4	4
$\lambda_{HS_2}^{(1)}\lambda_5$	4	4	4
$\lambda_5(2T_\nu^{(1)}+T_\nu^{(2)}+\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+4\lambda_{HS_2}^{(A)})$	0	0	0
$(\lambda_{X_{12}})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{X_7}^{(1)}(\lambda_{X_{12}})^*$	0	0	0
$\lambda_{X_{16}}(\lambda_{X_{12}})^*$	0	0	0
$\lambda_5(2T_\nu^{(1)}+T_\nu^{(2)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+4\lambda_{HS_2}^{(A)})$	0	0	0
$\lambda_5(T_\nu^{(1)}+T_\nu^{(2)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)})$	0	0	0
$(\lambda_7)^*\,\mathrm{Tr}(y_2(y_1)^\dagger)$	0	-1	0
$(\lambda_8)^*\,\mathrm{Tr}(y_1(y_2)^\dagger)$	0	3	0
$\lambda_5(\frac{3}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)})$	1	1	1
$\lambda_5(T_\nu^{(1)}+\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+2T_\nu^{(2)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)-4\lambda_{HS_1}^{(A)})$	0	0	0
$\lambda_5(T_\nu^{(1)}+2T_\nu^{(2)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)-4\lambda_{HS_1}^{(A)})$	0	0	0
$\lambda_5(T_\nu^{(1)}+2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)-4\lambda_{HS_1}^{(A)})$	0	0	0
$\lambda_5(T_\nu^{(1)}+\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+T_\nu^{(2)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)-\frac{4\lambda_{12}^{(A)}}{3}-2\lambda_{12}^{(\times)}-2\lambda_{HS_1}^{(A)}-\frac{4\lambda_{HS_2}^{(A)}}{3})$	0	0	0
$(\lambda_{X_{14}})^*(2(\lambda_{X_5}^{(1)})^*-(\lambda_{X_5}^{(2)})^*+4(\lambda_{X_{18}})^*)$	0	0	0
$\lambda_5(T_\nu^{(1)}+T_\nu^{(2)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)-\frac{4\lambda_{12}^{(A)}}{3}-2\lambda_{12}^{(\times)}-2\lambda_{HS_1}^{(A)}-\frac{4\lambda_{HS_2}^{(A)}}{3})$	0	0	0
$\lambda_5(T_\nu^{(1)}+T_\nu^{(2)}+\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)-\frac{4\lambda_{12}^{(A)}}{3}-2\lambda_{12}^{(\times)}-2\lambda_{HS_1}^{(A)}-\frac{4\lambda_{HS_2}^{(A)}}{3})$	0	0	0
$\lambda_{X_7}^{(2)}(\lambda_{X_{12}})^*$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_5(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_{HS_2}^{(A)})$	0	0	0
$(\lambda_{X_{14}})^*\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger)$	2	0	0
$(\lambda_{X_{14}})^*((\lambda_{X_5}^{(1)})^* + 2(\lambda_{X_{18}})^*)$	4	0	0
$g_2^2\lambda_5$	$-\frac{21}{2}$	$-\frac{21}{2}$	$-\frac{21}{2}$
$g_1^2\lambda_5$	$-\frac{9}{2}$	$-\frac{33}{2}$	$-\frac{9}{2}$
$\lambda_{12}^{(1)}\lambda_5$	2	2	2
$\lambda_1\lambda_5$	2	2	2
$\lambda_{HS_1}^{(1)}\lambda_5$	4	4	4
$\lambda_{HS_2}^{(1)}\lambda_5$	4	4	4
$\lambda_5(2T_\nu^{(1)} + T_\nu^{(2)} + \mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_{HS_2}^{(A)})$	0	0	0
$(\lambda_{X_{12}})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	4
$\lambda_{X_7}^{(1)}(\lambda_{X_{12}})^*$	0	0	4
$\lambda_{X_{16}}(\lambda_{X_{12}})^*$	0	0	8
$\lambda_5(2T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_{HS_2}^{(A)})$	0	0	0
$\lambda_5(T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + \frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)})$	0	0	0
$(\lambda_7)^*\,\mathrm{Tr}(y_2(y_1)^\dagger)$	0	0	0
$(\lambda_8)^*\,\mathrm{Tr}(y_1(y_2)^\dagger)$	0	0	0
$\lambda_5(\frac{3}{2}T_\nu^{(1)} + \frac{3}{2}T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + \frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)})$	0	0	0
$\lambda_5(T_\nu^{(1)} + \mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) - 4\lambda_{HS_1}^{(A)})$	1	0	0
$\lambda_5(T_\nu^{(1)} + 2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) - 4\lambda_{HS_1}^{(A)})$	0	1	0
$\lambda_5(T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) - 4\lambda_{HS_1}^{(A)})$	0	0	1
$\lambda_5(T_\nu^{(1)} + \mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\chi)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3})$	0	0	0
$(\lambda_{X_{14}})^*(2(\lambda_{X_5}^{(1)})^* - (\lambda_{X_5}^{(2)})^* + 4(\lambda_{X_{18}})^*)$	0	0	0
$\lambda_5(T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\chi)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3})$	0	0	0
$\lambda_5(T_\nu^{(1)} + T_\nu^{(2)} + \mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\chi)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3})$	0	0	0
$\lambda_{X_7}^{(2)}(\lambda_{X_{12}})^*$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda_5(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_{HS_2}^{(A)})$	0	0	0
$(\lambda_{X_{14}})^*\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger)$	2	0	0
$(\lambda_{X_{14}})^*((\lambda_{X_5}^{(1)})^* + 2(\lambda_{X_{18}})^*)$	0	0	0
$g_2^2\lambda_5$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$g_1^2\lambda_5$	$-\frac{9}{2}$	$-\frac{33}{2}$	$-\frac{9}{2}$
$\lambda_{12}^{(1)}\lambda_5$	2	2	2
$\lambda_1\lambda_5$	2	2	2
$\lambda_{HS_1}^{(1)}\lambda_5$	4	4	4
$\lambda_{HS_2}^{(1)}\lambda_5$	4	4	4
$\lambda_5(2T_\nu^{(1)} + T_\nu^{(2)} + \mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_{HS_2}^{(A)})$	0	0	0
$(\lambda_{X_{12}})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	2
$\lambda_{X_7}^{(1)}(\lambda_{X_{12}})^*$	0	0	4
$\lambda_{X_{16}}(\lambda_{X_{12}})^*$	0	0	8
$\lambda_5(2T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_{HS_2}^{(A)})$	0	0	0
$\lambda_5(T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + \frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)})$	0	0	0
$(\lambda_7)^*\,\mathrm{Tr}(y_2(y_1)^\dagger)$	0	0	0
$(\lambda_8)^*\,\mathrm{Tr}(y_1(y_2)^\dagger)$	0	0	0
$\lambda_5(\frac{3}{2}T_\nu^{(1)} + \frac{3}{2}T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + \frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)})$	0	0	0
$\lambda_5(T_\nu^{(1)} + \mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) - 4\lambda_{HS_1}^{(A)})$	0	0	0
$\lambda_5(T_\nu^{(1)} + 2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) - 4\lambda_{HS_1}^{(A)})$	0	0	0
$\lambda_5(T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) - 4\lambda_{HS_1}^{(A)})$	0	0	0
$\lambda_5(T_\nu^{(1)} + \mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\chi)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3})$	1	0	0
$(\lambda_{X_{14}})^*(2(\lambda_{X_5}^{(1)})^* - (\lambda_{X_5}^{(2)})^* + 4(\lambda_{X_{18}})^*)$	2	0	0

$\lambda_5(T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3})$	0	1	0
$\lambda_5(T_\nu^{(1)} + T_\nu^{(2)} + \mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3})$	0	0	1
$\lambda_{X_7}^{(2)}(\lambda_{X_{12}})^*$	0	0	-2

$\mathcal{S}$	E ($\alpha = -4$)
$\lambda_5(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_{HS_2}^{(A)})$	0
$(\lambda_{X_{14}})^*\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger)$	0
$(\lambda_{X_{14}})^*((\lambda_{X_5}^{(1)})^* + 2(\lambda_{X_{18}})^*)$	0
$g_2^2\lambda_5$	$-\frac{33}{2}$
$g_1^2\lambda_5$	$-\frac{33}{2}$
$\lambda_{12}^{(1)}\lambda_5$	2
$\lambda_1\lambda_5$	2
$\lambda_{HS_1}^{(1)}\lambda_5$	4
$\lambda_{HS_2}^{(1)}\lambda_5$	4
$\lambda_5(2T_\nu^{(1)} + T_\nu^{(2)} + \mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_{HS_2}^{(A)})$	0
$(\lambda_{X_{12}})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0
$\lambda_{X_7}^{(1)}(\lambda_{X_{12}})^*$	0
$\lambda_{X_{16}}(\lambda_{X_{12}})^*$	0
$\lambda_5(2T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_{HS_2}^{(A)})$	0
$\lambda_5(T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + \frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)})$	0
$(\lambda_7)^*\,\mathrm{Tr}(y_2(y_1)^\dagger)$	0
$(\lambda_8)^*\,\mathrm{Tr}(y_1(y_2)^\dagger)$	0
$\lambda_5(\frac{3}{2}T_\nu^{(1)} + \frac{3}{2}T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + \frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)})$	0
$\lambda_5(T_\nu^{(1)} + \mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) - 4\lambda_{HS_1}^{(A)})$	0
$\lambda_5(T_\nu^{(1)} + 2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) - 4\lambda_{HS_1}^{(A)})$	0
$\lambda_5(T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) - 4\lambda_{HS_1}^{(A)})$	0
$\lambda_5(T_\nu^{(1)} + \mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3})$	0
$(\lambda_{X_{14}})^*(2(\lambda_{X_5}^{(1)})^* - (\lambda_{X_5}^{(2)})^* + 4(\lambda_{X_{18}})^*)$	0
$\lambda_5(T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3})$	1
$\lambda_5(T_\nu^{(1)} + T_\nu^{(2)} + \mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3})$	0
$\lambda_{X_7}^{(2)}(\lambda_{X_{12}})^*$	0

$\beta_{\lambda_{X_1}}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda_{X_1}(4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	1	0
$\lambda_{X_2}\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	2	0
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger y_2(y_2^{\mathrm{conj}})^\dagger)$	0	$-\frac{23}{15}$	0
$g_2^2\lambda_{X_1}$	0	-24	0
$\lambda_{S_2}^{(1)}\lambda_{X_1}$	0	12	0
$\lambda_{S_2}^{(A)}\lambda_{X_1}$	0	-8	0
${\lambda_{X_2}}^2$	0	11	0
${\lambda_{X_7}^{(1)}}^2$	0	1	0
${\lambda_{X_{16}}}^2$	0	2	0
$\lambda_{X_1}(8T_\nu^{(2)} + 8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_7}^{(1)}\lambda_{X_7}^{(2)}$	0	0	0
${\lambda_{X_7}^{(2)}}^2$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_{X_1}(4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0

$\lambda_{X_2}\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$g_2^2\lambda_{X_1}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_1}$	0	0	0
$\lambda_{S_2}^{(A)}\lambda_{X_1}$	0	0	0
${\lambda_{X_2}}^2$	0	0	0
${\lambda_{X_7}^{(1)}}^2$	0	0	0
${\lambda_{X_{16}}}^2$	0	0	0
$\lambda_{X_1}(8T_\nu^{(2)}+8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_7}^{(1)}\lambda_{X_7}^{(2)}$	0	0	0
${\lambda_{X_7}^{(2)}}^2$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda_{X_1}(4T_\nu^{(2)}+4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_2}\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$g_2^2\lambda_{X_1}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_1}$	0	0	0
$\lambda_{S_2}^{(A)}\lambda_{X_1}$	0	0	0
${\lambda_{X_2}}^2$	0	0	0
${\lambda_{X_7}^{(1)}}^2$	0	0	0
${\lambda_{X_{16}}}^2$	0	0	0
$\lambda_{X_1}(8T_\nu^{(2)}+8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_7}^{(1)}\lambda_{X_7}^{(2)}$	0	0	0
${\lambda_{X_7}^{(2)}}^2$	0	0	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda_{X_1}(4T_\nu^{(2)}+4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_2}\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	4
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	-4
$g_2^2\lambda_{X_1}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_1}$	0	0	12
$\lambda_{S_2}^{(A)}\lambda_{X_1}$	0	0	0
${\lambda_{X_2}}^2$	0	0	9
${\lambda_{X_7}^{(1)}}^2$	0	0	3
${\lambda_{X_{16}}}^2$	0	0	2
$\lambda_{X_1}(8T_\nu^{(2)}+8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	1
$\lambda_{X_7}^{(1)}\lambda_{X_7}^{(2)}$	0	0	0
${\lambda_{X_7}^{(2)}}^2$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda_{X_1}(4T_\nu^{(2)}+4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	1
$\lambda_{X_2}\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	2
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	$-\frac{23}{15}$
$g_2^2\lambda_{X_1}$	0	0	-24
$\lambda_{S_2}^{(1)}\lambda_{X_1}$	0	0	12
$\lambda_{S_2}^{(A)}\lambda_{X_1}$	0	0	-8
${\lambda_{X_2}}^2$	0	0	11
${\lambda_{X_7}^{(1)}}^2$	0	0	3

${\lambda_{X_{16}}}^2$	0	0	2
$\lambda_{X_1}(8T_\nu^{(2)} + 8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_7}^{(1)}\lambda_{X_7}^{(2)}$	0	0	2
${\lambda_{X_7}^{(2)}}^2$	0	0	1

$\mathcal{S}$	E ($\alpha = -4$)
$\lambda_{X_1}(4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0
$\lambda_{X_2}\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger y_2(y_2^{\mathrm{conj}})^\dagger)$	0
$g_2^2\lambda_{X_1}$	0
$\lambda_{S_2}^{(1)}\lambda_{X_1}$	0
$\lambda_{S_2}^{(A)}\lambda_{X_1}$	0
${\lambda_{X_2}}^2$	0
${\lambda_{X_7}^{(1)}}^2$	0
${\lambda_{X_{16}}}^2$	0
$\lambda_{X_1}(8T_\nu^{(2)} + 8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0
$\lambda_{X_7}^{(1)}\lambda_{X_7}^{(2)}$	0
${\lambda_{X_7}^{(2)}}^2$	0

$\beta_{\lambda_{X_2}}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda_{S_2}^{(1)}(2\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger) + 22\lambda_{X_2})$	0	1	0
$\lambda_{X_1}(8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 88(\lambda_{X_2})^*)$	0	1	0
$\lambda_{X_2}(4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	1	0
$\mathrm{Tr}(y_2(y_2)^\dagger y_2(y_2^{\mathrm{conj}})^\dagger)$	0	$-\frac{46}{15}$	0
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0	$-\frac{46}{15}$	0
$g_2^2\lambda_{X_2}$	0	-24	0
$\lambda_{12}^{(1)}\lambda_{X_7}^{(1)}$	0	2	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{16}}$	0	4	0
$\lambda_{S_2}^{(1)}(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger) + 18\lambda_{X_2})$	0	0	0
$\lambda_{X_1}(16\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 72(\lambda_{X_2})^*)$	0	0	0
$\lambda_{X_2}(8T_\nu^{(2)} + 8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{12}^{(1)}(6\lambda_{X_7}^{(1)} + 2\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_7}^{(1)} + 4\lambda_{X_7}^{(2)})$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_{S_2}^{(1)}(2\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger) + 22\lambda_{X_2})$	0	0	0
$\lambda_{X_1}(8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 88(\lambda_{X_2})^*)$	0	0	0
$\lambda_{X_2}(4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\mathrm{Tr}(y_2(y_2)^\dagger y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$g_2^2\lambda_{X_2}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{16}}$	0	0	0
$\lambda_{S_2}^{(1)}(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger) + 18\lambda_{X_2})$	0	0	0
$\lambda_{X_1}(16\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 72(\lambda_{X_2})^*)$	0	0	0
$\lambda_{X_2}(8T_\nu^{(2)} + 8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{12}^{(1)}(6\lambda_{X_7}^{(1)} + 2\lambda_{X_7}^{(2)})$	0	0	0

$\lambda_{12}^{(\times)}(4\lambda_{X_7}^{(1)}+4\lambda_{X_7}^{(2)})$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda_{S_2}^{(1)}(2\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+22\lambda_{X_2})$	0	0	0
$\lambda_{X_1}(8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger)+88(\lambda_{X_2})^*)$	0	0	0
$\lambda_{X_2}(4T_\nu^{(2)}+4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\mathrm{Tr}(y_2(y_2)^\dagger y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$g_2^2\lambda_{X_2}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{16}}$	0	0	0
$\lambda_{S_2}^{(1)}(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+18\lambda_{X_2})$	0	0	0
$\lambda_{X_1}(16\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger)+72(\lambda_{X_2})^*)$	0	0	0
$\lambda_{X_2}(8T_\nu^{(2)}+8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{12}^{(1)}(6\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_7}^{(1)}+4\lambda_{X_7}^{(2)})$	0	0	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda_{S_2}^{(1)}(2\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+22\lambda_{X_2})$	0	0	0
$\lambda_{X_1}(8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger)+88(\lambda_{X_2})^*)$	0	0	0
$\lambda_{X_2}(4T_\nu^{(2)}+4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\mathrm{Tr}(y_2(y_2)^\dagger y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	-8
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0	0	-8
$g_2^2\lambda_{X_2}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_7}^{(1)}$	0	0	6
$\lambda_{HS_2}^{(1)}\lambda_{X_{16}}$	0	0	4
$\lambda_{S_2}^{(1)}(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+18\lambda_{X_2})$	0	0	1
$\lambda_{X_1}(16\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger)+72(\lambda_{X_2})^*)$	0	0	1
$\lambda_{X_2}(8T_\nu^{(2)}+8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	1
$\lambda_{12}^{(1)}(6\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_7}^{(1)}+4\lambda_{X_7}^{(2)})$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda_{S_2}^{(1)}(2\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+22\lambda_{X_2})$	0	0	1
$\lambda_{X_1}(8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger)+88(\lambda_{X_2})^*)$	0	0	1
$\lambda_{X_2}(4T_\nu^{(2)}+4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	1
$\mathrm{Tr}(y_2(y_2)^\dagger y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	$-\frac{46}{15}$
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0	0	$-\frac{46}{15}$
$g_2^2\lambda_{X_2}$	0	0	-24
$\lambda_{12}^{(1)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{16}}$	0	0	4
$\lambda_{S_2}^{(1)}(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+18\lambda_{X_2})$	0	0	0
$\lambda_{X_1}(16\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger)+72(\lambda_{X_2})^*)$	0	0	0
$\lambda_{X_2}(8T_\nu^{(2)}+8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{12}^{(1)}(6\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	1
$\lambda_{12}^{(\times)}(4\lambda_{X_7}^{(1)}+4\lambda_{X_7}^{(2)})$	0	0	1

$\mathcal{S}$	E ($\alpha=-4$)

$\lambda_{S_2}^{(1)}(2\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+22\lambda_{X_2})$	0
$\lambda_{X_1}(8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger)+88(\lambda_{X_2})^*)$	0
$\lambda_{X_2}(4T_\nu^{(2)}+4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0
$\mathrm{Tr}(y_2(y_2)^\dagger y_2(y_2^{\mathrm{conj}})^\dagger)$	0
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)$	0
$g_2^2\lambda_{X_2}$	0
$\lambda_{12}^{(1)}\lambda_{X_7}^{(1)}$	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{16}}$	0
$\lambda_{S_2}^{(1)}(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+18\lambda_{X_2})$	0
$\lambda_{X_1}(16\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger)+72(\lambda_{X_2})^*)$	0
$\lambda_{X_2}(8T_\nu^{(2)}+8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0
$\lambda_{12}^{(1)}(6\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0
$\lambda_{12}^{(\times)}(4\lambda_{X_7}^{(1)}+4\lambda_{X_7}^{(2)})$	0

$\beta^{(1)}_{\lambda_{X_5}^{(1)}}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_{12}^{(1)}(4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)+8\lambda_{X_5}^{(1)}+6\lambda_{X_9})$	1	0	0
$\lambda_{X_5}^{(1)}(4T_\nu^{(1)}+4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+2T_\nu^{(2)})$	1	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger y_2(y_2)^\dagger)$	-4	0	0
$(\lambda_5)^*(\lambda_{X_{14}})^*$	4	0	0
$\lambda_{X_6}(\lambda_{X_5}^{(1)})^*$	24	0	0
$g_2^2\lambda_{X_5}^{(1)}$	-12	0	0
$g_1^2\lambda_{X_5}^{(1)}$	-6	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{18}}$	4	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_5}^{(1)}$	2	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_5}^{(1)}$	8	0	0
$\lambda_{S_2}^{(A)}\lambda_{X_5}^{(1)}$	4	0	0
$\lambda_{12}^{(1)}(2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)+8\lambda_{X_5}^{(1)}+10\lambda_{X_9})$	0	0	0
$\lambda_{X_5}^{(1)}(2T_\nu^{(1)}+2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+4T_\nu^{(2)})$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_5}^{(1)}$	0	0	0
$\lambda_{X_5}^{(1)}(2T_\nu^{(1)}+2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+2T_\nu^{(2)})$	0	0	0
$\lambda_{X_6}(5(\lambda_{X_5}^{(1)})^*+(\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_5}^{(2)}+4\lambda_{X_9})$	0	0	0
$\lambda_{S_1}^{(A)}(-4\lambda_{X_5}^{(1)}-2\lambda_{X_5}^{(2)})$	0	0	0
$\lambda_{S_2}^{(1)}(8\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	0	0	0
$\lambda_{S_2}^{(A)}(4\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$\lambda_{12}^{(1)}(4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)+8\lambda_{X_5}^{(1)}+6\lambda_{X_9})$	0	0	0
$\lambda_{X_5}^{(1)}(4T_\nu^{(1)}+4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+2T_\nu^{(2)})$	0	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger y_2(y_2)^\dagger)$	0	0	0
$(\lambda_5)^*(\lambda_{X_{14}})^*$	0	0	0
$\lambda_{X_6}(\lambda_{X_5}^{(1)})^*$	0	0	0
$g_2^2\lambda_{X_5}^{(1)}$	0	0	0
$g_1^2\lambda_{X_5}^{(1)}$	0	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{18}}$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_5}^{(1)}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_5}^{(1)}$	0	0	0
$\lambda_{S_2}^{(A)}\lambda_{X_5}^{(1)}$	0	0	0

$\lambda_{12}^{(1)}(2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)+8\lambda_{X_5}^{(1)}+10\lambda_{X_9})$	0	0	0
$\lambda_{X_5}^{(1)}(2T_\nu^{(1)}+2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+4T_\nu^{(2)})$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_5}^{(1)}$	0	0	0
$\lambda_{X_5}^{(1)}(2T_\nu^{(1)}+2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+2T_\nu^{(2)})$	0	0	0
$\lambda_{X_6}(5(\lambda_{X_5}^{(1)})^*+(\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_5}^{(2)}+4\lambda_{X_9})$	0	0	0
$\lambda_{S_1}^{(A)}(-4\lambda_{X_5}^{(1)}-2\lambda_{X_5}^{(2)})$	0	0	0
$\lambda_{S_2}^{(1)}(8\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	0	0	0
$\lambda_{S_2}^{(A)}(4\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda_{12}^{(1)}(4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)+8\lambda_{X_5}^{(1)}+6\lambda_{X_9})$	0	0	0
$\lambda_{X_5}^{(1)}(4T_\nu^{(1)}+4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+2T_\nu^{(2)})$	0	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger y_2(y_2)^\dagger)$	0	0	0
$(\lambda_5)^*(\lambda_{X_{14}})^*$	0	0	0
$\lambda_{X_6}(\lambda_{X_5}^{(1)})^*$	0	0	0
$g_2^2\lambda_{X_5}^{(1)}$	0	0	0
$g_1^2\lambda_{X_5}^{(1)}$	0	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{18}}$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_5}^{(1)}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_5}^{(1)}$	0	0	0
$\lambda_{S_2}^{(A)}\lambda_{X_5}^{(1)}$	0	0	0
$\lambda_{12}^{(1)}(2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)+8\lambda_{X_5}^{(1)}+10\lambda_{X_9})$	0	0	0
$\lambda_{X_5}^{(1)}(2T_\nu^{(1)}+2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+4T_\nu^{(2)})$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_5}^{(1)}$	0	0	0
$\lambda_{X_5}^{(1)}(2T_\nu^{(1)}+2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+2T_\nu^{(2)})$	0	0	0
$\lambda_{X_6}(5(\lambda_{X_5}^{(1)})^*+(\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_5}^{(2)}+4\lambda_{X_9})$	0	0	0
$\lambda_{S_1}^{(A)}(-4\lambda_{X_5}^{(1)}-2\lambda_{X_5}^{(2)})$	0	0	0
$\lambda_{S_2}^{(1)}(8\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	0	0	0
$\lambda_{S_2}^{(A)}(4\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	0	0	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda_{12}^{(1)}(4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)+8\lambda_{X_5}^{(1)}+6\lambda_{X_9})$	0	0	0
$\lambda_{X_5}^{(1)}(4T_\nu^{(1)}+4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+2T_\nu^{(2)})$	0	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger y_2(y_2)^\dagger)$	-4	0	0
$(\lambda_5)^*(\lambda_{X_{14}})^*$	4	0	0
$\lambda_{X_6}(\lambda_{X_5}^{(1)})^*$	40	0	0
$g_2^2\lambda_{X_5}^{(1)}$	-12	0	0
$g_1^2\lambda_{X_5}^{(1)}$	-6	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{18}}$	4	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_5}^{(1)}$	2	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_5}^{(1)}$	4	0	0
$\lambda_{S_2}^{(A)}\lambda_{X_5}^{(1)}$	0	0	0
$\lambda_{12}^{(1)}(2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)+8\lambda_{X_5}^{(1)}+10\lambda_{X_9})$	1	0	0
$\lambda_{X_5}^{(1)}(2T_\nu^{(1)}+2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+4T_\nu^{(2)})$	1	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_5}^{(1)}$	-4	0	0
$\lambda_{X_5}^{(1)}(2T_\nu^{(1)}+2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+2T_\nu^{(2)})$	0	0	0
$\lambda_{X_6}(5(\lambda_{X_5}^{(1)})^*+(\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_5}^{(2)}+4\lambda_{X_9})$	0	0	0
$\lambda_{S_1}^{(A)}(-4\lambda_{X_5}^{(1)}-2\lambda_{X_5}^{(2)})$	0	0	0

$\lambda_{S_2}^{(1)}(8\lambda_{X_5}^{(1)} + 2\lambda_{X_5}^{(2)})$	0	0	0
$\lambda_{S_2}^{(A)}(4\lambda_{X_5}^{(1)} + 2\lambda_{X_5}^{(2)})$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda_{12}^{(1)}(4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger) + 8\lambda_{X_5}^{(1)} + 6\lambda_{X_9})$	0	0	0
$\lambda_{X_5}^{(1)}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)})$	0	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger y_2(y_2)^\dagger)$	$-\frac{22}{15}$	0	0
$(\lambda_5)^*(\lambda_{X_{14}})^*$	2	0	0
$\lambda_{X_6}(\lambda_{X_5}^{(1)})^*$	0	0	0
$g_2^2\lambda_{X_5}^{(1)}$	-24	0	0
$g_1^2\lambda_{X_5}^{(1)}$	-6	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{18}}$	4	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_5}^{(1)}$	2	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_5}^{(1)}$	0	0	0
$\lambda_{S_2}^{(A)}\lambda_{X_5}^{(1)}$	0	0	0
$\lambda_{12}^{(1)}(2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger) + 8\lambda_{X_5}^{(1)} + 10\lambda_{X_9})$	1	0	0
$\lambda_{X_5}^{(1)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 4T_\nu^{(2)})$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_5}^{(1)}$	0	0	0
$\lambda_{X_5}^{(1)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)})$	1	0	0
$\lambda_{X_6}(5(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	8	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_5}^{(2)} + 4\lambda_{X_9})$	1	0	0
$\lambda_{S_1}^{(A)}(-4\lambda_{X_5}^{(1)} - 2\lambda_{X_5}^{(2)})$	1	0	0
$\lambda_{S_2}^{(1)}(8\lambda_{X_5}^{(1)} + 2\lambda_{X_5}^{(2)})$	1	0	0
$\lambda_{S_2}^{(A)}(4\lambda_{X_5}^{(1)} + 2\lambda_{X_5}^{(2)})$	1	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$\lambda_{12}^{(1)}(4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger) + 8\lambda_{X_5}^{(1)} + 6\lambda_{X_9})$	0
$\lambda_{X_5}^{(1)}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)})$	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger y_2(y_2)^\dagger)$	0
$(\lambda_5)^*(\lambda_{X_{14}})^*$	0
$\lambda_{X_6}(\lambda_{X_5}^{(1)})^*$	0
$g_2^2\lambda_{X_5}^{(1)}$	0
$g_1^2\lambda_{X_5}^{(1)}$	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{18}}$	0
$\lambda_{S_1}^{(1)}\lambda_{X_5}^{(1)}$	0
$\lambda_{S_2}^{(1)}\lambda_{X_5}^{(1)}$	0
$\lambda_{S_2}^{(A)}\lambda_{X_5}^{(1)}$	0
$\lambda_{12}^{(1)}(2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger) + 8\lambda_{X_5}^{(1)} + 10\lambda_{X_9})$	0
$\lambda_{X_5}^{(1)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 4T_\nu^{(2)})$	0
$\lambda_{S_1}^{(A)}\lambda_{X_5}^{(1)}$	0
$\lambda_{X_5}^{(1)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)})$	0
$\lambda_{X_6}(5(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0
$\lambda_{12}^{(\times)}(4\lambda_{X_5}^{(2)} + 4\lambda_{X_9})$	0
$\lambda_{S_1}^{(A)}(-4\lambda_{X_5}^{(1)} - 2\lambda_{X_5}^{(2)})$	0
$\lambda_{S_2}^{(1)}(8\lambda_{X_5}^{(1)} + 2\lambda_{X_5}^{(2)})$	0
$\lambda_{S_2}^{(A)}(4\lambda_{X_5}^{(1)} + 2\lambda_{X_5}^{(2)})$	0

$\beta_{\lambda_{X_5}^{(2)}}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda_{12}^{(\times)}(4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger) + 16\lambda_{X_5}^{(1)} + 20\lambda_{X_5}^{(2)} + 8\lambda_{X_9})$	0	0	0
$\lambda_{X_5}^{(2)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)})$	0	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger y_2(y_2)^\dagger)$	0	0	0
$(\lambda_5)^*(\lambda_{X_{14}})^*$	0	0	0
$\lambda_{X_6}(\lambda_{X_5}^{(2)})^*$	0	0	0
$g_2^2\lambda_{X_5}^{(2)}$	0	0	0
$g_1^2\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{S_2}^{(A)}\lambda_{X_5}^{(2)}$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_{12}^{(\times)}(4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger) + 16\lambda_{X_5}^{(1)} + 20\lambda_{X_5}^{(2)} + 8\lambda_{X_9})$	0	0	0
$\lambda_{X_5}^{(2)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)})$	0	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger y_2(y_2)^\dagger)$	0	0	0
$(\lambda_5)^*(\lambda_{X_{14}})^*$	0	0	0
$\lambda_{X_6}(\lambda_{X_5}^{(2)})^*$	0	0	0
$g_2^2\lambda_{X_5}^{(2)}$	0	0	0
$g_1^2\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{S_2}^{(A)}\lambda_{X_5}^{(2)}$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_{12}^{(\times)}(4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger) + 16\lambda_{X_5}^{(1)} + 20\lambda_{X_5}^{(2)} + 8\lambda_{X_9})$	0	0	0
$\lambda_{X_5}^{(2)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)})$	0	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger y_2(y_2)^\dagger)$	0	0	0
$(\lambda_5)^*(\lambda_{X_{14}})^*$	0	0	0
$\lambda_{X_6}(\lambda_{X_5}^{(2)})^*$	0	0	0
$g_2^2\lambda_{X_5}^{(2)}$	0	0	0
$g_1^2\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{S_2}^{(A)}\lambda_{X_5}^{(2)}$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_{12}^{(\times)}(4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger) + 16\lambda_{X_5}^{(1)} + 20\lambda_{X_5}^{(2)} + 8\lambda_{X_9})$	0	0	0
$\lambda_{X_5}^{(2)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)})$	0	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger y_2(y_2)^\dagger)$	0	0	0
$(\lambda_5)^*(\lambda_{X_{14}})^*$	0	0	0
$\lambda_{X_6}(\lambda_{X_5}^{(2)})^*$	0	0	0

$g_2^2\lambda_{X_5}^{(2)}$	0	0	0
$g_1^2\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{S_2}^{(A)}\lambda_{X_5}^{(2)}$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda_{12}^{(\times)}(4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)+16\lambda_{X_5}^{(1)}+20\lambda_{X_5}^{(2)}+8\lambda_{X_9})$	1	0	0
$\lambda_{X_5}^{(2)}(2T_\nu^{(1)}+2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+2T_\nu^{(2)})$	1	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger y_2(y_2)^\dagger)$	$-\frac{8}{5}$	0	0
$(\lambda_5)^*(\lambda_{X_{14}})^*$	-2	0	0
$\lambda_{X_6}(\lambda_{X_5}^{(2)})^*$	16	0	0
$g_2^2\lambda_{X_5}^{(2)}$	-24	0	0
$g_1^2\lambda_{X_5}^{(2)}$	-6	0	0
$\lambda_{12}^{(1)}\lambda_{X_5}^{(2)}$	8	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_5}^{(2)}$	2	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_5}^{(2)}$	2	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_5}^{(2)}$	2	0	0
$\lambda_{S_2}^{(A)}\lambda_{X_5}^{(2)}$	-2	0	0

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_{12}^{(\times)}(4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)+16\lambda_{X_5}^{(1)}+20\lambda_{X_5}^{(2)}+8\lambda_{X_9})$	0
$\lambda_{X_5}^{(2)}(2T_\nu^{(1)}+2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+2T_\nu^{(2)})$	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger y_2(y_2)^\dagger)$	0
$(\lambda_5)^*(\lambda_{X_{14}})^*$	0
$\lambda_{X_6}(\lambda_{X_5}^{(2)})^*$	0
$g_2^2\lambda_{X_5}^{(2)}$	0
$g_1^2\lambda_{X_5}^{(2)}$	0
$\lambda_{12}^{(1)}\lambda_{X_5}^{(2)}$	0
$\lambda_{S_1}^{(1)}\lambda_{X_5}^{(2)}$	0
$\lambda_{S_1}^{(A)}\lambda_{X_5}^{(2)}$	0
$\lambda_{S_2}^{(1)}\lambda_{X_5}^{(2)}$	0
$\lambda_{S_2}^{(A)}\lambda_{X_5}^{(2)}$	0

$\beta_{\lambda_{X_6}}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_{X_6}(8T_\nu^{(1)}+8\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	1	0	0
$\lambda_{X_9}\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	4	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger y_1^{\mathrm{conj}}(y_1)^\dagger)$	-4	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_6}$	12	0	0
${\lambda_{X_5}^{(1)}}^2$	3	0	0
${\lambda_{X_9}}^2$	9	0	0
${\lambda_{X_{18}}}^2$	2	0	0
$\lambda_{X_6}(4T_\nu^{(1)}+4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0	0	0
$g_2^2\lambda_{X_6}$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_6}$	0	0	0

$\lambda_{X_5}^{(1)}\lambda_{X_5}^{(2)}$	0	0	0
${\lambda_{X_5}^{(2)}}^2$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_{X_6}(8T_\nu^{(1)} + 8\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_9}\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	0	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger y_1^{\mathrm{conj}}(y_1)^\dagger)$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_6}$	0	0	0
${\lambda_{X_5}^{(1)}}^2$	0	0	0
${\lambda_{X_9}}^2$	0	0	0
${\lambda_{X_{18}}}^2$	0	0	0
$\lambda_{X_6}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0	0	0
$g_2^2\lambda_{X_6}$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_6}$	0	0	0
$\lambda_{X_5}^{(1)}\lambda_{X_5}^{(2)}$	0	0	0
${\lambda_{X_5}^{(2)}}^2$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_{X_6}(8T_\nu^{(1)} + 8\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_9}\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	0	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger y_1^{\mathrm{conj}}(y_1)^\dagger)$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_6}$	0	0	0
${\lambda_{X_5}^{(1)}}^2$	0	0	0
${\lambda_{X_9}}^2$	0	0	0
${\lambda_{X_{18}}}^2$	0	0	0
$\lambda_{X_6}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0	0	0
$g_2^2\lambda_{X_6}$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_6}$	0	0	0
$\lambda_{X_5}^{(1)}\lambda_{X_5}^{(2)}$	0	0	0
${\lambda_{X_5}^{(2)}}^2$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_{X_6}(8T_\nu^{(1)} + 8\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_9}\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	2	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger y_1^{\mathrm{conj}}(y_1)^\dagger)$	$-\frac{23}{15}$	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_6}$	12	0	0
${\lambda_{X_5}^{(1)}}^2$	1	0	0
${\lambda_{X_9}}^2$	11	0	0
${\lambda_{X_{18}}}^2$	2	0	0
$\lambda_{X_6}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	1	0	0
$g_2^2\lambda_{X_6}$	-24	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_6}$	-8	0	0
$\lambda_{X_5}^{(1)}\lambda_{X_5}^{(2)}$	0	0	0
${\lambda_{X_5}^{(2)}}^2$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda_{X_6}(8T_\nu^{(1)} + 8\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0	0	0

$\lambda_{X_9}\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	2	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger y_1^{\mathrm{conj}}(y_1)^\dagger)$	$-\frac{23}{15}$	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_6}$	12	0	0
${\lambda_{X_5}^{(1)}}^2$	3	0	0
${\lambda_{X_9}}^2$	11	0	0
${\lambda_{X_{18}}}^2$	2	0	0
$\lambda_{X_6}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	1	0	0
$g_2^2\lambda_{X_6}$	-24	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_6}$	-8	0	0
$\lambda_{X_5}^{(1)}\lambda_{X_5}^{(2)}$	2	0	0
${\lambda_{X_5}^{(2)}}^2$	1	0	0

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_{X_6}(8T_\nu^{(1)} + 8\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0
$\lambda_{X_9}\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger y_1^{\mathrm{conj}}(y_1)^\dagger)$	0
$\lambda_{S_1}^{(1)}\lambda_{X_6}$	0
${\lambda_{X_5}^{(1)}}^2$	0
${\lambda_{X_9}}^2$	0
${\lambda_{X_{18}}}^2$	0
$\lambda_{X_6}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0
$g_2^2\lambda_{X_6}$	0
$\lambda_{S_1}^{(A)}\lambda_{X_6}$	0
$\lambda_{X_5}^{(1)}\lambda_{X_5}^{(2)}$	0
${\lambda_{X_5}^{(2)}}^2$	0

$\beta_{\lambda_{X_7}^{(1)}}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_{12}^{(1)}(2\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger) + 10\lambda_{X_2} + 8\lambda_{X_7}^{(1)})$	0	1	0
$\lambda_{X_7}^{(1)}(4T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	1	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2^{\mathrm{conj}})^\dagger)$	0	-4	0
$\lambda_{X_1}(\lambda_{X_7}^{(1)})^*$	0	40	0
$g_2^2\lambda_{X_7}^{(1)}$	0	-12	0
$g_1^2\lambda_{X_7}^{(1)}$	0	-6	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{16}}$	0	4	0
$\lambda_{S_1}^{(1)}\lambda_{X_7}^{(1)}$	0	4	0
$\lambda_{S_2}^{(1)}\lambda_{X_7}^{(1)}$	0	2	0
$\lambda_{S_2}^{(A)}\lambda_{X_7}^{(1)}$	0	-4	0
$\lambda_5\lambda_{X_{12}}$	0	4	0
$\lambda_{12}^{(1)}(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger) + 6\lambda_{X_2} + 8\lambda_{X_7}^{(1)})$	0	0	0
$\lambda_{X_7}^{(1)}(2T_\nu^{(1)} + 4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_{X_7}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_1}(5(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_2} + 4\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{S_1}^{(1)}(8\lambda_{X_7}^{(1)} + 2\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{S_1}^{(A)}(4\lambda_{X_7}^{(1)} + 2\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{S_2}^{(A)}(-4\lambda_{X_7}^{(1)} - 2\lambda_{X_7}^{(2)})$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_{12}^{(1)}(2\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger) + 10\lambda_{X_2} + 8\lambda_{X_7}^{(1)})$	0	0	0
$\lambda_{X_7}^{(1)}(4T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{X_1}(\lambda_{X_7}^{(1)})^*$	0	0	0
$g_2^2\lambda_{X_7}^{(1)}$	0	0	0
$g_1^2\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{16}}$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_{S_2}^{(A)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_5\lambda_{X_{12}}$	0	0	0
$\lambda_{12}^{(1)}(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger) + 6\lambda_{X_2} + 8\lambda_{X_7}^{(1)})$	0	0	0
$\lambda_{X_7}^{(1)}(2T_\nu^{(1)} + 4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_{X_7}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_1}(5(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_2} + 4\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{S_1}^{(1)}(8\lambda_{X_7}^{(1)} + 2\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{S_1}^{(A)}(4\lambda_{X_7}^{(1)} + 2\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{S_2}^{(A)}(-4\lambda_{X_7}^{(1)} - 2\lambda_{X_7}^{(2)})$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_{12}^{(1)}(2\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger) + 10\lambda_{X_2} + 8\lambda_{X_7}^{(1)})$	0	0	0
$\lambda_{X_7}^{(1)}(4T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{X_1}(\lambda_{X_7}^{(1)})^*$	0	0	0
$g_2^2\lambda_{X_7}^{(1)}$	0	0	0
$g_1^2\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{16}}$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_{S_2}^{(A)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_5\lambda_{X_{12}}$	0	0	0
$\lambda_{12}^{(1)}(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger) + 6\lambda_{X_2} + 8\lambda_{X_7}^{(1)})$	0	0	0
$\lambda_{X_7}^{(1)}(2T_\nu^{(1)} + 4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_{X_7}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_1}(5(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_2} + 4\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{S_1}^{(1)}(8\lambda_{X_7}^{(1)} + 2\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{S_1}^{(A)}(4\lambda_{X_7}^{(1)} + 2\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{S_2}^{(A)}(-4\lambda_{X_7}^{(1)} - 2\lambda_{X_7}^{(2)})$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_{12}^{(1)}(2\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger) + 10\lambda_{X_2} + 8\lambda_{X_7}^{(1)})$	0	0	0
$\lambda_{X_7}^{(1)}(4T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	-4
$\lambda_{X_1}(\lambda_{X_7}^{(1)})^*$	0	0	24

$g_2^2\lambda_{X_7}^{(1)}$	0	0	-12
$g_1^2\lambda_{X_7}^{(1)}$	0	0	-6
$\lambda_{HS_1}^{(1)}\lambda_{X_{16}}$	0	0	4
$\lambda_{S_1}^{(1)}\lambda_{X_7}^{(1)}$	0	0	8
$\lambda_{S_2}^{(1)}\lambda_{X_7}^{(1)}$	0	0	2
$\lambda_{S_2}^{(A)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_5\lambda_{X_{12}}$	0	0	4
$\lambda_{12}^{(1)}(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+6\lambda_{X_2}+8\lambda_{X_7}^{(1)})$	0	0	1
$\lambda_{X_7}^{(1)}(2T_\nu^{(1)}+4T_\nu^{(2)}+4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	1
$\lambda_{S_1}^{(A)}\lambda_{X_7}^{(1)}$	0	0	4
$\lambda_{X_7}^{(1)}(2T_\nu^{(1)}+2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_1}(5(\lambda_{X_7}^{(1)})^*+(\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_2}+4\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{S_1}^{(1)}(8\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{S_1}^{(A)}(4\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{S_2}^{(A)}(-4\lambda_{X_7}^{(1)}-2\lambda_{X_7}^{(2)})$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda_{12}^{(1)}(2\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+10\lambda_{X_2}+8\lambda_{X_7}^{(1)})$	0	0	1
$\lambda_{X_7}^{(1)}(4T_\nu^{(1)}+2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	$-\frac{22}{15}$
$\lambda_{X_1}(\lambda_{X_7}^{(1)})^*$	0	0	0
$g_2^2\lambda_{X_7}^{(1)}$	0	0	-24
$g_1^2\lambda_{X_7}^{(1)}$	0	0	-6
$\lambda_{HS_1}^{(1)}\lambda_{X_{16}}$	0	0	4
$\lambda_{S_1}^{(1)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_7}^{(1)}$	0	0	2
$\lambda_{S_2}^{(A)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_5\lambda_{X_{12}}$	0	0	2
$\lambda_{12}^{(1)}(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+6\lambda_{X_2}+8\lambda_{X_7}^{(1)})$	0	0	0
$\lambda_{X_7}^{(1)}(2T_\nu^{(1)}+4T_\nu^{(2)}+4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_{X_7}^{(1)}(2T_\nu^{(1)}+2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	1
$\lambda_{X_1}(5(\lambda_{X_7}^{(1)})^*+(\lambda_{X_7}^{(2)})^*)$	0	0	8
$\lambda_{12}^{(\times)}(4\lambda_{X_2}+4\lambda_{X_7}^{(2)})$	0	0	1
$\lambda_{S_1}^{(1)}(8\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	1
$\lambda_{S_1}^{(A)}(4\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	1
$\lambda_{S_2}^{(A)}(-4\lambda_{X_7}^{(1)}-2\lambda_{X_7}^{(2)})$	0	0	1

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_{12}^{(1)}(2\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+10\lambda_{X_2}+8\lambda_{X_7}^{(1)})$	0
$\lambda_{X_7}^{(1)}(4T_\nu^{(1)}+2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2^{\mathrm{conj}})^\dagger)$	0
$\lambda_{X_1}(\lambda_{X_7}^{(1)})^*$	0
$g_2^2\lambda_{X_7}^{(1)}$	0
$g_1^2\lambda_{X_7}^{(1)}$	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{16}}$	0
$\lambda_{S_1}^{(1)}\lambda_{X_7}^{(1)}$	0
$\lambda_{S_2}^{(1)}\lambda_{X_7}^{(1)}$	0
$\lambda_{S_2}^{(A)}\lambda_{X_7}^{(1)}$	0
$\lambda_5\lambda_{X_{12}}$	0

$\lambda_{12}^{(1)}(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+6\lambda_{X_2}+8\lambda_{X_7}^{(1)})$	0
$\lambda_{X_7}^{(1)}(2T_\nu^{(1)}+4T_\nu^{(2)}+4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0
$\lambda_{S_1}^{(A)}\lambda_{X_7}^{(1)}$	0
$\lambda_{X_7}^{(1)}(2T_\nu^{(1)}+2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0
$\lambda_{X_1}(5(\lambda_{X_7}^{(1)})^*+(\lambda_{X_7}^{(2)})^*)$	0
$\lambda_{12}^{(\times)}(4\lambda_{X_2}+4\lambda_{X_7}^{(2)})$	0
$\lambda_{S_1}^{(1)}(8\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0
$\lambda_{S_1}^{(A)}(4\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0
$\lambda_{S_2}^{(A)}(-4\lambda_{X_7}^{(1)}-2\lambda_{X_7}^{(2)})$	0

$\beta_{\lambda_{X_7}^{(2)}}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_{12}^{(\times)}(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+8\lambda_{X_2}+16\lambda_{X_7}^{(1)}+20\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{X_7}^{(2)}(2T_\nu^{(1)}+2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{X_1}(\lambda_{X_7}^{(2)})^*$	0	0	0
$g_2^2\lambda_{X_7}^{(2)}$	0	0	0
$g_1^2\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_{S_2}^{(A)}\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_5\lambda_{X_{12}}$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$\lambda_{12}^{(\times)}(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+8\lambda_{X_2}+16\lambda_{X_7}^{(1)}+20\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{X_7}^{(2)}(2T_\nu^{(1)}+2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{X_1}(\lambda_{X_7}^{(2)})^*$	0	0	0
$g_2^2\lambda_{X_7}^{(2)}$	0	0	0
$g_1^2\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_{S_2}^{(A)}\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_5\lambda_{X_{12}}$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda_{12}^{(\times)}(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+8\lambda_{X_2}+16\lambda_{X_7}^{(1)}+20\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{X_7}^{(2)}(2T_\nu^{(1)}+2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{X_1}(\lambda_{X_7}^{(2)})^*$	0	0	0
$g_2^2\lambda_{X_7}^{(2)}$	0	0	0
$g_1^2\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_7}^{(2)}$	0	0	0

$\lambda_{S_1}^{(A)}\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_{S_2}^{(A)}\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_5\lambda_{X_{12}}$	0	0	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda_{12}^{(\times)}(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+8\lambda_{X_2}+16\lambda_{X_7}^{(1)}+20\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{X_7}^{(2)}(2T_\nu^{(1)}+2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{X_1}(\lambda_{X_7}^{(2)})^*$	0	0	0
$g_2^2\lambda_{X_7}^{(2)}$	0	0	0
$g_1^2\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_{S_2}^{(A)}\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_5\lambda_{X_{12}}$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda_{12}^{(\times)}(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+8\lambda_{X_2}+16\lambda_{X_7}^{(1)}+20\lambda_{X_7}^{(2)})$	0	0	1
$\lambda_{X_7}^{(2)}(2T_\nu^{(1)}+2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	1
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	$-\frac{8}{5}$
$\lambda_{X_1}(\lambda_{X_7}^{(2)})^*$	0	0	16
$g_2^2\lambda_{X_7}^{(2)}$	0	0	-24
$g_1^2\lambda_{X_7}^{(2)}$	0	0	-6
$\lambda_{12}^{(1)}\lambda_{X_7}^{(2)}$	0	0	8
$\lambda_{S_1}^{(1)}\lambda_{X_7}^{(2)}$	0	0	2
$\lambda_{S_1}^{(A)}\lambda_{X_7}^{(2)}$	0	0	-2
$\lambda_{S_2}^{(1)}\lambda_{X_7}^{(2)}$	0	0	2
$\lambda_{S_2}^{(A)}\lambda_{X_7}^{(2)}$	0	0	2
$\lambda_5\lambda_{X_{12}}$	0	0	-2

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_{12}^{(\times)}(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+8\lambda_{X_2}+16\lambda_{X_7}^{(1)}+20\lambda_{X_7}^{(2)})$	0
$\lambda_{X_7}^{(2)}(2T_\nu^{(1)}+2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_2(y_2^{\mathrm{conj}})^\dagger)$	0
$\lambda_{X_1}(\lambda_{X_7}^{(2)})^*$	0
$g_2^2\lambda_{X_7}^{(2)}$	0
$g_1^2\lambda_{X_7}^{(2)}$	0
$\lambda_{12}^{(1)}\lambda_{X_7}^{(2)}$	0
$\lambda_{S_1}^{(1)}\lambda_{X_7}^{(2)}$	0
$\lambda_{S_1}^{(A)}\lambda_{X_7}^{(2)}$	0
$\lambda_{S_2}^{(1)}\lambda_{X_7}^{(2)}$	0
$\lambda_{S_2}^{(A)}\lambda_{X_7}^{(2)}$	0
$\lambda_5\lambda_{X_{12}}$	0

$\beta_{\lambda_{X_9}}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda_{S_1}^{(1)}(4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger) + 18\lambda_{X_9})$	1	0	0
$\lambda_{X_6}(16\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 72(\lambda_{X_9})^*)$	1	0	0
$\lambda_{X_9}(8T_\nu^{(1)} + 8\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	1	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_1^{\mathrm{conj}}(y_1)^\dagger)$	-8	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)$	-8	0	0
$\lambda_{12}^{(1)}\lambda_{X_5}^{(1)}$	6	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{18}}$	4	0	0
$\lambda_{S_1}^{(1)}(2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger) + 22\lambda_{X_9})$	0	0	0
$\lambda_{X_6}(8\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 88(\lambda_{X_9})^*)$	0	0	0
$\lambda_{X_9}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0	0	0
$g_2^2\lambda_{X_9}$	0	0	0
$\lambda_{12}^{(1)}(6\lambda_{X_5}^{(1)} + 2\lambda_{X_5}^{(2)})$	0	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_5}^{(1)} + 4\lambda_{X_5}^{(2)})$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_{S_1}^{(1)}(4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger) + 18\lambda_{X_9})$	0	0	0
$\lambda_{X_6}(16\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 72(\lambda_{X_9})^*)$	0	0	0
$\lambda_{X_9}(8T_\nu^{(1)} + 8\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_1^{\mathrm{conj}}(y_1)^\dagger)$	0	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_5}^{(1)}$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{18}}$	0	0	0
$\lambda_{S_1}^{(1)}(2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger) + 22\lambda_{X_9})$	0	0	0
$\lambda_{X_6}(8\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 88(\lambda_{X_9})^*)$	0	0	0
$\lambda_{X_9}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0	0	0
$g_2^2\lambda_{X_9}$	0	0	0
$\lambda_{12}^{(1)}(6\lambda_{X_5}^{(1)} + 2\lambda_{X_5}^{(2)})$	0	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_5}^{(1)} + 4\lambda_{X_5}^{(2)})$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_{S_1}^{(1)}(4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger) + 18\lambda_{X_9})$	0	0	0
$\lambda_{X_6}(16\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 72(\lambda_{X_9})^*)$	0	0	0
$\lambda_{X_9}(8T_\nu^{(1)} + 8\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_1^{\mathrm{conj}}(y_1)^\dagger)$	0	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_5}^{(1)}$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{18}}$	0	0	0
$\lambda_{S_1}^{(1)}(2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger) + 22\lambda_{X_9})$	0	0	0
$\lambda_{X_6}(8\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 88(\lambda_{X_9})^*)$	0	0	0
$\lambda_{X_9}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0	0	0
$g_2^2\lambda_{X_9}$	0	0	0
$\lambda_{12}^{(1)}(6\lambda_{X_5}^{(1)} + 2\lambda_{X_5}^{(2)})$	0	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_5}^{(1)} + 4\lambda_{X_5}^{(2)})$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_{S_1}^{(1)}(4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger) + 18\lambda_{X_9})$	0	0	0
$\lambda_{X_6}(16\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 72(\lambda_{X_9})^*)$	0	0	0
$\lambda_{X_9}(8T_\nu^{(1)} + 8\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0	0	0

$\mathrm{Tr}(y_1(y_1)^\dagger y_1^{\mathrm{conj}}(y_1)^\dagger)$	$-\frac{46}{15}$	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)$	$-\frac{46}{15}$	0	0
$\lambda_{12}^{(1)}\lambda_{X_5}^{(1)}$	2	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{18}}$	4	0	0
$\lambda_{S_1}^{(1)}(2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)+22\lambda_{X_9})$	1	0	0
$\lambda_{X_6}(8\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger)+88(\lambda_{X_9})^*)$	1	0	0
$\lambda_{X_9}(4T_\nu^{(1)}+4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	1	0	0
$g_2^2\lambda_{X_9}$	-24	0	0
$\lambda_{12}^{(1)}(6\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	0	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_5}^{(1)}+4\lambda_{X_5}^{(2)})$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda_{S_1}^{(1)}(4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)+18\lambda_{X_9})$	0	0	0
$\lambda_{X_6}(16\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger)+72(\lambda_{X_9})^*)$	0	0	0
$\lambda_{X_9}(8T_\nu^{(1)}+8\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0	0	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_1^{\mathrm{conj}}(y_1)^\dagger)$	$-\frac{46}{15}$	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)$	$-\frac{46}{15}$	0	0
$\lambda_{12}^{(1)}\lambda_{X_5}^{(1)}$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{18}}$	4	0	0
$\lambda_{S_1}^{(1)}(2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)+22\lambda_{X_9})$	1	0	0
$\lambda_{X_6}(8\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger)+88(\lambda_{X_9})^*)$	1	0	0
$\lambda_{X_9}(4T_\nu^{(1)}+4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	1	0	0
$g_2^2\lambda_{X_9}$	-24	0	0
$\lambda_{12}^{(1)}(6\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	1	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_5}^{(1)}+4\lambda_{X_5}^{(2)})$	1	0	0

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_{S_1}^{(1)}(4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)+18\lambda_{X_9})$	0
$\lambda_{X_6}(16\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger)+72(\lambda_{X_9})^*)$	0
$\lambda_{X_9}(8T_\nu^{(1)}+8\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0
$\mathrm{Tr}(y_1(y_1)^\dagger y_1^{\mathrm{conj}}(y_1)^\dagger)$	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)$	0
$\lambda_{12}^{(1)}\lambda_{X_5}^{(1)}$	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{18}}$	0
$\lambda_{S_1}^{(1)}(2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)+22\lambda_{X_9})$	0
$\lambda_{X_6}(8\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger)+88(\lambda_{X_9})^*)$	0
$\lambda_{X_9}(4T_\nu^{(1)}+4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0
$g_2^2\lambda_{X_9}$	0
$\lambda_{12}^{(1)}(6\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	0
$\lambda_{12}^{(\times)}(4\lambda_{X_5}^{(1)}+4\lambda_{X_5}^{(2)})$	0

$\beta_{\lambda_{X_{10}}}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_{X_{10}}(2T_\nu^{(2)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$g_2^2\lambda_{X_{10}}$	0	0	0
$g_1^2\lambda_{X_{10}}$	0	0	0
$\lambda_1\lambda_{X_{10}}$	0	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{10}}$	0	0	0
$\lambda_{HS_2}^{(A)}\lambda_{X_{10}}$	0	0	0

$\lambda_{S_2}^{(1)}\lambda_{X_{10}}$	0	0	0
$\lambda_{X_{10}}(3T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_{X_{10}}(2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	1
$g_2^2\lambda_{X_{10}}$	0	0	-9
$g_1^2\lambda_{X_{10}}$	0	0	-3
$\lambda_1\lambda_{X_{10}}$	0	0	2
$\lambda_{HS_2}^{(1)}\lambda_{X_{10}}$	0	0	8
$\lambda_{HS_2}^{(A)}\lambda_{X_{10}}$	0	0	-4
$\lambda_{S_2}^{(1)}\lambda_{X_{10}}$	0	0	2
$\lambda_{X_{10}}(3T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_{X_{10}}(2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$g_2^2\lambda_{X_{10}}$	0	0	-9
$g_1^2\lambda_{X_{10}}$	0	0	-3
$\lambda_1\lambda_{X_{10}}$	0	0	2
$\lambda_{HS_2}^{(1)}\lambda_{X_{10}}$	0	0	8
$\lambda_{HS_2}^{(A)}\lambda_{X_{10}}$	0	0	-4
$\lambda_{S_2}^{(1)}\lambda_{X_{10}}$	0	0	2
$\lambda_{X_{10}}(3T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	1

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_{X_{10}}(2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$g_2^2\lambda_{X_{10}}$	0	0	0
$g_1^2\lambda_{X_{10}}$	0	0	0
$\lambda_1\lambda_{X_{10}}$	0	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{10}}$	0	0	0
$\lambda_{HS_2}^{(A)}\lambda_{X_{10}}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_{10}}$	0	0	0
$\lambda_{X_{10}}(3T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda_{X_{10}}(2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$g_2^2\lambda_{X_{10}}$	0	0	0
$g_1^2\lambda_{X_{10}}$	0	0	0
$\lambda_1\lambda_{X_{10}}$	0	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{10}}$	0	0	0
$\lambda_{HS_2}^{(A)}\lambda_{X_{10}}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_{10}}$	0	0	0
$\lambda_{X_{10}}(3T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$\lambda_{X_{10}}(2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$g_2^2\lambda_{X_{10}}$	0
$g_1^2\lambda_{X_{10}}$	0
$\lambda_1\lambda_{X_{10}}$	0

$\lambda_{HS_2}^{(1)}\lambda_{X_{10}}$	0
$\lambda_{HS_2}^{(A)}\lambda_{X_{10}}$	0
$\lambda_{S_2}^{(1)}\lambda_{X_{10}}$	0
$\lambda_{X_{10}}(3T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0

$\beta_{\lambda_{X_{12}}}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$(\lambda_5)^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	2	0
$\lambda_{X_{12}}(2T_\nu^{(1)} + T_\nu^{(2)} + \mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	1	0
$\lambda_{X_7}^{(1)}(\lambda_5)^*$	0	4	0
$\lambda_{X_{16}}(\lambda_5)^*$	0	8	0
$g_2^2\lambda_{X_{12}}$	0	$-\frac{21}{2}$	0
$g_1^2\lambda_{X_{12}}$	0	$-\frac{9}{2}$	0
$\lambda_{12}^{(1)}\lambda_{X_{12}}$	0	2	0
$\lambda_1\lambda_{X_{12}}$	0	2	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{12}}$	0	4	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{12}}$	0	4	0
$\lambda_{HS_2}^{(A)}\lambda_{X_{12}}$	0	-4	0
$\lambda_{X_{12}}(T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{HS_1}^{(A)}\lambda_{X_{12}}$	0	0	0
$\lambda_{X_{12}}(T_\nu^{(1)} + T_\nu^{(2)} + \mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{X_7}^{(2)}(\lambda_5)^*$	0	0	0
$\lambda_{12}^{(A)}\lambda_{X_{12}}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{X_{12}}$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$(\lambda_5)^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{X_{12}}(2T_\nu^{(1)} + T_\nu^{(2)} + \mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{X_7}^{(1)}(\lambda_5)^*$	0	0	0
$\lambda_{X_{16}}(\lambda_5)^*$	0	0	0
$g_2^2\lambda_{X_{12}}$	0	0	0
$g_1^2\lambda_{X_{12}}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_{12}}$	0	0	0
$\lambda_1\lambda_{X_{12}}$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{12}}$	0	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{12}}$	0	0	0
$\lambda_{HS_2}^{(A)}\lambda_{X_{12}}$	0	0	0
$\lambda_{X_{12}}(T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{HS_1}^{(A)}\lambda_{X_{12}}$	0	0	0
$\lambda_{X_{12}}(T_\nu^{(1)} + T_\nu^{(2)} + \mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{X_7}^{(2)}(\lambda_5)^*$	0	0	0
$\lambda_{12}^{(A)}\lambda_{X_{12}}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{X_{12}}$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$(\lambda_5)^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{X_{12}}(2T_\nu^{(1)} + T_\nu^{(2)} + \mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{X_7}^{(1)}(\lambda_5)^*$	0	0	0
$\lambda_{X_{16}}(\lambda_5)^*$	0	0	0

$g_2^2\lambda_{X_{12}}$	0	0	0
$g_1^2\lambda_{X_{12}}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_{12}}$	0	0	0
$\lambda_1\lambda_{X_{12}}$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{12}}$	0	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{12}}$	0	0	0
$\lambda_{HS_2}^{(A)}\lambda_{X_{12}}$	0	0	0
$\lambda_{X_{12}}(T_\nu^{(1)}+2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{HS_1}^{(A)}\lambda_{X_{12}}$	0	0	0
$\lambda_{X_{12}}(T_\nu^{(1)}+T_\nu^{(2)}+\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{X_7}^{(2)}(\lambda_5)^*$	0	0	0
$\lambda_{12}^{(A)}\lambda_{X_{12}}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{X_{12}}$	0	0	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$(\lambda_5)^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	4
$\lambda_{X_{12}}(2T_\nu^{(1)}+T_\nu^{(2)}+\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{X_7}^{(1)}(\lambda_5)^*$	0	0	4
$\lambda_{X_{16}}(\lambda_5)^*$	0	0	8
$g_2^2\lambda_{X_{12}}$	0	0	$-\frac{21}{2}$
$g_1^2\lambda_{X_{12}}$	0	0	$-\frac{9}{2}$
$\lambda_{12}^{(1)}\lambda_{X_{12}}$	0	0	2
$\lambda_1\lambda_{X_{12}}$	0	0	2
$\lambda_{HS_1}^{(1)}\lambda_{X_{12}}$	0	0	4
$\lambda_{HS_2}^{(1)}\lambda_{X_{12}}$	0	0	4
$\lambda_{HS_2}^{(A)}\lambda_{X_{12}}$	0	0	0
$\lambda_{X_{12}}(T_\nu^{(1)}+2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	1
$\lambda_{HS_1}^{(A)}\lambda_{X_{12}}$	0	0	-4
$\lambda_{X_{12}}(T_\nu^{(1)}+T_\nu^{(2)}+\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{X_7}^{(2)}(\lambda_5)^*$	0	0	0
$\lambda_{12}^{(A)}\lambda_{X_{12}}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{X_{12}}$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$(\lambda_5)^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	2
$\lambda_{X_{12}}(2T_\nu^{(1)}+T_\nu^{(2)}+\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{X_7}^{(1)}(\lambda_5)^*$	0	0	4
$\lambda_{X_{16}}(\lambda_5)^*$	0	0	8
$g_2^2\lambda_{X_{12}}$	0	0	$-\frac{33}{2}$
$g_1^2\lambda_{X_{12}}$	0	0	$-\frac{9}{2}$
$\lambda_{12}^{(1)}\lambda_{X_{12}}$	0	0	2
$\lambda_1\lambda_{X_{12}}$	0	0	2
$\lambda_{HS_1}^{(1)}\lambda_{X_{12}}$	0	0	4
$\lambda_{HS_2}^{(1)}\lambda_{X_{12}}$	0	0	4
$\lambda_{HS_2}^{(A)}\lambda_{X_{12}}$	0	0	-2
$\lambda_{X_{12}}(T_\nu^{(1)}+2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{HS_1}^{(A)}\lambda_{X_{12}}$	0	0	-2
$\lambda_{X_{12}}(T_\nu^{(1)}+T_\nu^{(2)}+\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	1
$\lambda_{X_7}^{(2)}(\lambda_5)^*$	0	0	-2
$\lambda_{12}^{(A)}\lambda_{X_{12}}$	0	0	-2
$\lambda_{12}^{(\times)}\lambda_{X_{12}}$	0	0	-2

$\mathcal{S}$	E ($\alpha = -4$)
$(\lambda_5)^* \operatorname{Tr}(y_2 (y_2^{\text{conj}})^\dagger)$	0
$\lambda_{X_{12}}(2T_\nu^{(1)} + T_\nu^{(2)} + \operatorname{Tr}(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger) + 6\operatorname{Tr}(Y_d(Y_d)^\dagger) + 2\operatorname{Tr}(Y_e(Y_e)^\dagger) + 6\operatorname{Tr}(Y_u(Y_u)^\dagger))$	0
$\lambda_{X_7}^{(1)}(\lambda_5)^*$	0
$\lambda_{X_{16}}(\lambda_5)^*$	0
$g_2^2 \lambda_{X_{12}}$	0
$g_1^2 \lambda_{X_{12}}$	0
$\lambda_{12}^{(1)} \lambda_{X_{12}}$	0
$\lambda_1 \lambda_{X_{12}}$	0
$\lambda_{HS_1}^{(1)} \lambda_{X_{12}}$	0
$\lambda_{HS_2}^{(1)} \lambda_{X_{12}}$	0
$\lambda_{HS_2}^{(A)} \lambda_{X_{12}}$	0
$\lambda_{X_{12}}(T_\nu^{(1)} + 2T_\nu^{(2)} + 2\operatorname{Tr}(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger) + 6\operatorname{Tr}(Y_d(Y_d)^\dagger) + 2\operatorname{Tr}(Y_e(Y_e)^\dagger) + 6\operatorname{Tr}(Y_u(Y_u)^\dagger))$	0
$\lambda_{HS_1}^{(A)} \lambda_{X_{12}}$	0
$\lambda_{X_{12}}(T_\nu^{(1)} + T_\nu^{(2)} + \operatorname{Tr}(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger) + 6\operatorname{Tr}(Y_d(Y_d)^\dagger) + 2\operatorname{Tr}(Y_e(Y_e)^\dagger) + 6\operatorname{Tr}(Y_u(Y_u)^\dagger))$	0
$\lambda_{X_7}^{(2)}(\lambda_5)^*$	0
$\lambda_{12}^{(A)} \lambda_{X_{12}}$	0
$\lambda_{12}^{(\times)} \lambda_{X_{12}}$	0

$\beta_{\lambda_{X_{14}}}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$(\lambda_5)^* \operatorname{Tr}(y_1 (y_1^{\text{conj}})^\dagger)$	4	0	0
$\lambda_{X_{14}}(2T_\nu^{(1)} + 2\operatorname{Tr}(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger) + T_\nu^{(2)} + 6\operatorname{Tr}(Y_d(Y_d)^\dagger) + 2\operatorname{Tr}(Y_e(Y_e)^\dagger) + 6\operatorname{Tr}(Y_u(Y_u)^\dagger))$	1	0	0
$(\lambda_5)^*((\lambda_{X_5}^{(1)})^* + 2(\lambda_{X_{18}})^*)$	4	0	0
$g_2^2 \lambda_{X_{14}}$	$-\frac{21}{2}$	0	0
$g_1^2 \lambda_{X_{14}}$	$-\frac{9}{2}$	0	0
$\lambda_{12}^{(1)} \lambda_{X_{14}}$	2	0	0
$\lambda_1 \lambda_{X_{14}}$	2	0	0
$\lambda_{HS_1}^{(1)} \lambda_{X_{14}}$	4	0	0
$\lambda_{HS_2}^{(1)} \lambda_{X_{14}}$	4	0	0
$\lambda_{HS_2}^{(A)} \lambda_{X_{14}}$	4	0	0
$\lambda_{X_{14}}(T_\nu^{(1)} + \operatorname{Tr}(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger) + 2T_\nu^{(2)} + 6\operatorname{Tr}(Y_d(Y_d)^\dagger) + 2\operatorname{Tr}(Y_e(Y_e)^\dagger) + 6\operatorname{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{HS_1}^{(A)} \lambda_{X_{14}}$	0	0	0
$\lambda_{X_{14}}(T_\nu^{(1)} + \operatorname{Tr}(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger) + T_\nu^{(2)} + 6\operatorname{Tr}(Y_d(Y_d)^\dagger) + 2\operatorname{Tr}(Y_e(Y_e)^\dagger) + 6\operatorname{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$(\lambda_5)^*(2(\lambda_{X_5}^{(1)})^* - (\lambda_{X_5}^{(2)})^* + 4(\lambda_{X_{18}})^*)$	0	0	0
$\lambda_{12}^{(A)} \lambda_{X_{14}}$	0	0	0
$\lambda_{12}^{(\times)} \lambda_{X_{14}}$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$(\lambda_5)^* \operatorname{Tr}(y_1 (y_1^{\text{conj}})^\dagger)$	0	0	0
$\lambda_{X_{14}}(2T_\nu^{(1)} + 2\operatorname{Tr}(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger) + T_\nu^{(2)} + 6\operatorname{Tr}(Y_d(Y_d)^\dagger) + 2\operatorname{Tr}(Y_e(Y_e)^\dagger) + 6\operatorname{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$(\lambda_5)^*((\lambda_{X_5}^{(1)})^* + 2(\lambda_{X_{18}})^*)$	0	0	0
$g_2^2 \lambda_{X_{14}}$	0	0	0
$g_1^2 \lambda_{X_{14}}$	0	0	0
$\lambda_{12}^{(1)} \lambda_{X_{14}}$	0	0	0
$\lambda_1 \lambda_{X_{14}}$	0	0	0
$\lambda_{HS_1}^{(1)} \lambda_{X_{14}}$	0	0	0
$\lambda_{HS_2}^{(1)} \lambda_{X_{14}}$	0	0	0
$\lambda_{HS_2}^{(A)} \lambda_{X_{14}}$	0	0	0

$\lambda_{X_{14}}(T_\nu^{(1)} + \mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{HS_1}^{(A)}\lambda_{X_{14}}$	0	0	0
$\lambda_{X_{14}}(T_\nu^{(1)} + \mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$(\lambda_5)^*(2(\lambda_{X_5}^{(1)})^* - (\lambda_{X_5}^{(2)})^* + 4(\lambda_{X_{18}})^*)$	0	0	0
$\lambda_{12}^{(A)}\lambda_{X_{14}}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{X_{14}}$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$(\lambda_5)^*\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{X_{14}}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$(\lambda_5)^*((\lambda_{X_5}^{(1)})^* + 2(\lambda_{X_{18}})^*)$	0	0	0
$g_2^2\lambda_{X_{14}}$	0	0	0
$g_1^2\lambda_{X_{14}}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_{14}}$	0	0	0
$\lambda_1\lambda_{X_{14}}$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{14}}$	0	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{14}}$	0	0	0
$\lambda_{HS_2}^{(A)}\lambda_{X_{14}}$	0	0	0
$\lambda_{X_{14}}(T_\nu^{(1)} + \mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{HS_1}^{(A)}\lambda_{X_{14}}$	0	0	0
$\lambda_{X_{14}}(T_\nu^{(1)} + \mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$(\lambda_5)^*(2(\lambda_{X_5}^{(1)})^* - (\lambda_{X_5}^{(2)})^* + 4(\lambda_{X_{18}})^*)$	0	0	0
$\lambda_{12}^{(A)}\lambda_{X_{14}}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{X_{14}}$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$(\lambda_5)^*\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger)$	2	0	0
$\lambda_{X_{14}}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$(\lambda_5)^*((\lambda_{X_5}^{(1)})^* + 2(\lambda_{X_{18}})^*)$	4	0	0
$g_2^2\lambda_{X_{14}}$	$-\frac{21}{2}$	0	0
$g_1^2\lambda_{X_{14}}$	$-\frac{9}{2}$	0	0
$\lambda_{12}^{(1)}\lambda_{X_{14}}$	2	0	0
$\lambda_1\lambda_{X_{14}}$	2	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{14}}$	4	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{14}}$	4	0	0
$\lambda_{HS_2}^{(A)}\lambda_{X_{14}}$	0	0	0
$\lambda_{X_{14}}(T_\nu^{(1)} + \mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	0	0
$\lambda_{HS_1}^{(A)}\lambda_{X_{14}}$	4	0	0
$\lambda_{X_{14}}(T_\nu^{(1)} + \mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$(\lambda_5)^*(2(\lambda_{X_5}^{(1)})^* - (\lambda_{X_5}^{(2)})^* + 4(\lambda_{X_{18}})^*)$	0	0	0
$\lambda_{12}^{(A)}\lambda_{X_{14}}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{X_{14}}$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$(\lambda_5)^*\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger)$	2	0	0
$\lambda_{X_{14}}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + T_\nu^{(2)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$(\lambda_5)^*((\lambda_{X_5}^{(1)})^* + 2(\lambda_{X_{18}})^*)$	0	0	0
$g_2^2\lambda_{X_{14}}$	$-\frac{33}{2}$	0	0
$g_1^2\lambda_{X_{14}}$	$-\frac{9}{2}$	0	0
$\lambda_{12}^{(1)}\lambda_{X_{14}}$	2	0	0

$\lambda_1\lambda_{X_{14}}$	2	0	0
$\lambda^{(1)}_{HS_1}\lambda_{X_{14}}$	4	0	0
$\lambda^{(1)}_{HS_2}\lambda_{X_{14}}$	4	0	0
$\lambda^{(A)}_{HS_2}\lambda_{X_{14}}$	$-\frac{4}{3}$	0	0
$\lambda_{X_{14}}(T^{(1)}_\nu + \mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T^{(2)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda^{(A)}_{HS_1}\lambda_{X_{14}}$	$-\frac{4}{3}$	0	0
$\lambda_{X_{14}}(T^{(1)}_\nu + \mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + T^{(2)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	0	0
$(\lambda_5)^*(2(\lambda^{(1)}_{X_5})^* - (\lambda^{(2)}_{X_5})^* + 4(\lambda_{X_{18}})^*)$	2	0	0
$\lambda^{(A)}_{12}\lambda_{X_{14}}$	-2	0	0
$\lambda^{(\times)}_{12}\lambda_{X_{14}}$	-2	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$(\lambda_5)^*\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger)$	0
$\lambda_{X_{14}}(2T^{(1)}_\nu + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + T^{(2)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$(\lambda_5)^*((\lambda^{(1)}_{X_5})^* + 2(\lambda_{X_{18}})^*)$	0
$g_2^2\lambda_{X_{14}}$	0
$g_1^2\lambda_{X_{14}}$	0
$\lambda^{(1)}_{12}\lambda_{X_{14}}$	0
$\lambda_1\lambda_{X_{14}}$	0
$\lambda^{(1)}_{HS_1}\lambda_{X_{14}}$	0
$\lambda^{(1)}_{HS_2}\lambda_{X_{14}}$	0
$\lambda^{(A)}_{HS_2}\lambda_{X_{14}}$	0
$\lambda_{X_{14}}(T^{(1)}_\nu + \mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 2T^{(2)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$\lambda^{(A)}_{HS_1}\lambda_{X_{14}}$	0
$\lambda_{X_{14}}(T^{(1)}_\nu + \mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + T^{(2)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$(\lambda_5)^*(2(\lambda^{(1)}_{X_5})^* - (\lambda^{(2)}_{X_5})^* + 4(\lambda_{X_{18}})^*)$	0
$\lambda^{(A)}_{12}\lambda_{X_{14}}$	0
$\lambda^{(\times)}_{12}\lambda_{X_{14}}$	0

$\beta^{(1)}_{\lambda_{X_{15}}}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda_{X_{15}}(2T^{(1)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$g_2^2\lambda_{X_{15}}$	0	0	0
$g_1^2\lambda_{X_{15}}$	0	0	0
$\lambda_1\lambda_{X_{15}}$	0	0	0
$\lambda^{(1)}_{HS_1}\lambda_{X_{15}}$	0	0	0
$\lambda^{(A)}_{HS_1}\lambda_{X_{15}}$	0	0	0
$\lambda^{(1)}_{S_1}\lambda_{X_{15}}$	0	0	0
$\lambda_{X_{15}}(3T^{(1)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_{X_{15}}(2T^{(1)}_\nu + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	0	0
$g_2^2\lambda_{X_{15}}$	-9	0	0
$g_1^2\lambda_{X_{15}}$	-3	0	0
$\lambda_1\lambda_{X_{15}}$	2	0	0
$\lambda^{(1)}_{HS_1}\lambda_{X_{15}}$	8	0	0
$\lambda^{(A)}_{HS_1}\lambda_{X_{15}}$	4	0	0
$\lambda^{(1)}_{S_1}\lambda_{X_{15}}$	2	0	0

$\lambda_{X_{15}}(3T_\nu^{(1)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_{X_{15}}(2T_\nu^{(1)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$g_2^2\lambda_{X_{15}}$	-9	0	0
$g_1^2\lambda_{X_{15}}$	-3	0	0
$\lambda_1\lambda_{X_{15}}$	2	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{15}}$	8	0	0
$\lambda_{HS_1}^{(A)}\lambda_{X_{15}}$	4	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_{15}}$	2	0	0
$\lambda_{X_{15}}(3T_\nu^{(1)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_{X_{15}}(2T_\nu^{(1)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$g_2^2\lambda_{X_{15}}$	0	0	0
$g_1^2\lambda_{X_{15}}$	0	0	0
$\lambda_1\lambda_{X_{15}}$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{15}}$	0	0	0
$\lambda_{HS_1}^{(A)}\lambda_{X_{15}}$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_{15}}$	0	0	0
$\lambda_{X_{15}}(3T_\nu^{(1)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda_{X_{15}}(2T_\nu^{(1)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$g_2^2\lambda_{X_{15}}$	0	0	0
$g_1^2\lambda_{X_{15}}$	0	0	0
$\lambda_1\lambda_{X_{15}}$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{15}}$	0	0	0
$\lambda_{HS_1}^{(A)}\lambda_{X_{15}}$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_{15}}$	0	0	0
$\lambda_{X_{15}}(3T_\nu^{(1)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$\lambda_{X_{15}}(2T_\nu^{(1)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$g_2^2\lambda_{X_{15}}$	0
$g_1^2\lambda_{X_{15}}$	0
$\lambda_1\lambda_{X_{15}}$	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{15}}$	0
$\lambda_{HS_1}^{(A)}\lambda_{X_{15}}$	0
$\lambda_{S_1}^{(1)}\lambda_{X_{15}}$	0
$\lambda_{X_{15}}(3T_\nu^{(1)} + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0

$\beta_{\lambda_{X_{16}}}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda_{HS_2}^{(1)}(2\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger) + 10\lambda_{X_2} + 8\lambda_{X_{16}})$	0	1	0
$\lambda_{X_{16}}(2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	1	0

$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger Y_e(Y_e)^\dagger)$	0	-2	0
$\lambda_{X_1}(\lambda_{X_{16}})^*$	0	40	0
$g_2^2\lambda_{X_{16}}$	0	$-\frac{33}{2}$	0
$g_1^2\lambda_{X_{16}}$	0	$-\frac{3}{2}$	0
$\lambda_1\lambda_{X_{16}}$	0	6	0
$\lambda^{(1)}_{HS_1}\lambda^{(1)}_{X_7}$	0	2	0
$\lambda^{(1)}_{S_2}\lambda_{X_{16}}$	0	2	0
$\lambda^{(A)}_{S_2}\lambda_{X_{16}}$	0	-4	0
$\lambda_5\lambda_{X_{12}}$	0	4	0
$\lambda^{(1)}_{HS_2}(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+6\lambda_{X_2}+8\lambda_{X_{16}})$	0	0	0
$\lambda_{X_{16}}(4T_\nu^{(2)}+4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{X_7}+2\lambda^{(2)}_{X_7})$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$\lambda^{(1)}_{HS_2}(2\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+10\lambda_{X_2}+8\lambda_{X_{16}})$	0	0	0
$\lambda_{X_{16}}(2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger Y_e(Y_e)^\dagger)$	0	0	0
$\lambda_{X_1}(\lambda_{X_{16}})^*$	0	0	0
$g_2^2\lambda_{X_{16}}$	0	0	0
$g_1^2\lambda_{X_{16}}$	0	0	0
$\lambda_1\lambda_{X_{16}}$	0	0	0
$\lambda^{(1)}_{HS_1}\lambda^{(1)}_{X_7}$	0	0	0
$\lambda^{(1)}_{S_2}\lambda_{X_{16}}$	0	0	0
$\lambda^{(A)}_{S_2}\lambda_{X_{16}}$	0	0	0
$\lambda_5\lambda_{X_{12}}$	0	0	0
$\lambda^{(1)}_{HS_2}(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+6\lambda_{X_2}+8\lambda_{X_{16}})$	0	0	0
$\lambda_{X_{16}}(4T_\nu^{(2)}+4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{X_7}+2\lambda^{(2)}_{X_7})$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda^{(1)}_{HS_2}(2\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+10\lambda_{X_2}+8\lambda_{X_{16}})$	0	0	0
$\lambda_{X_{16}}(2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger Y_e(Y_e)^\dagger)$	0	0	0
$\lambda_{X_1}(\lambda_{X_{16}})^*$	0	0	0
$g_2^2\lambda_{X_{16}}$	0	0	0
$g_1^2\lambda_{X_{16}}$	0	0	0
$\lambda_1\lambda_{X_{16}}$	0	0	0
$\lambda^{(1)}_{HS_1}\lambda^{(1)}_{X_7}$	0	0	0
$\lambda^{(1)}_{S_2}\lambda_{X_{16}}$	0	0	0
$\lambda^{(A)}_{S_2}\lambda_{X_{16}}$	0	0	0
$\lambda_5\lambda_{X_{12}}$	0	0	0
$\lambda^{(1)}_{HS_2}(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+6\lambda_{X_2}+8\lambda_{X_{16}})$	0	0	0
$\lambda_{X_{16}}(4T_\nu^{(2)}+4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{X_7}+2\lambda^{(2)}_{X_7})$	0	0	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda^{(1)}_{HS_2}(2\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+10\lambda_{X_2}+8\lambda_{X_{16}})$	0	0	0
$\lambda_{X_{16}}(2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger Y_e(Y_e)^\dagger)$	0	0	-4
$\lambda_{X_1}(\lambda_{X_{16}})^*$	0	0	24

$g_2^2\lambda_{X_{16}}$	0	0	$-\frac{9}{2}$
$g_1^2\lambda_{X_{16}}$	0	0	$-\frac{3}{2}$
$\lambda_1\lambda_{X_{16}}$	0	0	6
$\lambda^{(1)}_{HS_1}\lambda^{(1)}_{X_7}$	0	0	6
$\lambda^{(1)}_{S_2}\lambda_{X_{16}}$	0	0	2
$\lambda^{(A)}_{S_2}\lambda_{X_{16}}$	0	0	0
$\lambda_5\lambda_{X_{12}}$	0	0	12
$\lambda^{(1)}_{HS_2}(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+6\lambda_{X_2}+8\lambda_{X_{16}})$	0	0	1
$\lambda_{X_{16}}(4T_\nu^{(2)}+4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	1
$\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{X_7}+2\lambda^{(2)}_{X_7})$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda^{(1)}_{HS_2}(2\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+10\lambda_{X_2}+8\lambda_{X_{16}})$	0	0	1
$\lambda_{X_{16}}(2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	1
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger Y_e(Y_e)^\dagger)$	0	0	-2
$\lambda_{X_1}(\lambda_{X_{16}})^*$	0	0	40
$g_2^2\lambda_{X_{16}}$	0	0	$-\frac{33}{2}$
$g_1^2\lambda_{X_{16}}$	0	0	$-\frac{3}{2}$
$\lambda_1\lambda_{X_{16}}$	0	0	6
$\lambda^{(1)}_{HS_1}\lambda^{(1)}_{X_7}$	0	0	0
$\lambda^{(1)}_{S_2}\lambda_{X_{16}}$	0	0	2
$\lambda^{(A)}_{S_2}\lambda_{X_{16}}$	0	0	-4
$\lambda_5\lambda_{X_{12}}$	0	0	4
$\lambda^{(1)}_{HS_2}(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+6\lambda_{X_2}+8\lambda_{X_{16}})$	0	0	0
$\lambda_{X_{16}}(4T_\nu^{(2)}+4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{X_7}+2\lambda^{(2)}_{X_7})$	0	0	1

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda^{(1)}_{HS_2}(2\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+10\lambda_{X_2}+8\lambda_{X_{16}})$	0
$\lambda_{X_{16}}(2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger Y_e(Y_e)^\dagger)$	0
$\lambda_{X_1}(\lambda_{X_{16}})^*$	0
$g_2^2\lambda_{X_{16}}$	0
$g_1^2\lambda_{X_{16}}$	0
$\lambda_1\lambda_{X_{16}}$	0
$\lambda^{(1)}_{HS_1}\lambda^{(1)}_{X_7}$	0
$\lambda^{(1)}_{S_2}\lambda_{X_{16}}$	0
$\lambda^{(A)}_{S_2}\lambda_{X_{16}}$	0
$\lambda_5\lambda_{X_{12}}$	0
$\lambda^{(1)}_{HS_2}(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+6\lambda_{X_2}+8\lambda_{X_{16}})$	0
$\lambda_{X_{16}}(4T_\nu^{(2)}+4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{X_7}+2\lambda^{(2)}_{X_7})$	0

$\beta^{(1)}_{\lambda^{(1)}_{X_{17}}}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda^{(1)}_{X_{17}}(T_\nu^{(1)}+T_\nu^{(2)}+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_9(8(\lambda^{(1)}_{X_{17}})^*-4(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*)$	0	0	0
$g_2^2\lambda^{(1)}_{X_{17}}$	0	0	0

$g_1^2\lambda_{X_{17}}^{(1)}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_{17}}^{(1)}$	0	0	0
$\lambda_{12}^{(A)}(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} - \frac{\lambda_{X_{17}}^{(1)}}{2})$	0	0	0
$\lambda_1(4\lambda_{X_{17}}^{(1)} - 2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})$	0	0	0
$\lambda_{HS_1}^{(1)}(4\lambda_{X_{17}}^{(1)} - 6\lambda_{11})$	0	0	0
$\lambda_{HS_1}^{(A)}(-\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} + \frac{\lambda_{11}}{2} - 2\lambda_{X_{17}}^{(1)})$	0	0	0
$\lambda_{HS_2}^{(1)}(4\lambda_{X_{17}}^{(1)} - 6\lambda_{10})$	0	0	0
$\lambda_{HS_2}^{(A)}(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} - \frac{\lambda_{10}}{2} + 2\lambda_{X_{17}}^{(1)})$	0	0	0
$\lambda_{X_{17}}^{(1)}(\frac{3}{2}T_\nu^{(1)} + \frac{3}{2}T_\nu^{(2)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_{X_{17}}^{(1)}(T_\nu^{(1)} + T_\nu^{(2)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	1	0
$\lambda_9(8(\lambda_{X_{17}}^{(1)})^* - 4(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})^*)$	0	1	0
$g_2^2\lambda_{X_{17}}^{(1)}$	0	-9	0
$g_1^2\lambda_{X_{17}}^{(1)}$	0	-3	0
$\lambda_{12}^{(1)}\lambda_{X_{17}}^{(1)}$	0	2	0
$\lambda_{12}^{(A)}(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} - \frac{\lambda_{X_{17}}^{(1)}}{2})$	0	1	0
$\lambda_1(4\lambda_{X_{17}}^{(1)} - 2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})$	0	1	0
$\lambda_{HS_1}^{(1)}(4\lambda_{X_{17}}^{(1)} - 6\lambda_{11})$	0	1	0
$\lambda_{HS_1}^{(A)}(-\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} + \frac{\lambda_{11}}{2} - 2\lambda_{X_{17}}^{(1)})$	0	1	0
$\lambda_{HS_2}^{(1)}(4\lambda_{X_{17}}^{(1)} - 6\lambda_{10})$	0	1	0
$\lambda_{HS_2}^{(A)}(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} - \frac{\lambda_{10}}{2} + 2\lambda_{X_{17}}^{(1)})$	0	1	0
$\lambda_{X_{17}}^{(1)}(\frac{3}{2}T_\nu^{(1)} + \frac{3}{2}T_\nu^{(2)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_{X_{17}}^{(1)}(T_\nu^{(1)} + T_\nu^{(2)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_9(8(\lambda_{X_{17}}^{(1)})^* - 4(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})^*)$	0	1	0
$g_2^2\lambda_{X_{17}}^{(1)}$	0	-9	0
$g_1^2\lambda_{X_{17}}^{(1)}$	0	-3	0
$\lambda_{12}^{(1)}\lambda_{X_{17}}^{(1)}$	0	2	0
$\lambda_{12}^{(A)}(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} - \frac{\lambda_{X_{17}}^{(1)}}{2})$	0	1	0
$\lambda_1(4\lambda_{X_{17}}^{(1)} - 2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})$	0	1	0
$\lambda_{HS_1}^{(1)}(4\lambda_{X_{17}}^{(1)} - 6\lambda_{11})$	0	1	0
$\lambda_{HS_1}^{(A)}(-\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} + \frac{\lambda_{11}}{2} - 2\lambda_{X_{17}}^{(1)})$	0	1	0
$\lambda_{HS_2}^{(1)}(4\lambda_{X_{17}}^{(1)} - 6\lambda_{10})$	0	1	0
$\lambda_{HS_2}^{(A)}(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} - \frac{\lambda_{10}}{2} + 2\lambda_{X_{17}}^{(1)})$	0	1	0
$\lambda_{X_{17}}^{(1)}(\frac{3}{2}T_\nu^{(1)} + \frac{3}{2}T_\nu^{(2)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	1	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_{X_{17}}^{(1)}(T_\nu^{(1)} + T_\nu^{(2)} + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_9(8(\lambda_{X_{17}}^{(1)})^* - 4(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})^*)$	0	0	0
$g_2^2\lambda_{X_{17}}^{(1)}$	0	0	0
$g_1^2\lambda_{X_{17}}^{(1)}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_{17}}^{(1)}$	0	0	0
$\lambda_{12}^{(A)}(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} - \frac{\lambda_{X_{17}}^{(1)}}{2})$	0	0	0

$\lambda_1(4\lambda^{(1)}_{X_{17}} - 2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})$	0	0	0
$\lambda^{(1)}_{HS_1}(4\lambda^{(1)}_{X_{17}} - 6\lambda_{11})$	0	0	0
$\lambda^{(A)}_{HS_1}(-\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + \frac{\lambda_{11}}{2} - 2\lambda^{(1)}_{X_{17}})$	0	0	0
$\lambda^{(1)}_{HS_2}(4\lambda^{(1)}_{X_{17}} - 6\lambda_{10})$	0	0	0
$\lambda^{(A)}_{HS_2}(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - \frac{\lambda_{10}}{2} + 2\lambda^{(1)}_{X_{17}})$	0	0	0
$\lambda^{(1)}_{X_{17}}(\frac{3}{2}T^{(1)}_\nu + \frac{3}{2}T^{(2)}_\nu + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda^{(1)}_{X_{17}}(T^{(1)}_\nu + T^{(2)}_\nu + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_9(8(\lambda^{(1)}_{X_{17}})^* - 4(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*)$	0	0	0
$g_2^2\lambda^{(1)}_{X_{17}}$	0	0	0
$g_1^2\lambda^{(1)}_{X_{17}}$	0	0	0
$\lambda^{(1)}_{12}\lambda^{(1)}_{X_{17}}$	0	0	0
$\lambda^{(A)}_{12}(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - \frac{\lambda^{(1)}_{X_{17}}}{2})$	0	0	0
$\lambda_1(4\lambda^{(1)}_{X_{17}} - 2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})$	0	0	0
$\lambda^{(1)}_{HS_1}(4\lambda^{(1)}_{X_{17}} - 6\lambda_{11})$	0	0	0
$\lambda^{(A)}_{HS_1}(-\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + \frac{\lambda_{11}}{2} - 2\lambda^{(1)}_{X_{17}})$	0	0	0
$\lambda^{(1)}_{HS_2}(4\lambda^{(1)}_{X_{17}} - 6\lambda_{10})$	0	0	0
$\lambda^{(A)}_{HS_2}(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - \frac{\lambda_{10}}{2} + 2\lambda^{(1)}_{X_{17}})$	0	0	0
$\lambda^{(1)}_{X_{17}}(\frac{3}{2}T^{(1)}_\nu + \frac{3}{2}T^{(2)}_\nu + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$\lambda^{(1)}_{X_{17}}(T^{(1)}_\nu + T^{(2)}_\nu + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0
$\lambda_9(8(\lambda^{(1)}_{X_{17}})^* - 4(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*)$	0
$g_2^2\lambda^{(1)}_{X_{17}}$	0
$g_1^2\lambda^{(1)}_{X_{17}}$	0
$\lambda^{(1)}_{12}\lambda^{(1)}_{X_{17}}$	0
$\lambda^{(A)}_{12}(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - \frac{\lambda^{(1)}_{X_{17}}}{2})$	0
$\lambda_1(4\lambda^{(1)}_{X_{17}} - 2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})$	0
$\lambda^{(1)}_{HS_1}(4\lambda^{(1)}_{X_{17}} - 6\lambda_{11})$	0
$\lambda^{(A)}_{HS_1}(-\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + \frac{\lambda_{11}}{2} - 2\lambda^{(1)}_{X_{17}})$	0
$\lambda^{(1)}_{HS_2}(4\lambda^{(1)}_{X_{17}} - 6\lambda_{10})$	0
$\lambda^{(A)}_{HS_2}(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - \frac{\lambda_{10}}{2} + 2\lambda^{(1)}_{X_{17}})$	0
$\lambda^{(1)}_{X_{17}}(\frac{3}{2}T^{(1)}_\nu + \frac{3}{2}T^{(2)}_\nu + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0

$\beta^{(1)}_{\lambda_{X_{18}}}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda^{(1)}_{HS_1}(4\,\text{Tr}(y_1^{\text{conj}}(y_1)^\dagger) + 6\lambda_{X_9} + 8\lambda_{X_{18}})$	1	0	0
$\lambda_{X_{18}}(4T^{(1)}_\nu + 4\,\text{Tr}(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger) + 6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	1	0	0
$\text{Tr}(y_1^{\text{conj}}(y_1)^\dagger Y_e(Y_e)^\dagger)$	-4	0	0
$(\lambda_5)^*(\lambda_{X_{14}})^*$	12	0	0
$\lambda_{X_6}(\lambda_{X_{18}})^*$	24	0	0
$g_2^2\lambda_{X_{18}}$	$-\frac{9}{2}$	0	0
$g_1^2\lambda_{X_{18}}$	$-\frac{3}{2}$	0	0
$\lambda_1\lambda_{X_{18}}$	6	0	0

$\lambda^{(1)}_{HS_2}\lambda^{(1)}_{X_5}$	6	0	0
$\lambda^{(1)}_{S_1}\lambda_{X_{18}}$	2	0	0
$\lambda^{(1)}_{HS_1}(2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)+10\lambda_{X_9}+8\lambda_{X_{18}})$	0	0	0
$\lambda_{X_{18}}(2T_\nu^{(1)}+2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda^{(A)}_{S_1}\lambda_{X_{18}}$	0	0	0
$\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{X_5}+2\lambda^{(2)}_{X_5})$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$\lambda^{(1)}_{HS_1}(4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)+6\lambda_{X_9}+8\lambda_{X_{18}})$	0	0	0
$\lambda_{X_{18}}(4T_\nu^{(1)}+4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger Y_e(Y_e)^\dagger)$	0	0	0
$(\lambda_5)^*(\lambda_{X_{14}})^*$	0	0	0
$\lambda_{X_6}(\lambda_{X_{18}})^*$	0	0	0
$g_2^2\lambda_{X_{18}}$	0	0	0
$g_1^2\lambda_{X_{18}}$	0	0	0
$\lambda_1\lambda_{X_{18}}$	0	0	0
$\lambda^{(1)}_{HS_2}\lambda^{(1)}_{X_5}$	0	0	0
$\lambda^{(1)}_{S_1}\lambda_{X_{18}}$	0	0	0
$\lambda^{(1)}_{HS_1}(2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)+10\lambda_{X_9}+8\lambda_{X_{18}})$	0	0	0
$\lambda_{X_{18}}(2T_\nu^{(1)}+2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda^{(A)}_{S_1}\lambda_{X_{18}}$	0	0	0
$\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{X_5}+2\lambda^{(2)}_{X_5})$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda^{(1)}_{HS_1}(4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)+6\lambda_{X_9}+8\lambda_{X_{18}})$	0	0	0
$\lambda_{X_{18}}(4T_\nu^{(1)}+4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger Y_e(Y_e)^\dagger)$	0	0	0
$(\lambda_5)^*(\lambda_{X_{14}})^*$	0	0	0
$\lambda_{X_6}(\lambda_{X_{18}})^*$	0	0	0
$g_2^2\lambda_{X_{18}}$	0	0	0
$g_1^2\lambda_{X_{18}}$	0	0	0
$\lambda_1\lambda_{X_{18}}$	0	0	0
$\lambda^{(1)}_{HS_2}\lambda^{(1)}_{X_5}$	0	0	0
$\lambda^{(1)}_{S_1}\lambda_{X_{18}}$	0	0	0
$\lambda^{(1)}_{HS_1}(2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)+10\lambda_{X_9}+8\lambda_{X_{18}})$	0	0	0
$\lambda_{X_{18}}(2T_\nu^{(1)}+2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda^{(A)}_{S_1}\lambda_{X_{18}}$	0	0	0
$\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{X_5}+2\lambda^{(2)}_{X_5})$	0	0	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda^{(1)}_{HS_1}(4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)+6\lambda_{X_9}+8\lambda_{X_{18}})$	0	0	0
$\lambda_{X_{18}}(4T_\nu^{(1)}+4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger Y_e(Y_e)^\dagger)$	-2	0	0
$(\lambda_5)^*(\lambda_{X_{14}})^*$	4	0	0
$\lambda_{X_6}(\lambda_{X_{18}})^*$	40	0	0
$g_2^2\lambda_{X_{18}}$	$-\frac{33}{2}$	0	0
$g_1^2\lambda_{X_{18}}$	$-\frac{3}{2}$	0	0
$\lambda_1\lambda_{X_{18}}$	6	0	0
$\lambda^{(1)}_{HS_2}\lambda^{(1)}_{X_5}$	2	0	0
$\lambda^{(1)}_{S_1}\lambda_{X_{18}}$	2	0	0

$\lambda_{HS_1}^{(1)}(2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)+10\lambda_{X_9}+8\lambda_{X_{18}})$	1	0	0
$\lambda_{X_{18}}(2T_\nu^{(1)}+2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_{18}}$	-4	0	0
$\lambda_{HS_2}^{(1)}(6\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda_{HS_1}^{(1)}(4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)+6\lambda_{X_9}+8\lambda_{X_{18}})$	0	0	0
$\lambda_{X_{18}}(4T_\nu^{(1)}+4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger Y_e(Y_e)^\dagger)$	-2	0	0
$(\lambda_5)^*(\lambda_{X_{14}})^*$	4	0	0
$\lambda_{X_6}(\lambda_{X_{18}})^*$	40	0	0
$g_2^2\lambda_{X_{18}}$	$-\frac{33}{2}$	0	0
$g_1^2\lambda_{X_{18}}$	$-\frac{3}{2}$	0	0
$\lambda_1\lambda_{X_{18}}$	6	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_5}^{(1)}$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_{18}}$	2	0	0
$\lambda_{HS_1}^{(1)}(2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)+10\lambda_{X_9}+8\lambda_{X_{18}})$	1	0	0
$\lambda_{X_{18}}(2T_\nu^{(1)}+2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_{18}}$	-4	0	0
$\lambda_{HS_2}^{(1)}(6\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	1	0	0

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_{HS_1}^{(1)}(4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)+6\lambda_{X_9}+8\lambda_{X_{18}})$	0
$\lambda_{X_{18}}(4T_\nu^{(1)}+4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger Y_e(Y_e)^\dagger)$	0
$(\lambda_5)^*(\lambda_{X_{14}})^*$	0
$\lambda_{X_6}(\lambda_{X_{18}})^*$	0
$g_2^2\lambda_{X_{18}}$	0
$g_1^2\lambda_{X_{18}}$	0
$\lambda_1\lambda_{X_{18}}$	0
$\lambda_{HS_2}^{(1)}\lambda_{X_5}^{(1)}$	0
$\lambda_{S_1}^{(1)}\lambda_{X_{18}}$	0
$\lambda_{HS_1}^{(1)}(2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)+10\lambda_{X_9}+8\lambda_{X_{18}})$	0
$\lambda_{X_{18}}(2T_\nu^{(1)}+2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$\lambda_{S_1}^{(A)}\lambda_{X_{18}}$	0
$\lambda_{HS_2}^{(1)}(6\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	0

$\beta_{m_1^2}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$m_1^2(4T_\nu^{(1)}+4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger)+4\lambda_{S_1}^{(1)})$	1	0	0
$\mu_{\mathrm{T3GeneralS1Sq}}(8\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger)+12(\lambda_{X_9})^*)$	1	0	0
$(\mu_{\mathrm{T3GeneralS1Sq}})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	8	0	0
$\lambda_{X_9}(\mu_{\mathrm{T3GeneralS1Sq}})^*$	12	0	0
$\lambda_{12}^{(1)}m_2^2$	6	6	6
$T_\nu^{(1)}$	-8	-8	-8
T_ν	-8	0	0
$m_1^2(4T_\nu^{(1)}+4\lambda_{S_1}^{(1)})$	0	1	1
$\mu_{\mathrm{T3GeneralS2Sq}}(\lambda_{X_7}^{(1)})^*$	0	12	0
$\lambda_{X_7}^{(1)}(\mu_{\mathrm{T3GeneralS2Sq}})^*$	0	12	0

$g_1^2 m_1^2$	0	-6	-24
$m_1^2(2T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$	0	0	0
$g_2^2 m_1^2$	0	0	0
$\mu_{\text{T3GeneralS1S2}}(\lambda_{11})^*$	0	0	0
$\lambda_{11}(\mu_{\text{T3GeneralS1S2}})^*$	0	0	0
$m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$	0	0	0
$m_1^2(2T_\nu^{(1)} + 2\,\text{Tr}(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger) + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	0	0	0
$\mu_{\text{T3GeneralS1Sq}}(4\,\text{Tr}(y_1(y_1^{\text{conj}})^\dagger) + 20(\lambda_{X_9})^*)$	0	0	0
$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	0	0	0
$m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0
$\mu_{\text{T3GeneralS2Sq}}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}(\mu_{\text{T3GeneralS2Sq}})^*$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$m_1^2(4T_\nu^{(1)} + 4\,\text{Tr}(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger) + 4\lambda_{S_1}^{(1)})$	0	0	0
$\mu_{\text{T3GeneralS1Sq}}(8\,\text{Tr}(y_1(y_1^{\text{conj}})^\dagger) + 12(\lambda_{X_9})^*)$	0	0	0
$(\mu_{\text{T3GeneralS1Sq}})^*\,\text{Tr}(y_1^{\text{conj}}(y_1)^\dagger)$	0	0	0
$\lambda_{X_9}(\mu_{\text{T3GeneralS1Sq}})^*$	0	0	0
$\lambda_{12}^{(1)} m_2^2$	4	4	4
$T_\nu^{(1)}$	-4	-16	-4
T_ν	0	0	0
$m_1^2(4T_\nu^{(1)} + 4\lambda_{S_1}^{(1)})$	0	0	0
$\mu_{\text{T3GeneralS2Sq}}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_7}^{(1)}(\mu_{\text{T3GeneralS2Sq}})^*$	0	0	0
$g_1^2 m_1^2$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{27}{2}$
$m_1^2(2T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$	1	1	1
$g_2^2 m_1^2$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$\mu_{\text{T3GeneralS1S2}}(\lambda_{11})^*$	0	6	0
$\lambda_{11}(\mu_{\text{T3GeneralS1S2}})^*$	0	6	0
$m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$	0	0	0
$m_1^2(2T_\nu^{(1)} + 2\,\text{Tr}(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger) + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	0	0	0
$\mu_{\text{T3GeneralS1Sq}}(4\,\text{Tr}(y_1(y_1^{\text{conj}})^\dagger) + 20(\lambda_{X_9})^*)$	0	0	0
$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	0	0	0
$m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0
$\mu_{\text{T3GeneralS2Sq}}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}(\mu_{\text{T3GeneralS2Sq}})^*$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$m_1^2(4T_\nu^{(1)} + 4\,\text{Tr}(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger) + 4\lambda_{S_1}^{(1)})$	0	0	0
$\mu_{\text{T3GeneralS1Sq}}(8\,\text{Tr}(y_1(y_1^{\text{conj}})^\dagger) + 12(\lambda_{X_9})^*)$	0	0	0
$(\mu_{\text{T3GeneralS1Sq}})^*\,\text{Tr}(y_1^{\text{conj}}(y_1)^\dagger)$	0	0	0
$\lambda_{X_9}(\mu_{\text{T3GeneralS1Sq}})^*$	0	0	0
$\lambda_{12}^{(1)} m_2^2$	4	4	4
$T_\nu^{(1)}$	-6	-24	-6
T_ν	0	0	0
$m_1^2(4T_\nu^{(1)} + 4\lambda_{S_1}^{(1)})$	0	0	0
$\mu_{\text{T3GeneralS2Sq}}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_7}^{(1)}(\mu_{\text{T3GeneralS2Sq}})^*$	0	0	0
$g_1^2 m_1^2$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{27}{2}$
$m_1^2(2T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$	0	0	0
$g_2^2 m_1^2$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$

$\mu_{\text{T3GeneralS1S2}}(\lambda_{11})^*$	0	6	0
$\lambda_{11}(\mu_{\text{T3GeneralS1S2}})^*$	0	6	0
$m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$	1	1	1
$m_1^2(2T_\nu^{(1)} + 2\,\text{Tr}(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger) + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	0	0	0
$\mu_{\text{T3GeneralS1Sq}}(4\,\text{Tr}(y_1(y_1^{\text{conj}})^\dagger) + 20(\lambda_{X_9})^*)$	0	0	0
$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	0	0	0
$m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0
$\mu_{\text{T3GeneralS2Sq}}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}(\mu_{\text{T3GeneralS2Sq}})^*$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$m_1^2(4T_\nu^{(1)} + 4\,\text{Tr}(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger) + 4\lambda_{S_1}^{(1)})$	0	0	0
$\mu_{\text{T3GeneralS1Sq}}(8\,\text{Tr}(y_1(y_1^{\text{conj}})^\dagger) + 12(\lambda_{X_9})^*)$	0	0	0
$(\mu_{\text{T3GeneralS1Sq}})^*\,\text{Tr}(y_1^{\text{conj}}(y_1)^\dagger)$	4	0	0
$\lambda_{X_9}(\mu_{\text{T3GeneralS1Sq}})^*$	20	0	0
$\lambda_{12}^{(1)} m_2^2$	2	2	2
$T_\nu^{(1)}$	-4	-4	-4
T_ν	-4	0	0
$m_1^2(4T_\nu^{(1)} + 4\lambda_{S_1}^{(1)})$	0	0	0
$\mu_{\text{T3GeneralS2Sq}}(\lambda_{X_7}^{(1)})^*$	0	0	4
$\lambda_{X_7}^{(1)}(\mu_{\text{T3GeneralS2Sq}})^*$	0	0	4
$g_1^2 m_1^2$	0	-6	-6
$m_1^2(2T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$	0	0	0
$g_2^2 m_1^2$	-12	-12	-12
$\mu_{\text{T3GeneralS1S2}}(\lambda_{11})^*$	0	0	0
$\lambda_{11}(\mu_{\text{T3GeneralS1S2}})^*$	0	0	0
$m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$	0	0	0
$m_1^2(2T_\nu^{(1)} + 2\,\text{Tr}(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger) + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	1	0	0
$\mu_{\text{T3GeneralS1Sq}}(4\,\text{Tr}(y_1(y_1^{\text{conj}})^\dagger) + 20(\lambda_{X_9})^*)$	1	0	0
$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	0	1	1
$m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0
$\mu_{\text{T3GeneralS2Sq}}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}(\mu_{\text{T3GeneralS2Sq}})^*$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$m_1^2(4T_\nu^{(1)} + 4\,\text{Tr}(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger) + 4\lambda_{S_1}^{(1)})$	0	0	0
$\mu_{\text{T3GeneralS1Sq}}(8\,\text{Tr}(y_1(y_1^{\text{conj}})^\dagger) + 12(\lambda_{X_9})^*)$	0	0	0
$(\mu_{\text{T3GeneralS1Sq}})^*\,\text{Tr}(y_1^{\text{conj}}(y_1)^\dagger)$	4	0	0
$\lambda_{X_9}(\mu_{\text{T3GeneralS1Sq}})^*$	20	0	0
$\lambda_{12}^{(1)} m_2^2$	0	0	0
$T_\nu^{(1)}$	-4	-4	-4
T_ν	-4	0	0
$m_1^2(4T_\nu^{(1)} + 4\lambda_{S_1}^{(1)})$	0	0	0
$\mu_{\text{T3GeneralS2Sq}}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_7}^{(1)}(\mu_{\text{T3GeneralS2Sq}})^*$	0	0	12
$g_1^2 m_1^2$	0	-6	-6
$m_1^2(2T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$	0	0	0
$g_2^2 m_1^2$	-12	-12	-12
$\mu_{\text{T3GeneralS1S2}}(\lambda_{11})^*$	0	0	0
$\lambda_{11}(\mu_{\text{T3GeneralS1S2}})^*$	0	0	0
$m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$	0	0	0

$m_1^2(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	1	0	0
$\mu_{\mathrm{T3GeneralS1Sq}}(4\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 20(\lambda_{X_9})^*)$	1	0	0
$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	0	1	1
$m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	1	1	1
$\mu_{\mathrm{T3GeneralS2Sq}}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	4
$\lambda_{X_7}^{(2)}(\mu_{\mathrm{T3GeneralS2Sq}})^*$	0	0	4

$\mathcal{S}$	E ($\alpha = -4$)
$m_1^2(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 4\lambda_{S_1}^{(1)})$	0
$\mu_{\mathrm{T3GeneralS1Sq}}(8\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 12(\lambda_{X_9})^*)$	0
$(\mu_{\mathrm{T3GeneralS1Sq}})^*\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)$	0
$\lambda_{X_9}(\mu_{\mathrm{T3GeneralS1Sq}})^*$	0
$\lambda_{12}^{(1)} m_2^2$	0
$T_\nu^{(1)}$	-4
T_ν	0
$m_1^2(4T_\nu^{(1)} + 4\lambda_{S_1}^{(1)})$	0
$\mu_{\mathrm{T3GeneralS2Sq}}(\lambda_{X_7}^{(1)})^*$	0
$\lambda_{X_7}^{(1)}(\mu_{\mathrm{T3GeneralS2Sq}})^*$	0
$g_1^2 m_1^2$	-24
$m_1^2(2T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$	0
$g_2^2 m_1^2$	-12
$\mu_{\mathrm{T3GeneralS1S2}}(\lambda_{11})^*$	0
$\lambda_{11}(\mu_{\mathrm{T3GeneralS1S2}})^*$	0
$m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$	0
$m_1^2(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger) + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	0
$\mu_{\mathrm{T3GeneralS1Sq}}(4\,\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger) + 20(\lambda_{X_9})^*)$	0
$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	1
$m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	1
$\mu_{\mathrm{T3GeneralS2Sq}}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0
$\lambda_{X_7}^{(2)}(\mu_{\mathrm{T3GeneralS2Sq}})^*$	0

$\beta_{m_2^2}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$m_2^2(2T_\nu^{(2)} + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$	1	0	1
$\mu_{\mathrm{T3GeneralS1Sq}}(\lambda_{X_5}^{(1)})^*$	4	0	0
$\lambda_{X_5}^{(1)}(\mu_{\mathrm{T3GeneralS1Sq}})^*$	4	0	0
$g_2^2 m_2^2$	-12	-12	-12
$g_1^2 m_2^2$	-6	0	-6
$\lambda_{12}^{(1)} m_1^2$	2	2	2
$T_\nu^{(2)}$	-4	-4	-4
$m_2^2(2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$	0	1	0
$\mu_{\mathrm{T3GeneralS2Sq}}(4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 20(\lambda_{X_2})^*)$	0	1	0
$(\mu_{\mathrm{T3GeneralS2Sq}})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	4	0
$\lambda_{X_2}(\mu_{\mathrm{T3GeneralS2Sq}})^*$	0	20	0
T_ν	0	-4	0
$m_2^2(2T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$	0	0	0
$\mu_{\mathrm{T3GeneralS1S2}}(\lambda_{10})^*$	0	0	0
$\lambda_{10}(\mu_{\mathrm{T3GeneralS1S2}})^*$	0	0	0
$m_2^2(3T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$	0	0	0
$m_2^2(4T_\nu^{(2)} + 4\lambda_{S_2}^{(1)})$	0	0	0

$m_2^2(4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 4\lambda_{S_2}^{(1)})$	0	0	0
$\mu_{\mathrm{T3GeneralS2Sq}}(8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 12(\lambda_{X_2})^*)$	0	0	0
$\mu_{\mathrm{T3GeneralS1Sq}}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}(\mu_{\mathrm{T3GeneralS1Sq}})^*$	0	0	0
$m_1^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$m_2^2(2T_\nu^{(2)} + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$	0	0	0
$\mu_{\mathrm{T3GeneralS1Sq}}(\lambda_{X_5}^{(1)})^*$	0	0	0
$\lambda_{X_5}^{(1)}(\mu_{\mathrm{T3GeneralS1Sq}})^*$	0	0	0
$g_2^2 m_2^2$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2 m_2^2$	$-\frac{27}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$\lambda_{12}^{(1)} m_1^2$	4	4	4
$T_\nu^{(2)}$	-4	-16	-4
$m_2^2(2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$	0	0	0
$\mu_{\mathrm{T3GeneralS2Sq}}(4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 20(\lambda_{X_2})^*)$	0	0	0
$(\mu_{\mathrm{T3GeneralS2Sq}})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{X_2}(\mu_{\mathrm{T3GeneralS2Sq}})^*$	0	0	0
T_ν	0	0	0
$m_2^2(2T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$	1	1	1
$\mu_{\mathrm{T3GeneralS1S2}}(\lambda_{10})^*$	0	6	0
$\lambda_{10}(\mu_{\mathrm{T3GeneralS1S2}})^*$	0	6	0
$m_2^2(3T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$	0	0	0
$m_2^2(4T_\nu^{(2)} + 4\lambda_{S_2}^{(1)})$	0	0	0
$m_2^2(4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 4\lambda_{S_2}^{(1)})$	0	0	0
$\mu_{\mathrm{T3GeneralS2Sq}}(8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 12(\lambda_{X_2})^*)$	0	0	0
$\mu_{\mathrm{T3GeneralS1Sq}}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}(\mu_{\mathrm{T3GeneralS1Sq}})^*$	0	0	0
$m_1^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$m_2^2(2T_\nu^{(2)} + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$	0	0	0
$\mu_{\mathrm{T3GeneralS1Sq}}(\lambda_{X_5}^{(1)})^*$	0	0	0
$\lambda_{X_5}^{(1)}(\mu_{\mathrm{T3GeneralS1Sq}})^*$	0	0	0
$g_2^2 m_2^2$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2 m_2^2$	$-\frac{27}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$\lambda_{12}^{(1)} m_1^2$	4	4	4
$T_\nu^{(2)}$	-6	-24	-6
$m_2^2(2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$	0	0	0
$\mu_{\mathrm{T3GeneralS2Sq}}(4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 20(\lambda_{X_2})^*)$	0	0	0
$(\mu_{\mathrm{T3GeneralS2Sq}})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	0
$\lambda_{X_2}(\mu_{\mathrm{T3GeneralS2Sq}})^*$	0	0	0
T_ν	0	0	0
$m_2^2(2T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$	0	0	0
$\mu_{\mathrm{T3GeneralS1S2}}(\lambda_{10})^*$	0	6	0
$\lambda_{10}(\mu_{\mathrm{T3GeneralS1S2}})^*$	0	6	0
$m_2^2(3T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$	1	1	1
$m_2^2(4T_\nu^{(2)} + 4\lambda_{S_2}^{(1)})$	0	0	0
$m_2^2(4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger) + 4\lambda_{S_2}^{(1)})$	0	0	0
$\mu_{\mathrm{T3GeneralS2Sq}}(8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger) + 12(\lambda_{X_2})^*)$	0	0	0
$\mu_{\mathrm{T3GeneralS1Sq}}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0

$\lambda^{(2)}_{X_5}(\mu_{\mathrm{T3GeneralS1Sq}})^*$	0	0	0
$m_1^2(6\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12})$	0	0	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$m_2^2(2T_\nu^{(2)}+8\lambda^{(1)}_{S_2}+4\lambda^{(A)}_{S_2})$	0	0	0
$\mu_{\mathrm{T3GeneralS1Sq}}(\lambda^{(1)}_{X_5})^*$	12	0	0
$\lambda^{(1)}_{X_5}(\mu_{\mathrm{T3GeneralS1Sq}})^*$	12	0	0
$g_2^2m_2^2$	0	0	0
$g_1^2m_2^2$	-6	-24	0
$\lambda^{(1)}_{12}m_1^2$	6	6	6
$T_\nu^{(2)}$	-8	-8	-8
$m_2^2(2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+8\lambda^{(1)}_{S_2}+4\lambda^{(A)}_{S_2})$	0	0	0
$\mu_{\mathrm{T3GeneralS2Sq}}(4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger)+20(\lambda_{X_2})^*)$	0	0	0
$(\mu_{\mathrm{T3GeneralS2Sq}})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	8
$\lambda_{X_2}(\mu_{\mathrm{T3GeneralS2Sq}})^*$	0	0	12
T_ν	0	0	-8
$m_2^2(2T_\nu^{(2)}+6\lambda^{(1)}_{S_2})$	0	0	0
$\mu_{\mathrm{T3GeneralS1S2}}(\lambda_{10})^*$	0	0	0
$\lambda_{10}(\mu_{\mathrm{T3GeneralS1S2}})^*$	0	0	0
$m_2^2(3T_\nu^{(2)}+6\lambda^{(1)}_{S_2})$	0	0	0
$m_2^2(4T_\nu^{(2)}+4\lambda^{(1)}_{S_2})$	1	1	0
$m_2^2(4T_\nu^{(2)}+4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+4\lambda^{(1)}_{S_2})$	0	0	1
$\mu_{\mathrm{T3GeneralS2Sq}}(8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger)+12(\lambda_{X_2})^*)$	0	0	1
$\mu_{\mathrm{T3GeneralS1Sq}}(3(\lambda^{(1)}_{X_5})^*+(\lambda^{(2)}_{X_5})^*)$	0	0	0
$\lambda^{(2)}_{X_5}(\mu_{\mathrm{T3GeneralS1Sq}})^*$	0	0	0
$m_1^2(6\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12})$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$m_2^2(2T_\nu^{(2)}+8\lambda^{(1)}_{S_2}+4\lambda^{(A)}_{S_2})$	1	1	0
$\mu_{\mathrm{T3GeneralS1Sq}}(\lambda^{(1)}_{X_5})^*$	0	0	0
$\lambda^{(1)}_{X_5}(\mu_{\mathrm{T3GeneralS1Sq}})^*$	12	0	0
$g_2^2m_2^2$	-12	-12	-12
$g_1^2m_2^2$	-6	-24	0
$\lambda^{(1)}_{12}m_1^2$	0	0	0
$T_\nu^{(2)}$	-4	-4	-4
$m_2^2(2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+8\lambda^{(1)}_{S_2}+4\lambda^{(A)}_{S_2})$	0	0	1
$\mu_{\mathrm{T3GeneralS2Sq}}(4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger)+20(\lambda_{X_2})^*)$	0	0	1
$(\mu_{\mathrm{T3GeneralS2Sq}})^*\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)$	0	0	4
$\lambda_{X_2}(\mu_{\mathrm{T3GeneralS2Sq}})^*$	0	0	20
T_ν	0	0	-4
$m_2^2(2T_\nu^{(2)}+6\lambda^{(1)}_{S_2})$	0	0	0
$\mu_{\mathrm{T3GeneralS1S2}}(\lambda_{10})^*$	0	0	0
$\lambda_{10}(\mu_{\mathrm{T3GeneralS1S2}})^*$	0	0	0
$m_2^2(3T_\nu^{(2)}+6\lambda^{(1)}_{S_2})$	0	0	0
$m_2^2(4T_\nu^{(2)}+4\lambda^{(1)}_{S_2})$	0	0	0
$m_2^2(4T_\nu^{(2)}+4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger)+4\lambda^{(1)}_{S_2})$	0	0	0
$\mu_{\mathrm{T3GeneralS2Sq}}(8\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger)+12(\lambda_{X_2})^*)$	0	0	0
$\mu_{\mathrm{T3GeneralS1Sq}}(3(\lambda^{(1)}_{X_5})^*+(\lambda^{(2)}_{X_5})^*)$	4	0	0
$\lambda^{(2)}_{X_5}(\mu_{\mathrm{T3GeneralS1Sq}})^*$	4	0	0
$m_1^2(6\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12})$	1	1	1

$\mathcal{S}$	E ($\alpha=-4$)
$m_2^2(2T_\nu^{(2)}+8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$	1
$\mu_{\text{T3GeneralS1Sq}}(\lambda_{X_5}^{(1)})^*$	0
$\lambda_{X_5}^{(1)}(\mu_{\text{T3GeneralS1Sq}})^*$	0
$g_2^2m_2^2$	-12
$g_1^2m_2^2$	-6
$\lambda_{12}^{(1)}m_1^2$	0
$T_\nu^{(2)}$	-4
$m_2^2(2T_\nu^{(2)}+2\,\text{Tr}(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger)+8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$	0
$\mu_{\text{T3GeneralS2Sq}}(4\,\text{Tr}(y_2^{\text{conj}}(y_2)^\dagger)+20(\lambda_{X_2})^*)$	0
$(\mu_{\text{T3GeneralS2Sq}})^*\,\text{Tr}(y_2(y_2^{\text{conj}})^\dagger)$	0
$\lambda_{X_2}(\mu_{\text{T3GeneralS2Sq}})^*$	0
T_ν	0
$m_2^2(2T_\nu^{(2)}+6\lambda_{S_2}^{(1)})$	0
$\mu_{\text{T3GeneralS1S2}}(\lambda_{10})^*$	0
$\lambda_{10}(\mu_{\text{T3GeneralS1S2}})^*$	0
$m_2^2(3T_\nu^{(2)}+6\lambda_{S_2}^{(1)})$	0
$m_2^2(4T_\nu^{(2)}+4\lambda_{S_2}^{(1)})$	0
$m_2^2(4T_\nu^{(2)}+4\,\text{Tr}(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger)+4\lambda_{S_2}^{(1)})$	0
$\mu_{\text{T3GeneralS2Sq}}(8\,\text{Tr}(y_2^{\text{conj}}(y_2)^\dagger)+12(\lambda_{X_2})^*)$	0
$\mu_{\text{T3GeneralS1Sq}}(3(\lambda_{X_5}^{(1)})^*+(\lambda_{X_5}^{(2)})^*)$	0
$\lambda_{X_5}^{(2)}(\mu_{\text{T3GeneralS1Sq}})^*$	0
$m_1^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$	1

$\beta^{(1)}_{\mu_{\text{T3GeneralS1S2}}}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\mu_{\text{T3GeneralS1S2}}(T_\nu^{(1)}+T_\nu^{(2)})$	0	0	0
$\lambda_9(\mu_{\text{T3GeneralS1S2}})^*$	0	0	0
$g_2^2\mu_{\text{T3GeneralS1S2}}$	0	0	0
$g_1^2\mu_{\text{T3GeneralS1S2}}$	0	0	0
$\lambda_{12}^{(1)}\mu_{\text{T3GeneralS1S2}}$	0	0	0
$\lambda_{12}^{(A)}\mu_{\text{T3GeneralS1S2}}$	0	0	0
$\lambda_{10}m_2^2$	0	0	0
$\lambda_{11}m_1^2$	0	0	0
$\mu_{\text{T3GeneralS1S2}}(\frac{3}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)})$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$\mu_{\text{T3GeneralS1S2}}(T_\nu^{(1)}+T_\nu^{(2)})$	0	1	0
$\lambda_9(\mu_{\text{T3GeneralS1S2}})^*$	0	12	0
$g_2^2\mu_{\text{T3GeneralS1S2}}$	0	$-\frac{9}{2}$	0
$g_1^2\mu_{\text{T3GeneralS1S2}}$	0	$-\frac{3}{2}$	0
$\lambda_{12}^{(1)}\mu_{\text{T3GeneralS1S2}}$	0	2	0
$\lambda_{12}^{(A)}\mu_{\text{T3GeneralS1S2}}$	0	$-\frac{3}{2}$	0
$\lambda_{10}m_2^2$	0	6	0
$\lambda_{11}m_1^2$	0	6	0
$\mu_{\text{T3GeneralS1S2}}(\frac{3}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)})$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)

$\mu_{\text{T3GeneralS1S2}}(T_\nu^{(1)} + T_\nu^{(2)})$	0	0	0
$\lambda_9(\mu_{\text{T3GeneralS1S2}})^*$	0	12	0
$g_2^2\mu_{\text{T3GeneralS1S2}}$	0	$-\frac{9}{2}$	0
$g_1^2\mu_{\text{T3GeneralS1S2}}$	0	$-\frac{3}{2}$	0
$\lambda_{12}^{(1)}\mu_{\text{T3GeneralS1S2}}$	0	2	0
$\lambda_{12}^{(A)}\mu_{\text{T3GeneralS1S2}}$	0	$-\frac{3}{2}$	0
$\lambda_{10}m_2^2$	0	6	0
$\lambda_{11}m_1^2$	0	6	0
$\mu_{\text{T3GeneralS1S2}}(\frac{3}{2}T_\nu^{(1)} + \frac{3}{2}T_\nu^{(2)})$	0	1	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\mu_{\text{T3GeneralS1S2}}(T_\nu^{(1)} + T_\nu^{(2)})$	0	0	0
$\lambda_9(\mu_{\text{T3GeneralS1S2}})^*$	0	0	0
$g_2^2\mu_{\text{T3GeneralS1S2}}$	0	0	0
$g_1^2\mu_{\text{T3GeneralS1S2}}$	0	0	0
$\lambda_{12}^{(1)}\mu_{\text{T3GeneralS1S2}}$	0	0	0
$\lambda_{12}^{(A)}\mu_{\text{T3GeneralS1S2}}$	0	0	0
$\lambda_{10}m_2^2$	0	0	0
$\lambda_{11}m_1^2$	0	0	0
$\mu_{\text{T3GeneralS1S2}}(\frac{3}{2}T_\nu^{(1)} + \frac{3}{2}T_\nu^{(2)})$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\mu_{\text{T3GeneralS1S2}}(T_\nu^{(1)} + T_\nu^{(2)})$	0	0	0
$\lambda_9(\mu_{\text{T3GeneralS1S2}})^*$	0	0	0
$g_2^2\mu_{\text{T3GeneralS1S2}}$	0	0	0
$g_1^2\mu_{\text{T3GeneralS1S2}}$	0	0	0
$\lambda_{12}^{(1)}\mu_{\text{T3GeneralS1S2}}$	0	0	0
$\lambda_{12}^{(A)}\mu_{\text{T3GeneralS1S2}}$	0	0	0
$\lambda_{10}m_2^2$	0	0	0
$\lambda_{11}m_1^2$	0	0	0
$\mu_{\text{T3GeneralS1S2}}(\frac{3}{2}T_\nu^{(1)} + \frac{3}{2}T_\nu^{(2)})$	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$\mu_{\text{T3GeneralS1S2}}(T_\nu^{(1)} + T_\nu^{(2)})$	0
$\lambda_9(\mu_{\text{T3GeneralS1S2}})^*$	0
$g_2^2\mu_{\text{T3GeneralS1S2}}$	0
$g_1^2\mu_{\text{T3GeneralS1S2}}$	0
$\lambda_{12}^{(1)}\mu_{\text{T3GeneralS1S2}}$	0
$\lambda_{12}^{(A)}\mu_{\text{T3GeneralS1S2}}$	0
$\lambda_{10}m_2^2$	0
$\lambda_{11}m_1^2$	0
$\mu_{\text{T3GeneralS1S2}}(\frac{3}{2}T_\nu^{(1)} + \frac{3}{2}T_\nu^{(2)})$	0

$\beta^{(1)}_{\mu_{\text{T3GeneralS1Sq}}}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$m_1^2(4\,\text{Tr}(y_1^{\text{conj}}(y_1)^\dagger) + 6\lambda_{X_9})$	1	0	0
$\text{Tr}(M_F(M_F)^\dagger(y_1^{\text{conj}})^T(y_1)^*)$	-8	0	0
$\mu_{\text{T3GeneralS1Sq}}(4T_\nu^{(1)} + 4\,\text{Tr}(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger))$	1	0	0

$\lambda_{X_6}(\mu_{\mathrm{T3GeneralS1Sq}})^*$	24	0	0
$\lambda^{(1)}_{S_1}\mu_{\mathrm{T3GeneralS1Sq}}$	2	0	0
$\lambda^{(1)}_{X_5} m_2^2$	6	0	0
$m_1^2(2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger) + 10\lambda_{X_9})$	0	0	0
$\mu_{\mathrm{T3GeneralS1Sq}}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0	0	0
$g_2^2\mu_{\mathrm{T3GeneralS1Sq}}$	0	0	0
$\lambda^{(A)}_{S_1}\mu_{\mathrm{T3GeneralS1Sq}}$	0	0	0
$m_2^2(6\lambda^{(1)}_{X_5} + 2\lambda^{(2)}_{X_5})$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$m_1^2(4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger) + 6\lambda_{X_9})$	0	0	0
$\mathrm{Tr}(M_F(M_F)^\dagger(y_1^{\mathrm{conj}})^T(y_1)^*)$	0	0	0
$\mu_{\mathrm{T3GeneralS1Sq}}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_6}(\mu_{\mathrm{T3GeneralS1Sq}})^*$	0	0	0
$\lambda^{(1)}_{S_1}\mu_{\mathrm{T3GeneralS1Sq}}$	0	0	0
$\lambda^{(1)}_{X_5} m_2^2$	0	0	0
$m_1^2(2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger) + 10\lambda_{X_9})$	0	0	0
$\mu_{\mathrm{T3GeneralS1Sq}}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0	0	0
$g_2^2\mu_{\mathrm{T3GeneralS1Sq}}$	0	0	0
$\lambda^{(A)}_{S_1}\mu_{\mathrm{T3GeneralS1Sq}}$	0	0	0
$m_2^2(6\lambda^{(1)}_{X_5} + 2\lambda^{(2)}_{X_5})$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$m_1^2(4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger) + 6\lambda_{X_9})$	0	0	0
$\mathrm{Tr}(M_F(M_F)^\dagger(y_1^{\mathrm{conj}})^T(y_1)^*)$	0	0	0
$\mu_{\mathrm{T3GeneralS1Sq}}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_6}(\mu_{\mathrm{T3GeneralS1Sq}})^*$	0	0	0
$\lambda^{(1)}_{S_1}\mu_{\mathrm{T3GeneralS1Sq}}$	0	0	0
$\lambda^{(1)}_{X_5} m_2^2$	0	0	0
$m_1^2(2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger) + 10\lambda_{X_9})$	0	0	0
$\mu_{\mathrm{T3GeneralS1Sq}}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0	0	0
$g_2^2\mu_{\mathrm{T3GeneralS1Sq}}$	0	0	0
$\lambda^{(A)}_{S_1}\mu_{\mathrm{T3GeneralS1Sq}}$	0	0	0
$m_2^2(6\lambda^{(1)}_{X_5} + 2\lambda^{(2)}_{X_5})$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$m_1^2(4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger) + 6\lambda_{X_9})$	0	0	0
$\mathrm{Tr}(M_F(M_F)^\dagger(y_1^{\mathrm{conj}})^T(y_1)^*)$	-4	0	0
$\mu_{\mathrm{T3GeneralS1Sq}}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_6}(\mu_{\mathrm{T3GeneralS1Sq}})^*$	40	0	0
$\lambda^{(1)}_{S_1}\mu_{\mathrm{T3GeneralS1Sq}}$	2	0	0
$\lambda^{(1)}_{X_5} m_2^2$	2	0	0
$m_1^2(2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger) + 10\lambda_{X_9})$	1	0	0
$\mu_{\mathrm{T3GeneralS1Sq}}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	1	0	0
$g_2^2\mu_{\mathrm{T3GeneralS1Sq}}$	-12	0	0
$\lambda^{(A)}_{S_1}\mu_{\mathrm{T3GeneralS1Sq}}$	-4	0	0
$m_2^2(6\lambda^{(1)}_{X_5} + 2\lambda^{(2)}_{X_5})$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$m_1^2(4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger) + 6\lambda_{X_9})$	0	0	0
$\mathrm{Tr}(M_F(M_F)^\dagger(y_1^{\mathrm{conj}})^T(y_1)^*)$	-4	0	0
$\mu_{\mathrm{T3GeneralS1Sq}}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_6}(\mu_{\mathrm{T3GeneralS1Sq}})^*$	40	0	0
$\lambda_{S_1}^{(1)}\mu_{\mathrm{T3GeneralS1Sq}}$	2	0	0
$\lambda_{X_5}^{(1)}m_2^2$	0	0	0
$m_1^2(2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger) + 10\lambda_{X_9})$	1	0	0
$\mu_{\mathrm{T3GeneralS1Sq}}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	1	0	0
$g_2^2\mu_{\mathrm{T3GeneralS1Sq}}$	-12	0	0
$\lambda_{S_1}^{(A)}\mu_{\mathrm{T3GeneralS1Sq}}$	-4	0	0
$m_2^2(6\lambda_{X_5}^{(1)} + 2\lambda_{X_5}^{(2)})$	1	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$m_1^2(4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger) + 6\lambda_{X_9})$	0
$\mathrm{Tr}(M_F(M_F)^\dagger(y_1^{\mathrm{conj}})^T(y_1)^*)$	0
$\mu_{\mathrm{T3GeneralS1Sq}}(4T_\nu^{(1)} + 4\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0
$\lambda_{X_6}(\mu_{\mathrm{T3GeneralS1Sq}})^*$	0
$\lambda_{S_1}^{(1)}\mu_{\mathrm{T3GeneralS1Sq}}$	0
$\lambda_{X_5}^{(1)}m_2^2$	0
$m_1^2(2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger) + 10\lambda_{X_9})$	0
$\mu_{\mathrm{T3GeneralS1Sq}}(2T_\nu^{(1)} + 2\,\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger))$	0
$g_2^2\mu_{\mathrm{T3GeneralS1Sq}}$	0
$\lambda_{S_1}^{(A)}\mu_{\mathrm{T3GeneralS1Sq}}$	0
$m_2^2(6\lambda_{X_5}^{(1)} + 2\lambda_{X_5}^{(2)})$	0

$\beta^{(1)}_{\mu_{\mathrm{T3GeneralS2Sq}}}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$m_2^2(2\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger) + 10\lambda_{X_2})$	0	1	0
$\mathrm{Tr}(M_F(y_2^{\mathrm{conj}})^\dagger y_2(M_F)^\dagger)$	0	-4	0
$\mu_{\mathrm{T3GeneralS2Sq}}(2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	1	0
$\lambda_{X_1}(\mu_{\mathrm{T3GeneralS2Sq}})^*$	0	40	0
$g_2^2\mu_{\mathrm{T3GeneralS2Sq}}$	0	-12	0
$\lambda_{S_2}^{(1)}\mu_{\mathrm{T3GeneralS2Sq}}$	0	2	0
$\lambda_{S_2}^{(A)}\mu_{\mathrm{T3GeneralS2Sq}}$	0	-4	0
$\lambda_{X_7}^{(1)}m_1^2$	0	2	0
$m_2^2(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger) + 6\lambda_{X_2})$	0	0	0
$\mu_{\mathrm{T3GeneralS2Sq}}(4T_\nu^{(2)} + 4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$m_1^2(6\lambda_{X_7}^{(1)} + 2\lambda_{X_7}^{(2)})$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$m_2^2(2\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger) + 10\lambda_{X_2})$	0	0	0
$\mathrm{Tr}(M_F(y_2^{\mathrm{conj}})^\dagger y_2(M_F)^\dagger)$	0	0	0
$\mu_{\mathrm{T3GeneralS2Sq}}(2T_\nu^{(2)} + 2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_1}(\mu_{\mathrm{T3GeneralS2Sq}})^*$	0	0	0
$g_2^2\mu_{\mathrm{T3GeneralS2Sq}}$	0	0	0
$\lambda_{S_2}^{(1)}\mu_{\mathrm{T3GeneralS2Sq}}$	0	0	0
$\lambda_{S_2}^{(A)}\mu_{\mathrm{T3GeneralS2Sq}}$	0	0	0
$\lambda_{X_7}^{(1)}m_1^2$	0	0	0

$m_2^2(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+6\lambda_{X_2})$	0	0	0
$\mu_{\mathrm{T3GeneralS2Sq}}(4T_\nu^{(2)}+4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$m_1^2(6\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$m_2^2(2\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+10\lambda_{X_2})$	0	0	0
$\mathrm{Tr}(M_F(y_2^{\mathrm{conj}})^\dagger y_2(M_F)^\dagger)$	0	0	0
$\mu_{\mathrm{T3GeneralS2Sq}}(2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_1}(\mu_{\mathrm{T3GeneralS2Sq}})^*$	0	0	0
$g_2^2\mu_{\mathrm{T3GeneralS2Sq}}$	0	0	0
$\lambda_{S_2}^{(1)}\mu_{\mathrm{T3GeneralS2Sq}}$	0	0	0
$\lambda_{S_2}^{(A)}\mu_{\mathrm{T3GeneralS2Sq}}$	0	0	0
$\lambda_{X_7}^{(1)}m_1^2$	0	0	0
$m_2^2(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+6\lambda_{X_2})$	0	0	0
$\mu_{\mathrm{T3GeneralS2Sq}}(4T_\nu^{(2)}+4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$m_1^2(6\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$m_2^2(2\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+10\lambda_{X_2})$	0	0	0
$\mathrm{Tr}(M_F(y_2^{\mathrm{conj}})^\dagger y_2(M_F)^\dagger)$	0	0	-8
$\mu_{\mathrm{T3GeneralS2Sq}}(2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$\lambda_{X_1}(\mu_{\mathrm{T3GeneralS2Sq}})^*$	0	0	24
$g_2^2\mu_{\mathrm{T3GeneralS2Sq}}$	0	0	0
$\lambda_{S_2}^{(1)}\mu_{\mathrm{T3GeneralS2Sq}}$	0	0	2
$\lambda_{S_2}^{(A)}\mu_{\mathrm{T3GeneralS2Sq}}$	0	0	0
$\lambda_{X_7}^{(1)}m_1^2$	0	0	6
$m_2^2(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+6\lambda_{X_2})$	0	0	1
$\mu_{\mathrm{T3GeneralS2Sq}}(4T_\nu^{(2)}+4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	1
$m_1^2(6\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$m_2^2(2\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+10\lambda_{X_2})$	0	0	1
$\mathrm{Tr}(M_F(y_2^{\mathrm{conj}})^\dagger y_2(M_F)^\dagger)$	0	0	-4
$\mu_{\mathrm{T3GeneralS2Sq}}(2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	1
$\lambda_{X_1}(\mu_{\mathrm{T3GeneralS2Sq}})^*$	0	0	40
$g_2^2\mu_{\mathrm{T3GeneralS2Sq}}$	0	0	-12
$\lambda_{S_2}^{(1)}\mu_{\mathrm{T3GeneralS2Sq}}$	0	0	2
$\lambda_{S_2}^{(A)}\mu_{\mathrm{T3GeneralS2Sq}}$	0	0	-4
$\lambda_{X_7}^{(1)}m_1^2$	0	0	0
$m_2^2(4\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+6\lambda_{X_2})$	0	0	0
$\mu_{\mathrm{T3GeneralS2Sq}}(4T_\nu^{(2)}+4\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0	0	0
$m_1^2(6\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	1

$\mathcal{S}$	E ($\alpha=-4$)
$m_2^2(2\,\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)+10\lambda_{X_2})$	0
$\mathrm{Tr}(M_F(y_2^{\mathrm{conj}})^\dagger y_2(M_F)^\dagger)$	0
$\mu_{\mathrm{T3GeneralS2Sq}}(2T_\nu^{(2)}+2\,\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger))$	0
$\lambda_{X_1}(\mu_{\mathrm{T3GeneralS2Sq}})^*$	0
$g_2^2\mu_{\mathrm{T3GeneralS2Sq}}$	0

$\lambda_{S_2}^{(1)}\mu_{\text{T3GeneralS2Sq}}$	0
$\lambda_{S_2}^{(A)}\mu_{\text{T3GeneralS2Sq}}$	0
$\lambda_{X_7}^{(1)}m_1^2$	0
$m_2^2(4\,\text{Tr}(y_2(y_2^{\text{conj}})^\dagger)+6\lambda_{X_2})$	0
$\mu_{\text{T3GeneralS2Sq}}(4T_\nu^{(2)}+4\,\text{Tr}(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger))$	0
$m_1^2(6\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0

$\beta_{y_1}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\text{Tr}(y_1(y_1^{\text{conj}})^\dagger)(y_1^{\text{conj}})_{ij}$	4	0	0
$T_\nu^{(1)}(y_1)_{ij}$	2	2	2
$(y_1(y_1^{\text{conj}})^\dagger y_1^{\text{conj}})_{ij}$	$\frac{5}{2}$	0	0
$(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger y_1)_{ij}$	$\frac{5}{2}$	0	0
$(y_2(y_2)^\dagger y_1)_{ij}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$(Y_e(Y_e)^\dagger y_1)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(y_1(y_1)^\dagger y_1)_{ij}$	1	1	1
$g_2^2(y_1)_{ij}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2(y_1)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{15}{2}$
$T_\nu(y_1)_{ij}$	2	0	0
$(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger y_1)_{ij}$	0	$\frac{3}{4}$	0
$\text{Tr}(y_1(y_2)^\dagger)(y_2)_{ij}$	0	0	0
$(y_1(y_2)^\dagger y_2)_{ij}$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$\text{Tr}(y_1(y_1^{\text{conj}})^\dagger)(y_1^{\text{conj}})_{ij}$	0	0	0
$T_\nu^{(1)}(y_1)_{ij}$	1	1	1
$(y_1(y_1^{\text{conj}})^\dagger y_1^{\text{conj}})_{ij}$	0	0	0
$(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger y_1)_{ij}$	0	0	0
$(y_2(y_2)^\dagger y_1)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(Y_e(Y_e)^\dagger y_1)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(y_1(y_1)^\dagger y_1)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(y_1)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_1^2(y_1)_{ij}$	$-\frac{15}{4}$	$-\frac{3}{4}$	$-\frac{15}{4}$
$T_\nu(y_1)_{ij}$	0	0	0
$(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger y_1)_{ij}$	0	0	0
$\text{Tr}(y_1(y_2)^\dagger)(y_2)_{ij}$	0	1	0
$(y_1(y_2)^\dagger y_2)_{ij}$	0	1	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\text{Tr}(y_1(y_1^{\text{conj}})^\dagger)(y_1^{\text{conj}})_{ij}$	0	0	0
$T_\nu^{(1)}(y_1)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$(y_1(y_1^{\text{conj}})^\dagger y_1^{\text{conj}})_{ij}$	0	0	0
$(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger y_1)_{ij}$	0	0	0
$(y_2(y_2)^\dagger y_1)_{ij}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$(Y_e(Y_e)^\dagger y_1)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(y_1(y_1)^\dagger y_1)_{ij}$	$\frac{5}{4}$	$\frac{5}{4}$	$\frac{5}{4}$
$g_2^2(y_1)_{ij}$	$-\frac{33}{4}$	$-\frac{33}{4}$	$-\frac{33}{4}$
$g_1^2(y_1)_{ij}$	$-\frac{15}{4}$	$-\frac{3}{4}$	$-\frac{15}{4}$
$T_\nu(y_1)_{ij}$	0	0	0
$(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger y_1)_{ij}$	0	0	0

$\mathrm{Tr}(y_1(y_2)^\dagger)(y_2)_{ij}$	0	$\frac{3}{2}$	0
$(y_1(y_2)^\dagger y_2)_{ij}$	0	$\frac{1}{2}$	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger)(y_1^{\mathrm{conj}})_{ij}$	2	0	0
$T_\nu^{(1)}(y_1)_{ij}$	1	1	1
$(y_1(y_1^{\mathrm{conj}})^\dagger y_1^{\mathrm{conj}})_{ij}$	$\frac{29}{12}$	0	0
$(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger y_1)_{ij}$	$\frac{29}{12}$	0	0
$(y_2(y_2)^\dagger y_1)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(Y_e(Y_e)^\dagger y_1)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(y_1(y_1)^\dagger y_1)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(y_1)_{ij}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2(y_1)_{ij}$	$-\frac{3}{2}$	$-\frac{15}{2}$	$-\frac{3}{2}$
$T_\nu(y_1)_{ij}$	1	0	0
$(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger y_1)_{ij}$	0	0	$\frac{1}{2}$
$\mathrm{Tr}(y_1(y_2)^\dagger)(y_2)_{ij}$	0	0	0
$(y_1(y_2)^\dagger y_2)_{ij}$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger)(y_1^{\mathrm{conj}})_{ij}$	2	0	0
$T_\nu^{(1)}(y_1)_{ij}$	1	1	1
$(y_1(y_1^{\mathrm{conj}})^\dagger y_1^{\mathrm{conj}})_{ij}$	$\frac{29}{12}$	0	0
$(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger y_1)_{ij}$	$\frac{29}{12}$	0	0
$(y_2(y_2)^\dagger y_1)_{ij}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$(Y_e(Y_e)^\dagger y_1)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(y_1(y_1)^\dagger y_1)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(y_1)_{ij}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2(y_1)_{ij}$	$-\frac{3}{2}$	$-\frac{15}{2}$	$-\frac{3}{2}$
$T_\nu(y_1)_{ij}$	1	0	0
$(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger y_1)_{ij}$	0	0	$\frac{3}{4}$
$\mathrm{Tr}(y_1(y_2)^\dagger)(y_2)_{ij}$	0	0	0
$(y_1(y_2)^\dagger y_2)_{ij}$	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$\mathrm{Tr}(y_1(y_1^{\mathrm{conj}})^\dagger)(y_1^{\mathrm{conj}})_{ij}$	0
$T_\nu^{(1)}(y_1)_{ij}$	1
$(y_1(y_1^{\mathrm{conj}})^\dagger y_1^{\mathrm{conj}})_{ij}$	0
$(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger y_1)_{ij}$	0
$(y_2(y_2)^\dagger y_1)_{ij}$	$\frac{3}{4}$
$(Y_e(Y_e)^\dagger y_1)_{ij}$	$\frac{1}{2}$
$(y_1(y_1)^\dagger y_1)_{ij}$	$\frac{3}{2}$
$g_2^2(y_1)_{ij}$	$-\frac{9}{2}$
$g_1^2(y_1)_{ij}$	$-\frac{15}{2}$
$T_\nu(y_1)_{ij}$	0
$(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger y_1)_{ij}$	0
$\mathrm{Tr}(y_1(y_2)^\dagger)(y_2)_{ij}$	0
$(y_1(y_2)^\dagger y_2)_{ij}$	0

$\beta^{(1)}_{y_1^{\mathrm{conj}}}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)(y_1)_{ij}$	4	0	0
$T_\nu^{(1)}(y_1^{\mathrm{conj}})_{ij}$	2	0	0
$(y_1(y_1)^\dagger y_1^{\mathrm{conj}})_{ij}$	$\frac{5}{2}$	0	0
$(y_1^{\mathrm{conj}}(y_1)^\dagger y_1)_{ij}$	$\frac{5}{2}$	0	0
$(y_2(y_2)^\dagger y_1^{\mathrm{conj}})_{ij}$	$\frac{3}{4}$	0	0
$(Y_e(Y_e)^\dagger y_1^{\mathrm{conj}})_{ij}$	$\frac{1}{2}$	0	0
$(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger y_1^{\mathrm{conj}})_{ij}$	1	0	0
$g_2^2(y_1^{\mathrm{conj}})_{ij}$	$-\frac{9}{2}$	0	0
$g_1^2(y_1^{\mathrm{conj}})_{ij}$	$-\frac{3}{2}$	0	0
$T_\nu(y_1^{\mathrm{conj}})_{ij}$	2	0	0
$((y_2(y_2)^\dagger y_1^{\mathrm{conj}})_{ij} + (Y_e(Y_e)^\dagger y_1^{\mathrm{conj}})_{ij} + 3(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger y_1^{\mathrm{conj}})_{ij})$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)(y_1)_{ij}$	0	0	0
$T_\nu^{(1)}(y_1^{\mathrm{conj}})_{ij}$	0	0	0
$(y_1(y_1)^\dagger y_1^{\mathrm{conj}})_{ij}$	0	0	0
$(y_1^{\mathrm{conj}}(y_1)^\dagger y_1)_{ij}$	0	0	0
$(y_2(y_2)^\dagger y_1^{\mathrm{conj}})_{ij}$	0	0	0
$(Y_e(Y_e)^\dagger y_1^{\mathrm{conj}})_{ij}$	0	0	0
$(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger y_1^{\mathrm{conj}})_{ij}$	0	0	0
$g_2^2(y_1^{\mathrm{conj}})_{ij}$	0	0	0
$g_1^2(y_1^{\mathrm{conj}})_{ij}$	0	0	0
$T_\nu(y_1^{\mathrm{conj}})_{ij}$	0	0	0
$((y_2(y_2)^\dagger y_1^{\mathrm{conj}})_{ij} + (Y_e(Y_e)^\dagger y_1^{\mathrm{conj}})_{ij} + 3(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger y_1^{\mathrm{conj}})_{ij})$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)(y_1)_{ij}$	0	0	0
$T_\nu^{(1)}(y_1^{\mathrm{conj}})_{ij}$	0	0	0
$(y_1(y_1)^\dagger y_1^{\mathrm{conj}})_{ij}$	0	0	0
$(y_1^{\mathrm{conj}}(y_1)^\dagger y_1)_{ij}$	0	0	0
$(y_2(y_2)^\dagger y_1^{\mathrm{conj}})_{ij}$	0	0	0
$(Y_e(Y_e)^\dagger y_1^{\mathrm{conj}})_{ij}$	0	0	0
$(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger y_1^{\mathrm{conj}})_{ij}$	0	0	0
$g_2^2(y_1^{\mathrm{conj}})_{ij}$	0	0	0
$g_1^2(y_1^{\mathrm{conj}})_{ij}$	0	0	0
$T_\nu(y_1^{\mathrm{conj}})_{ij}$	0	0	0
$((y_2(y_2)^\dagger y_1^{\mathrm{conj}})_{ij} + (Y_e(Y_e)^\dagger y_1^{\mathrm{conj}})_{ij} + 3(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger y_1^{\mathrm{conj}})_{ij})$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\mathrm{Tr}(y_1^{\mathrm{conj}}(y_1)^\dagger)(y_1)_{ij}$	2	0	0
$T_\nu^{(1)}(y_1^{\mathrm{conj}})_{ij}$	1	0	0
$(y_1(y_1)^\dagger y_1^{\mathrm{conj}})_{ij}$	$\frac{29}{12}$	0	0
$(y_1^{\mathrm{conj}}(y_1)^\dagger y_1)_{ij}$	$\frac{29}{12}$	0	0
$(y_2(y_2)^\dagger y_1^{\mathrm{conj}})_{ij}$	0	0	0
$(Y_e(Y_e)^\dagger y_1^{\mathrm{conj}})_{ij}$	0	0	0
$(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger y_1^{\mathrm{conj}})_{ij}$	0	0	0
$g_2^2(y_1^{\mathrm{conj}})_{ij}$	$-\frac{9}{2}$	0	0
$g_1^2(y_1^{\mathrm{conj}})_{ij}$	$-\frac{3}{2}$	0	0
$T_\nu(y_1^{\mathrm{conj}})_{ij}$	1	0	0

$((y_2(y_2)^\dagger y_1^{\rm conj})_{ij} + (Y_e(Y_e)^\dagger y_1^{\rm conj})_{ij} + 3(y_1^{\rm conj}(y_1^{\rm conj})^\dagger y_1^{\rm conj})_{ij})$	$\frac{1}{2}$	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\mathrm{Tr}(y_1^{\rm conj}(y_1)^\dagger)(y_1)_{ij}$	2	0	0
$T_\nu^{(1)}(y_1^{\rm conj})_{ij}$	1	0	0
$(y_1(y_1)^\dagger y_1^{\rm conj})_{ij}$	$\frac{29}{12}$	0	0
$(y_1^{\rm conj}(y_1)^\dagger y_1)_{ij}$	$\frac{29}{12}$	0	0
$(y_2(y_2)^\dagger y_1^{\rm conj})_{ij}$	$\frac{3}{4}$	0	0
$(Y_e(Y_e)^\dagger y_1^{\rm conj})_{ij}$	$\frac{1}{2}$	0	0
$(y_1^{\rm conj}(y_1^{\rm conj})^\dagger y_1^{\rm conj})_{ij}$	$\frac{3}{2}$	0	0
$g_2^2(y_1^{\rm conj})_{ij}$	$-\frac{9}{2}$	0	0
$g_1^2(y_1^{\rm conj})_{ij}$	$-\frac{3}{2}$	0	0
$T_\nu(y_1^{\rm conj})_{ij}$	1	0	0
$((y_2(y_2)^\dagger y_1^{\rm conj})_{ij} + (Y_e(Y_e)^\dagger y_1^{\rm conj})_{ij} + 3(y_1^{\rm conj}(y_1^{\rm conj})^\dagger y_1^{\rm conj})_{ij})$	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$\mathrm{Tr}(y_1^{\rm conj}(y_1)^\dagger)(y_1)_{ij}$	0
$T_\nu^{(1)}(y_1^{\rm conj})_{ij}$	0
$(y_1(y_1)^\dagger y_1^{\rm conj})_{ij}$	0
$(y_1^{\rm conj}(y_1)^\dagger y_1)_{ij}$	0
$(y_2(y_2)^\dagger y_1^{\rm conj})_{ij}$	0
$(Y_e(Y_e)^\dagger y_1^{\rm conj})_{ij}$	0
$(y_1^{\rm conj}(y_1^{\rm conj})^\dagger y_1^{\rm conj})_{ij}$	0
$g_2^2(y_1^{\rm conj})_{ij}$	0
$g_1^2(y_1^{\rm conj})_{ij}$	0
$T_\nu(y_1^{\rm conj})_{ij}$	0
$((y_2(y_2)^\dagger y_1^{\rm conj})_{ij} + (Y_e(Y_e)^\dagger y_1^{\rm conj})_{ij} + 3(y_1^{\rm conj}(y_1^{\rm conj})^\dagger y_1^{\rm conj})_{ij})$	0

$\beta_{y_2}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$T_\nu^{(2)}(y_2)_{ij}$	1	1	1
$(y_1(y_1)^\dagger y_2)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(y_1^{\rm conj}(y_1^{\rm conj})^\dagger y_2)_{ij}$	$\frac{1}{2}$	0	0
$(Y_e(Y_e)^\dagger y_2)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(y_2(y_2)^\dagger y_2)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(y_2)_{ij}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2(y_2)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{15}{2}$
$\mathrm{Tr}(y_2(y_2^{\rm conj})^\dagger)(y_2^{\rm conj})_{ij}$	0	2	0
$(y_2(y_2^{\rm conj})^\dagger y_2^{\rm conj})_{ij}$	0	$\frac{29}{12}$	0
$(y_2^{\rm conj}(y_2^{\rm conj})^\dagger y_2)_{ij}$	0	$\frac{29}{12}$	0
$T_\nu(y_2)_{ij}$	0	1	0
$\mathrm{Tr}(y_2(y_1)^\dagger)(y_1)_{ij}$	0	0	0
$(y_2(y_1)^\dagger y_1)_{ij}$	0	0	0
$(5(y_2(y_2^{\rm conj})^\dagger y_2^{\rm conj})_{ij} + 5(y_2^{\rm conj}(y_2^{\rm conj})^\dagger y_2)_{ij} + (Y_e(Y_e)^\dagger y_2)_{ij} + 2(y_2(y_2)^\dagger y_2)_{ij})$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$T_\nu^{(2)}(y_2)_{ij}$	1	1	1
$(y_1(y_1)^\dagger y_2)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$

$(y_1^{\rm conj}(y_1^{\rm conj})^\dagger y_2)_{ij}$	0	0	0
$(Y_e(Y_e)^\dagger y_2)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(y_2(y_2)^\dagger y_2)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(y_2)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_1^2(y_2)_{ij}$	$-\frac{15}{4}$	$-\frac{3}{4}$	$-\frac{15}{4}$
$\mathrm{Tr}(y_2(y_2^{\rm conj})^\dagger)(y_2^{\rm conj})_{ij}$	0	0	0
$(y_2(y_2^{\rm conj})^\dagger y_2^{\rm conj})_{ij}$	0	0	0
$(y_2^{\rm conj}(y_2^{\rm conj})^\dagger y_2)_{ij}$	0	0	0
$T_\nu(y_2)_{ij}$	0	0	0
$\mathrm{Tr}(y_2(y_1)^\dagger)(y_1)_{ij}$	0	1	0
$(y_2(y_1)^\dagger y_1)_{ij}$	0	1	0
$(5(y_2(y_2^{\rm conj})^\dagger y_2^{\rm conj})_{ij} + 5(y_2^{\rm conj}(y_2^{\rm conj})^\dagger y_2)_{ij} + (Y_e(Y_e)^\dagger y_2)_{ij} + 2(y_2(y_2)^\dagger y_2)_{ij})$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$T_\nu^{(2)}(y_2)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$(y_1(y_1)^\dagger y_2)_{ij}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$(y_1^{\rm conj}(y_1^{\rm conj})^\dagger y_2)_{ij}$	0	0	0
$(Y_e(Y_e)^\dagger y_2)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(y_2(y_2)^\dagger y_2)_{ij}$	$\frac{5}{4}$	$\frac{5}{4}$	$\frac{5}{4}$
$g_2^2(y_2)_{ij}$	$-\frac{33}{4}$	$-\frac{33}{4}$	$-\frac{33}{4}$
$g_1^2(y_2)_{ij}$	$-\frac{15}{4}$	$-\frac{3}{4}$	$-\frac{15}{4}$
$\mathrm{Tr}(y_2(y_2^{\rm conj})^\dagger)(y_2^{\rm conj})_{ij}$	0	0	0
$(y_2(y_2^{\rm conj})^\dagger y_2^{\rm conj})_{ij}$	0	0	0
$(y_2^{\rm conj}(y_2^{\rm conj})^\dagger y_2)_{ij}$	0	0	0
$T_\nu(y_2)_{ij}$	0	0	0
$\mathrm{Tr}(y_2(y_1)^\dagger)(y_1)_{ij}$	0	$\frac{3}{2}$	0
$(y_2(y_1)^\dagger y_1)_{ij}$	0	$\frac{1}{2}$	0
$(5(y_2(y_2^{\rm conj})^\dagger y_2^{\rm conj})_{ij} + 5(y_2^{\rm conj}(y_2^{\rm conj})^\dagger y_2)_{ij} + (Y_e(Y_e)^\dagger y_2)_{ij} + 2(y_2(y_2)^\dagger y_2)_{ij})$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$T_\nu^{(2)}(y_2)_{ij}$	2	2	2
$(y_1(y_1)^\dagger y_2)_{ij}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$(y_1^{\rm conj}(y_1^{\rm conj})^\dagger y_2)_{ij}$	$\frac{3}{4}$	0	0
$(Y_e(Y_e)^\dagger y_2)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	0
$(y_2(y_2)^\dagger y_2)_{ij}$	1	1	0
$g_2^2(y_2)_{ij}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2(y_2)_{ij}$	$-\frac{3}{2}$	$-\frac{15}{2}$	$-\frac{3}{2}$
$\mathrm{Tr}(y_2(y_2^{\rm conj})^\dagger)(y_2^{\rm conj})_{ij}$	0	0	4
$(y_2(y_2^{\rm conj})^\dagger y_2^{\rm conj})_{ij}$	0	0	0
$(y_2^{\rm conj}(y_2^{\rm conj})^\dagger y_2)_{ij}$	0	0	0
$T_\nu(y_2)_{ij}$	0	0	2
$\mathrm{Tr}(y_2(y_1)^\dagger)(y_1)_{ij}$	0	0	0
$(y_2(y_1)^\dagger y_1)_{ij}$	0	0	0
$(5(y_2(y_2^{\rm conj})^\dagger y_2^{\rm conj})_{ij} + 5(y_2^{\rm conj}(y_2^{\rm conj})^\dagger y_2)_{ij} + (Y_e(Y_e)^\dagger y_2)_{ij} + 2(y_2(y_2)^\dagger y_2)_{ij})$	0	0	$\frac{1}{2}$

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$T_\nu^{(2)}(y_2)_{ij}$	1	1	1
$(y_1(y_1)^\dagger y_2)_{ij}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$(y_1^{\rm conj}(y_1^{\rm conj})^\dagger y_2)_{ij}$	$\frac{3}{4}$	0	0
$(Y_e(Y_e)^\dagger y_2)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(y_2(y_2)^\dagger y_2)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$

$g_2^2(y_2)_{ij}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2(y_2)_{ij}$	$-\frac{3}{2}$	$-\frac{15}{2}$	$-\frac{3}{2}$
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)(y_2^{\mathrm{conj}})_{ij}$	0	0	2
$(y_2(y_2^{\mathrm{conj}})^\dagger y_2^{\mathrm{conj}})_{ij}$	0	0	$\frac{29}{12}$
$(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger y_2)_{ij}$	0	0	$\frac{29}{12}$
$T_\nu(y_2)_{ij}$	0	0	1
$\mathrm{Tr}(y_2(y_1)^\dagger)(y_1)_{ij}$	0	0	0
$(y_2(y_1)^\dagger y_1)_{ij}$	0	0	0
$(5(y_2(y_2^{\mathrm{conj}})^\dagger y_2^{\mathrm{conj}})_{ij} + 5(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger y_2)_{ij} + (Y_e(Y_e)^\dagger y_2)_{ij} + 2(y_2(y_2)^\dagger y_2)_{ij})$	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$T_\nu^{(2)}(y_2)_{ij}$	1
$(y_1(y_1)^\dagger y_2)_{ij}$	$\frac{3}{4}$
$(y_1^{\mathrm{conj}}(y_1^{\mathrm{conj}})^\dagger y_2)_{ij}$	0
$(Y_e(Y_e)^\dagger y_2)_{ij}$	$\frac{1}{2}$
$(y_2(y_2)^\dagger y_2)_{ij}$	$\frac{3}{2}$
$g_2^2(y_2)_{ij}$	$-\frac{9}{2}$
$g_1^2(y_2)_{ij}$	$-\frac{15}{2}$
$\mathrm{Tr}(y_2(y_2^{\mathrm{conj}})^\dagger)(y_2^{\mathrm{conj}})_{ij}$	0
$(y_2(y_2^{\mathrm{conj}})^\dagger y_2^{\mathrm{conj}})_{ij}$	0
$(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger y_2)_{ij}$	0
$T_\nu(y_2)_{ij}$	0
$\mathrm{Tr}(y_2(y_1)^\dagger)(y_1)_{ij}$	0
$(y_2(y_1)^\dagger y_1)_{ij}$	0
$(5(y_2(y_2^{\mathrm{conj}})^\dagger y_2^{\mathrm{conj}})_{ij} + 5(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger y_2)_{ij} + (Y_e(Y_e)^\dagger y_2)_{ij} + 2(y_2(y_2)^\dagger y_2)_{ij})$	0

$\beta^{(1)}_{y_2^{\mathrm{conj}}}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger)(y_2)_{ij}$	0	2	0
$T_\nu^{(2)}(y_2^{\mathrm{conj}})_{ij}$	0	1	0
$(y_1(y_1)^\dagger y_2^{\mathrm{conj}})_{ij}$	0	$\frac{1}{2}$	0
$(y_2(y_2)^\dagger y_2^{\mathrm{conj}})_{ij}$	0	$\frac{29}{12}$	0
$(y_2^{\mathrm{conj}}(y_2)^\dagger y_2)_{ij}$	0	$\frac{29}{12}$	0
$(Y_e(Y_e)^\dagger y_2^{\mathrm{conj}})_{ij}$	0	$\frac{1}{2}$	0
$(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger y_2^{\mathrm{conj}})_{ij}$	0	$\frac{3}{2}$	0
$g_2^2(y_2^{\mathrm{conj}})_{ij}$	0	$-\frac{9}{2}$	0
$g_1^2(y_2^{\mathrm{conj}})_{ij}$	0	$-\frac{3}{2}$	0
$T_\nu(y_2^{\mathrm{conj}})_{ij}$	0	1	0
$(5(y_2(y_2)^\dagger y_2^{\mathrm{conj}})_{ij} + 5(y_2^{\mathrm{conj}}(y_2)^\dagger y_2)_{ij} + (Y_e(Y_e)^\dagger y_2^{\mathrm{conj}})_{ij} + 2(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger y_2^{\mathrm{conj}})_{ij})$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\mathrm{Tr}(y_2^{\mathrm{conj}}(y_2)^\dagger)(y_2)_{ij}$	0	0	0
$T_\nu^{(2)}(y_2^{\mathrm{conj}})_{ij}$	0	0	0
$(y_1(y_1)^\dagger y_2^{\mathrm{conj}})_{ij}$	0	0	0
$(y_2(y_2)^\dagger y_2^{\mathrm{conj}})_{ij}$	0	0	0
$(y_2^{\mathrm{conj}}(y_2)^\dagger y_2)_{ij}$	0	0	0
$(Y_e(Y_e)^\dagger y_2^{\mathrm{conj}})_{ij}$	0	0	0
$(y_2^{\mathrm{conj}}(y_2^{\mathrm{conj}})^\dagger y_2^{\mathrm{conj}})_{ij}$	0	0	0
$g_2^2(y_2^{\mathrm{conj}})_{ij}$	0	0	0
$g_1^2(y_2^{\mathrm{conj}})_{ij}$	0	0	0

$T_\nu(y_2^{\text{conj}})_{ij}$	0	0	0
$(5(y_2(y_2)^\dagger y_2^{\text{conj}})_{ij} + 5(y_2^{\text{conj}}(y_2)^\dagger y_2)_{ij} + (Y_e(Y_e)^\dagger y_2^{\text{conj}})_{ij} + 2(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger y_2^{\text{conj}})_{ij})$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\text{Tr}(y_2^{\text{conj}}(y_2)^\dagger)(y_2)_{ij}$	0	0	0
$T_\nu^{(2)}(y_2^{\text{conj}})_{ij}$	0	0	0
$(y_1(y_1)^\dagger y_2^{\text{conj}})_{ij}$	0	0	0
$(y_2(y_2)^\dagger y_2^{\text{conj}})_{ij}$	0	0	0
$(y_2^{\text{conj}}(y_2)^\dagger y_2)_{ij}$	0	0	0
$(Y_e(Y_e)^\dagger y_2^{\text{conj}})_{ij}$	0	0	0
$(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger y_2^{\text{conj}})_{ij}$	0	0	0
$g_2^2(y_2^{\text{conj}})_{ij}$	0	0	0
$g_1^2(y_2^{\text{conj}})_{ij}$	0	0	0
$T_\nu(y_2^{\text{conj}})_{ij}$	0	0	0
$(5(y_2(y_2)^\dagger y_2^{\text{conj}})_{ij} + 5(y_2^{\text{conj}}(y_2)^\dagger y_2)_{ij} + (Y_e(Y_e)^\dagger y_2^{\text{conj}})_{ij} + 2(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger y_2^{\text{conj}})_{ij})$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\text{Tr}(y_2^{\text{conj}}(y_2)^\dagger)(y_2)_{ij}$	0	0	4
$T_\nu^{(2)}(y_2^{\text{conj}})_{ij}$	0	0	2
$(y_1(y_1)^\dagger y_2^{\text{conj}})_{ij}$	0	0	$\frac{3}{4}$
$(y_2(y_2)^\dagger y_2^{\text{conj}})_{ij}$	0	0	0
$(y_2^{\text{conj}}(y_2)^\dagger y_2)_{ij}$	0	0	0
$(Y_e(Y_e)^\dagger y_2^{\text{conj}})_{ij}$	0	0	0
$(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger y_2^{\text{conj}})_{ij}$	0	0	0
$g_2^2(y_2^{\text{conj}})_{ij}$	0	0	$-\frac{9}{2}$
$g_1^2(y_2^{\text{conj}})_{ij}$	0	0	$-\frac{3}{2}$
$T_\nu(y_2^{\text{conj}})_{ij}$	0	0	2
$(5(y_2(y_2)^\dagger y_2^{\text{conj}})_{ij} + 5(y_2^{\text{conj}}(y_2)^\dagger y_2)_{ij} + (Y_e(Y_e)^\dagger y_2^{\text{conj}})_{ij} + 2(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger y_2^{\text{conj}})_{ij})$	0	0	$\frac{1}{2}$

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\text{Tr}(y_2^{\text{conj}}(y_2)^\dagger)(y_2)_{ij}$	0	0	2
$T_\nu^{(2)}(y_2^{\text{conj}})_{ij}$	0	0	1
$(y_1(y_1)^\dagger y_2^{\text{conj}})_{ij}$	0	0	$\frac{3}{4}$
$(y_2(y_2)^\dagger y_2^{\text{conj}})_{ij}$	0	0	$\frac{29}{12}$
$(y_2^{\text{conj}}(y_2)^\dagger y_2)_{ij}$	0	0	$\frac{29}{12}$
$(Y_e(Y_e)^\dagger y_2^{\text{conj}})_{ij}$	0	0	$\frac{1}{2}$
$(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger y_2^{\text{conj}})_{ij}$	0	0	$\frac{3}{2}$
$g_2^2(y_2^{\text{conj}})_{ij}$	0	0	$-\frac{9}{2}$
$g_1^2(y_2^{\text{conj}})_{ij}$	0	0	$-\frac{3}{2}$
$T_\nu(y_2^{\text{conj}})_{ij}$	0	0	1
$(5(y_2(y_2)^\dagger y_2^{\text{conj}})_{ij} + 5(y_2^{\text{conj}}(y_2)^\dagger y_2)_{ij} + (Y_e(Y_e)^\dagger y_2^{\text{conj}})_{ij} + 2(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger y_2^{\text{conj}})_{ij})$	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$\text{Tr}(y_2^{\text{conj}}(y_2)^\dagger)(y_2)_{ij}$	0
$T_\nu^{(2)}(y_2^{\text{conj}})_{ij}$	0
$(y_1(y_1)^\dagger y_2^{\text{conj}})_{ij}$	0
$(y_2(y_2)^\dagger y_2^{\text{conj}})_{ij}$	0
$(y_2^{\text{conj}}(y_2)^\dagger y_2)_{ij}$	0
$(Y_e(Y_e)^\dagger y_2^{\text{conj}})_{ij}$	0

$(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger y_2^{\text{conj}})_{ij}$	0
$g_2^2(y_2^{\text{conj}})_{ij}$	0
$g_1^2(y_2^{\text{conj}})_{ij}$	0
$T_\nu(y_2^{\text{conj}})_{ij}$	0
$(5(y_2(y_2)^\dagger y_2^{\text{conj}})_{ij} + 5(y_2^{\text{conj}}(y_2)^\dagger y_2)_{ij} + (Y_e(Y_e)^\dagger y_2^{\text{conj}})_{ij} + 2(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger y_2^{\text{conj}})_{ij})$	0

$\beta_{Y_d}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8	-8	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1	1	1

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8	-8	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1	1	1

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8	-8	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1	1	1

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8	-8	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1	1	1

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8	-8	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1	1	1

$\mathcal{S}$	E ($\alpha=-4$)
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1

$\beta^{(1)}_{Y_e}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$(y_1(y_1)^\dagger Y_e)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger Y_e)_{ij}$	$\frac{1}{2}$	0	0
$(y_2(y_2)^\dagger Y_e)_{ij}$	$\frac{3}{4}$	0	$\frac{3}{4}$
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$	0	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1	1	1
$((y_2(y_2)^\dagger Y_e)_{ij}+(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger Y_e)_{ij}+2(Y_e(Y_e)^\dagger Y_e)_{ij})$	0	$\frac{3}{4}$	0
$((y_2(y_2)^\dagger Y_e)_{ij}+(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger Y_e)_{ij}+3(Y_e(Y_e)^\dagger Y_e)_{ij})$	0	0	0
$(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger Y_e)_{ij}$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$(y_1(y_1)^\dagger Y_e)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger Y_e)_{ij}$	0	0	0
$(y_2(y_2)^\dagger Y_e)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1	1	1
$((y_2(y_2)^\dagger Y_e)_{ij}+(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger Y_e)_{ij}+2(Y_e(Y_e)^\dagger Y_e)_{ij})$	0	0	0
$((y_2(y_2)^\dagger Y_e)_{ij}+(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger Y_e)_{ij}+3(Y_e(Y_e)^\dagger Y_e)_{ij})$	0	0	0
$(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger Y_e)_{ij}$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$(y_1(y_1)^\dagger Y_e)_{ij}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger Y_e)_{ij}$	0	0	0
$(y_2(y_2)^\dagger Y_e)_{ij}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1	1	1
$((y_2(y_2)^\dagger Y_e)_{ij}+(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger Y_e)_{ij}+2(Y_e(Y_e)^\dagger Y_e)_{ij})$	0	0	0
$((y_2(y_2)^\dagger Y_e)_{ij}+(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger Y_e)_{ij}+3(Y_e(Y_e)^\dagger Y_e)_{ij})$	0	0	0
$(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger Y_e)_{ij}$	0	0	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$(y_1(y_1)^\dagger Y_e)_{ij}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$

$(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger Y_e)_{ij}$	$\frac{3}{4}$	0	0
$(y_2(y_2)^\dagger Y_e)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	0
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	0
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1	1	1
$((y_2(y_2)^\dagger Y_e)_{ij} + (y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger Y_e)_{ij} + 2(Y_e(Y_e)^\dagger Y_e)_{ij})$	0	0	0
$((y_2(y_2)^\dagger Y_e)_{ij} + (y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger Y_e)_{ij} + 3(Y_e(Y_e)^\dagger Y_e)_{ij})$	0	0	$\frac{1}{2}$
$(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger Y_e)_{ij}$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$(y_1(y_1)^\dagger Y_e)_{ij}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger Y_e)_{ij}$	$\frac{3}{4}$	0	0
$(y_2(y_2)^\dagger Y_e)_{ij}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1	1	1
$((y_2(y_2)^\dagger Y_e)_{ij} + (y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger Y_e)_{ij} + 2(Y_e(Y_e)^\dagger Y_e)_{ij})$	0	0	0
$((y_2(y_2)^\dagger Y_e)_{ij} + (y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger Y_e)_{ij} + 3(Y_e(Y_e)^\dagger Y_e)_{ij})$	0	0	0
$(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger Y_e)_{ij}$	0	0	$\frac{3}{4}$

$\mathcal{S}$	E ($\alpha = -4$)
$(y_1(y_1)^\dagger Y_e)_{ij}$	$\frac{3}{4}$
$(y_1^{\text{conj}}(y_1^{\text{conj}})^\dagger Y_e)_{ij}$	0
$(y_2(y_2)^\dagger Y_e)_{ij}$	$\frac{3}{4}$
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1
$((y_2(y_2)^\dagger Y_e)_{ij} + (y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger Y_e)_{ij} + 2(Y_e(Y_e)^\dagger Y_e)_{ij})$	0
$((y_2(y_2)^\dagger Y_e)_{ij} + (y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger Y_e)_{ij} + 3(Y_e(Y_e)^\dagger Y_e)_{ij})$	0
$(y_2^{\text{conj}}(y_2^{\text{conj}})^\dagger Y_e)_{ij}$	0

$\beta_{Y_u}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8	-8	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1	1	1

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8	-8	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$

$T(Y_u)_{ij}$	1	1	1

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8	-8	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1	1	1

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8	-8	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1	1	1

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8	-8	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1	1	1

$\mathcal{S}$	E ($\alpha = -4$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1